LINEAR LATTICE PDE WITH BLOCK CONSTRAINTS

Abstract. We study scalar and gauge-field operators on separated multiscale lattices with prescribed block averages. The estimates retain the physical fine and coarse pairings, the actual determining rows and the boundary convention of each operator. Scalar regularity and weighted row lifts lead to constrained inverse and reconstruction operators. Local carrier expansions control changes of determining region and the admitted covariance matrix functions. Physical fluctuation gaps use the full specified Lagrangian Hessian, including the constraint curvature and its multiplier. The coefficient classes are stated independently of a nonlinear construction of a background.

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1. Introduction

Block constraints give an elliptic problem a mass at the scale of each assigned block. In a gauge theory, a second constraint removes the gauge directions. These operations interact at a seam between two scales and at the boundary of the action region. An inverse estimate must retain the actual rows and adjoints there. Replacing the boundary problem by a periodic one, or a physical transpose by an unweighted array transpose, changes the operator.

We develop the linear estimates in four stages. Scalar heat and singular-integral estimates supply the local regularity. Energetic row lifts and normalized Gram bounds then give the constrained quadratic estimates. Exact Schur and current identities construct the Green and reconstruction operators. Finally, local carrier expansions preserve the estimates before summation and allow conditioning and matrix functions on their declared complex domains.

The block maps and their finite geometric identities are the input from [1]. Scalar covariant bounds originate in [2, pp. 573–574]. Flat and mixed-scale constrained propagators are developed in [3, Sections D–F] and [4, Sections A–C]. Regular-background propagators and local expansions are treated in [6, Sections C–E]. The derivative ranges of these results differ. In particular, the

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covariant-Laplacian member of [6, eq. (3.42)] is excluded in that paper's Theorems 3.12 and 3.13. We state each operator's output range separately.

The results are uniform in the mesh, depth and admitted volume. The geometric and coefficient parameters are fixed in the stated order. We do not construct the nonlinear critical background. For physical fluctuation statements its complete Hessian identification is part of the input, including both the inner constraint multiplier and the next-fibre pivot correction. The consequences concern the specified forms and do not establish admission of later nonlinear substitutions or retained histories.

2. Geometry, pairings and coefficient classes

Fix an odd integer $L \geq 3$, an integer $k \geq 0$ and a compatible four-dimensional torus. For the block maps, compatibility means that every period is an integer multiple of the largest mesh used and contains at least two sites at that mesh. A next-lattice assertion includes its next mesh in this requirement. Period-two axes retain their distinct stored oriented bonds. Purely scalar assertions need only the divisibility conditions stated in their own setting. In units of the top lattice, put $\eta = L^{-k}$ and $\xi_j = L^{j-k}$. The fine lattice has spacing η. A positive bond is stored once; the value on its reverse is determined by orientation and transport. We use the real space $\mathfrak{g}$ of traceless Hermitian 2×2 matrices, identified with $\mathfrak{su}(2)$ by $A \mapsto iA$. Thus $e^{i\eta A}$ is unitary for real $A \in \mathfrak{g}$. The color pairing is $\langle A, B\rangle = \frac{1}{2}\operatorname{Tr}(AB)$, extended bilinearly to the complexification.

Let $\Omega_0 \supset \cdots \supset \Omega_k$ use the nested centred tilings, with Ω_j a union of native level-j blocks of side ξ_j, and put $\Omega_{k+1} = \varnothing$. The assigned blocks B_y, called pillars, partition Ω_0 with a half-open convention. A pillar assigned to level j has side $\xi_y = \xi_j$. The separation assumption is

$$\operatorname{dist}_\infty(\Omega_{j+1}, \Omega_j^c) > \rho\xi_j, \qquad 0 \leq j < k. \tag{2.i}$$

Here ρ is fixed sufficiently large. It includes the margins needed for all stated interior boxes and complete averaging carriers. A field row at level j joins two level-j sites and has length $\xi_c = \xi_j$. Interior and outward rows are retained; an inward bond entering Ω_{j+1} supplies no row. At the outer endpoint of an outward row, the own assigned site level is $j - 1$. A level-j endpoint average there includes the additional transported average of those finer site values.

There are two boundary conventions. The proper problem uses the original active graph and the action of its inner plaquettes. The compressed problem extends an unknown by zero and restricts the original ambient operator, including stencils crossing the boundary. These forms can differ. Each result below states which one it uses. On a periodic lattice all adjoints are periodic. Interior estimates use complete stencils on the specified larger box. They assert no regularity across an arbitrary rough exterior boundary.

2.1. Physical coordinates and adjoints.

Fine site and bond fields have pairing $\langle u, v\rangle_f = \eta^4 \sum_x \langle u(x), v(x)\rangle$. The sum is over the appropriate sites or stored positive bonds. Site-top and field-row pairings are

$$\langle b, c\rangle_s = \sum_y \xi_y^4 \langle b_y, c_y\rangle, \qquad \langle B, C\rangle_r = \sum_c \xi_c^4 \langle B_c, C_c\rangle. \tag{2.ii}$$

We use T^{T} for the transpose between the declared bilinear pairings. This transpose is holomorphic in complex coefficients. At real coefficients it is also the real Hilbert adjoint. Positivity and operator norms use the corresponding Hermitian inner products. A plain array transpose does not include the factors in eq. 2.ii.

For a connection U, the site gradient is

$$(D_U\lambda)(b) = \eta^{-1}\{\lambda(b_-) - \operatorname{Ad}(U_b)\lambda(b_+)\}. \tag{2.iii}$$

The positive site Laplacian is $D_U^{\mathsf{T}} D_U$. Component covariant differences of a bond field are written ∇_U. For a site-based component put $T_\mu v(x) = \operatorname{Ad} U(x, x+\eta e_\mu) v(x+\eta e_\mu)$ and $\nabla_{\mu,U} v = (T_\mu - I)v/\eta$.

The covariant curl on a positive $\mu\nu$ plaquette is

$$(\mathrm{d}_U v)_{\mu\nu} = \eta^{-1}\{(T_\mu - I)v_\nu - (T_\nu - I)v_\mu\}. \tag{2.iv}$$

Its transpose uses the declared plaquette set and physical η^4 measure. In particular it is different from the component differences ∇_U.

Write q for the actual transported own-pillar site average, and Q for the per-length derivative of the straightened field average. The raw constraint derivative is $\mathsf{m}Q$, where $(\mathsf{m}B)_c = \xi_c B_c$. Thus a raw datum is $b_c = \xi_c B_c$. Its counting-measure multiplier $\widehat{\omega}$ corresponds to the physical per-length multiplier by

$$\sum_c \langle \widehat{\omega}_c, b_c\rangle = \langle \omega, B\rangle_r, \qquad \widehat{\omega}_c = \xi_c^3 \omega_c. \tag{2.v}$$

The analogous conversion is used at every constrained Hessian. In particular, a coarse-bond current cannot be substituted for a new-row multiplier without the derivative of the relevant constraint chart.

At a real connection, the site average is an arithmetic mean of adjoint transports in each disjoint pillar. Its physical transpose satisfies $qq^{\mathsf{T}} = I$. Indeed the transpose inserts the inverse transport, and the fine density η^4/ξ_y^4 sums to one on B_y. The same identity holds for the holomorphic orthogonal transports on their admitted complex branch. The exact straightened row identity is

$$QD_U = \overline{D}\, q. \tag{2.vi}$$

Here $\overline{D}$ includes the extended endpoint average just described. An ordinary difference of two own-top values would give a different map at an outward row. The block-averaging identities used in defining these maps are the supplied geometric input; the inverse and coercivity estimates for them are proved below.

2.2. **The layer metric and summation.** Put $\xi(x) = \xi_y$ on B_y and extend it by η outside Ω_0. Equivalently, the metric scale is ξ_j in layer $\Omega_j \setminus \Omega_{j+1}$ and is η outside Ω_1. Define

$$d(x,z) = \inf_{\gamma: x\to z} \int_\gamma \frac{ds_\infty}{\xi(s)}. \tag{2.vii}$$

Paths are taken in the torus or in a lift to its periodic cover. Changing the assignment of a cell face does not change the infimum. Distances between assigned cells mean distances between their fixed centers, with a bounded enlargement when a carrier is used. The scale outside the active region prevents a path from acquiring a cheaper metric by leaving that region.

Lemma 2.1 (Growth and scale ratios)**.** For fixed L and $\rho \ge 1$, the assigned centers obey

$$\#\{z : d(y,z) \le R\} \le C_L(1+R)^5 \exp\{8(\log L)R/\rho\}, \qquad R \ge 0. \tag{2.viii}$$

If y, z have levels j, i, then $d(y,z) \ge \rho(|i-j|-1)_+$. Consequently, for any real p,

$$(\xi_y/\xi_z)^p \le L^{|p|} \exp\{|p|(\log L)d(y,z)/\rho\}. \tag{2.ix}$$

The constants are independent of depth and compatible volume. ∎

Proof. A path between levels i and j crosses every complete intervening layer. By eq. 2.i, each such crossing has physical length greater than $\rho\xi_r$ within layer r. It therefore costs at least ρ in the metric. The crossing arcs have disjoint interiors. At most one layer is lost at an endpoint, proving the level bound and eq. 2.ix.

A path of metric length at most $R+1$ starting at level j uses no scale greater than $L\xi_j L^{(R+1)/\rho}$. Its physical length is at most $(R+1)L\xi_j L^{(R+1)/\rho}$. At any reachable level, the smallest possible mesh is $\xi_j L^{-1-(R+1)/\rho}$. Counting that level's grid points in the resulting physical sup-norm ball gives $C_L(1+R)^4 L^{8R/\rho}$. There are at most $C(1+(R+1)/\rho)$ reachable levels. Summing proves eq. 2.viii. On a torus, choose for each reached center the lift reached by its path. Distinct periodic

centers give distinct lifted grid points. Counting the whole cover ball therefore remains an upper bound. □

For species s,t with positive amplitudes $a_s(y), a_t(y)$, let $k_{st}(y,z)$ be the norm of $P_y K P_z$ between the normalized cell sup norms $a_t(z)^{-1}\|\cdot\|_\infty$ and $a_s(y)^{-1}\|\cdot\|_\infty$. Every physical integration weight is included before this norm is taken. Define

$$\|K\|_{\delta;s\leftarrow t} = \max\left\{\sup_y \sum_z e^{\delta d(y,z)} k_{st}(y,z), \sup_z \sum_y e^{\delta d(y,z)} k_{st}(y,z)\right\}. \tag{2.x}$$

Typical amplitudes are ξ^{-1} for a potential, ξ^{-3} for a force or physical row multiplier, and one for a site gauge parameter or raw datum.

Lemma 2.2 (Weighted composition)**.** The norms in eq. 2.x are submultiplicative for compatible species. An entry estimate $k_{st}(y,z) \le C(\xi_y/\xi_z)^p e^{-md(y,z)}$ implies a uniform bound in that norm whenever

$$m - \delta > (8 + |p|)(\log L)/\rho. \tag{2.xi}$$

■

Proof. Insert the cell partition between two operators. The triangle inequality gives $k^{KL}_{su}(y,z) \le \sum_w k^K_{st}(y,w) k^L_{tu}(w,z)$. The triangle inequality for d then proves the row and column bounds for the product. For the second assertion, sum over unit metric annuli and use eq. 2.viii and eq. 2.ix. The resulting series is bounded by a constant times

$$\sum_{n\ge 0} (n+2)^5 e^{-[m-\delta-(8+|p|)\log L/\rho]n}.$$

It converges under eq. 2.xi. The same argument centered at z proves the column estimate. □

A cell norm does not remove the Coulomb singularity of a fine kernel inside a cell. A pointwise kernel estimate will therefore be stated separately from a cell estimate. We choose geometric separation with a strict reserve in eq. 2.xi for the finitely many scale exponents that occur below.

2.3. **Coefficient hypotheses.** The basic real class consists of unitary connections on the declared graph whose actual averaging branches are admitted and whose plaquette curvature satisfies

$$F_U(p) = (i\eta^2)^{-1} \log U(\partial p), \qquad \sup_p \mathsf{L}(p)^2 |F_U(p)| \le \kappa. \tag{2.xii}$$

Here L is the assigned scale on a complete stencil; its values on that stencil are comparable within fixed geometric factors. The logarithm is the branch near the identity. The curvature ceiling κ is small and fixed. A theorem using stencils crossing an action boundary requires eq. 2.xii on those stencils as well.

Some results require the stronger regular class. On each stated complete box of side a fixed multiple of ξ_y, it admits a real gauge $U = e^{i\eta a}$ with $\xi_y\|a\|_\infty + \xi_y^2\|\nabla a\|_\infty \le \kappa_1$. When a $C^{1,\alpha}$ potential is needed, it is stated explicitly. For the strong scalar estimate we require either the explicit coefficient condition eq. 3.xix or an admitted real Coulomb frame with the hypotheses of Corollary 3.12. On sufficiently enlarged complete real cubes, Proposition 3.13 constructs the Coulomb frame from the curvature bound. The actual Wilson current then gives the additional regularity. An independent current parameter cannot replace the actual current in this deduction.

For a complex tube, write $\mathbf{U} = e^{i\eta A'} U$ about an admitted real center, and impose the stated bounds on $\xi_y|A'|$ and $\xi_y^2|\nabla_U A'|$. Independent complex currents $\mathbf{J}$ satisfy the separate bound

$$\sup_b \mathsf{L}(b)^3 |\mathbf{J}(b)| \le \gamma.$$

Their identification with the derivative of the inner Wilson action is an additional hypothesis. An on-shell assertion explicitly requires that identification on every stencil it uses.

The physical Hessian class specifies the complete Lagrangian form, including the mean-constraint multiplier and the actual outer pivot curvature. This is input data independent of the existence of a nonlinear critical field. The estimates below prove its linear consequences. They neither construct that field nor identify a different form by a Ward identity alone.

Unless stated otherwise, constants may depend on fixed L, the positive row-weight bounds and the declared collar geometry. They do not depend on η, depth, the shapes or number of components, or compatible volume. Constants indexed by p or α may also depend on those fixed exponents. A radius or constant that depends on a localization length is identified where it occurs.

3. Scalar kernels and interior regularity

For this section let $\eta > 0$ be the mesh and use measure η^4 at each site and each positively oriented bond. Write

$$\partial_\mu u(x) = \frac{u(x+\eta e_\mu) - u(x)}{\eta}, \qquad \partial_\mu^* u(x) = \frac{u(x-\eta e_\mu) - u(x)}{\eta}, \qquad \Delta_\eta = \sum_{\mu=1}^{4} \partial_\mu^* \partial_\mu.$$

Here D^a denotes any product of a of these differences and $du = (\partial_\mu u)_\mu$. A convolution kernel is a density for $\eta^4 \sum_x$. In particular, the kernel of the identity is $\eta^{-4} 1_{\{0\}}$. We use $|x|_1$ in kernel estimates and the distance it induces on a periodic lattice. Changing to Euclidean distance changes only dimension constants.

3.1. **Heat and free Green kernels.**

Proposition 3.1 (Free kernel bounds)**.** Let p_t^η be the heat kernel on $\eta\mathbb{Z}^4$, and put $s = t + \eta^2$. It has mass one and is nonnegative. For every fixed nonnegative integer a,

$$|D^a p_t^\eta(x)| \le C_a s^{-(4+a)/2} \exp(-c|x|_1/\sqrt{s}), \qquad t \ge 0. \tag{3.i}$$

For $\mu \ge 0$ define $G_\mu^\eta = \int_0^\infty e^{-\mu^2 t} p_t^\eta \, dt$, and put $m = \min(\mu, \eta^{-1})$. Then

$$|D^a G_\mu^\eta(x)| \le C_a (|x|_1 + \eta)^{-2-a} \exp(-cm|x|_1). \tag{3.ii}$$

The convolution by G_μ^η inverts $\Delta_\eta + \mu^2$ on compactly supported tests. For $\mu > 0$, periodization gives the massive inverse on every compatible rectangular torus. ∎

Proof. The generator is $\eta^{-2}(\sum_\mu (T_{\eta e_\mu} + T_{-\eta e_\mu}) - 8)$, where $T_y f(x) = f(x+y)$. Its exponential series gives positivity and preservation of constants. The Fourier density is the integral of $e^{ik\cdot x} e^{-t\lambda_\eta(k)}$ over $[-\pi/\eta, \pi/\eta]^4$ with factor $(2\pi)^{-4}$, where $\lambda_\eta(k) = 4\eta^{-2} \sum_\mu \sin^2(\eta k_\mu/2)$. Difference multipliers are $(e^{\pm i\eta k_\mu} - 1)/\eta$.

Use the same integral with k replaced by $k + i\sigma$, where $\sigma_j = c_0 \operatorname{sgn}(x_j)(s^{-1/2} + m)$. Each integrand is entire and $2\pi/\eta$-periodic in each variable, including $e^{ik\cdot x}$, so opposite sides in the contour shift cancel. Choose c_0 small so that $\eta|\sigma_j| \le 1$ and

$$\operatorname{Re} \lambda_\eta(k + i\sigma) \ge \lambda_\eta(k) - C|\sigma|^2, \qquad Ct|\sigma|^2 \le C' + \mu^2 t/4.$$

The first inequality follows from the real part of the cosine formula. The second uses $t/s \le 1$ and $m \le \mu$. The shifted difference factors have modulus at most $C(|k| + s^{-1/2} + m)$. If $t \ge \eta^2$, integration against $e^{-ct|k|^2}$ gives $C_a s^{-(4+a)/2}(1 + m\sqrt{s})^a$. If $t \le \eta^2$, the cube volume and $|k| \le 2\pi/\eta$ give the same bound. The polynomial is bounded by $C_a e^{\mu^2 t/4}$ since $m\eta \le 1$. Consequently

$$e^{-\mu^2 t} |D^a p_t^\eta(x)| \le C_a s^{-(4+a)/2} e^{-c|x|_1/\sqrt{s} - cm|x|_1 - \mu^2 t/2}.$$

Taking $\mu = 0$ proves the heat estimate. To obtain the Green estimate, integrate the power of s at $|x|_1 \le \eta$ and substitute $v = |x|_1/\sqrt{s}$ at $|x|_1 > \eta$. The estimates justify each fixed difference under the integral. Integrating $-\partial_t(e^{-\mu^2 t} p_t^\eta)$ gives the identity density. For $\mu = 0$, the large-time boundary term vanishes against any finitely supported function by the heat supremum bound. For

$\mu > 0$, both periodization and convolution are absolutely summable. Periodization commutes with the differences and sends the identity density to the periodic identity density. □

Summing the exponential bound gives, for $1 \le q \le \infty$,

$$\|D^a p_t^\eta\|_q \le C_{a,q}(t+\eta^2)^{-a/2-2(1-1/q)}. \tag{3.iii}$$

Indeed a ball of radius $r \ge \eta$ has physical mass at most Cr^4. The same summation outside a ball of radius r gives an additional factor $C_N(1+r/\sqrt{t+\eta^2})^{-N}$ in the ℓ^1 estimate, for any fixed N. In particular, for $R \ge \eta$, $b > 0$ and $0 \le a \le 3$,

$$\sum_{|x|_1 \ge bR} \eta^4 |D^a G^\eta_{1/R}(x)| \le C_{a,b} R^{2-a}. \tag{3.iv}$$

This follows directly from eq. 3.ii and shell summation, also when $bR < \eta$. For a periodized kernel, the same bound holds with periodic distance, since every lift of a point in the indicated sum lies at distance at least bR.

3.2. **Finite-exponent and Hölder estimates.**

Proposition 3.2 (Second differences on L^p)**.** For $1 < p < \infty$, every mesh $\eta > 0$, and every $\mu \ge 0$,

$$\|D^2(\Delta_\eta + \mu^2)^{-1} f\|_p \le C_p \|f\|_p. \tag{3.v}$$

This holds on the full lattice and on every compatible rectangular torus, with the same constant. At $\mu = 0$ the full-lattice operator is the differentiated Green convolution on finite supports. On a torus it is the inverse on mean-zero inputs, extended to vanish on constants. ■

Proof. The Fourier multiplier has modulus at most one, since the modulus of each first-difference multiplier is bounded by $\lambda_\eta^{1/2}$. This gives the L^2 bound. The physical kernel $k = D^2 G^\eta_\mu$ satisfies

$$|k(z)| \le C(|z|_1 + \eta)^{-4}, \qquad |Dk(z)| \le C(|z|_1 + \eta)^{-5}.$$

If $y \ne y'$, telescope along a shortest coordinate path from y to y'. Writing $\delta = |y - y'|_1$, every point in the telescope remains at distance at least $\frac{3}{4}|x-y|_1$ from x when $|x-y|_1 > 4\delta$. Summation in shells gives

$$\eta^4 \sum_{|x-y|_1 > 4\delta} |k(x-y) - k(x-y')| \le C. \tag{3.vi}$$

Use dyadic cubes of side $2^j\eta$ in the full lattice, down to singletons. For a finitely supported f and level $\lambda > 0$, select the maximal cubes C with mean $|f| > \lambda$. Their total volume is at most $\lambda^{-1}\|f\|_1$. Replace f on each by its mean to obtain g, and let $b_C = (f - \mathrm{mean}_C\, f)1_C$. The parent-to-child volume ratio is 16, hence $\|g\|_\infty \le 16\lambda$ and $\|g\|_2^2 \le 16\lambda\|f\|_1$. Also $\sum_C \|b_C\|_1 \le 2\|f\|_1$. Outside fixed enlargements of the selected cubes, use $\sum_{x\in C} b_C(x) = 0$ on each cube. Inserting the kernel at a reference point of C and applying eq. 3.vi bounds the L^1 norm of the bad output there by $C\|f\|_1$. The L^2 good bound and the volume of the enlargements give $\mathrm{vol}\{|Tf| > \lambda\} \le C\lambda^{-1}\|f\|_1$.

For $1 < p < 2$, at level λ apply the weak bound to $f1_{\{|f|>\lambda\}}$ and the L^2 bound to its complement. Integration of the distribution function gives inner integrals $\int_0^{|f(x)|} \lambda^{p-2}\, d\lambda$ and $\int_{|f(x)|}^\infty \lambda^{p-3}\, d\lambda$, respectively, which yield $C_p\|f\|_p^p$. Duality gives $p > 2$ because the adjoint multiplier has the same bounds.

At fixed mesh and positive mass the kernel is absolutely summable. To pass to periodic data, first truncate the kernel at a fixed radius. Apply the full-lattice estimate to periodic data restricted to a growing rectangular union of period cells, and discard cells within that radius of its boundary. The proportion discarded tends to zero. The truncated operator norm differs from the proved bound by at most the absolute tail of the kernel. Let the radius tend to infinity. On the finite torus the massive multipliers converge to the stated mass-zero multiplier, including zero on the constant mode. □

For nested cubes separated by a distance bR, with the outer cube of radius comparable to R, a sampled smooth cutoff and eq. 3.v give

$$\|D^2u\|_{p,\mathrm{in}} \le C_{p,b}(\|\Delta_\eta u\|_{p,\mathrm{out}} + R^{-1}\|Du\|_{p,\mathrm{out}} + R^{-2}\|u\|_{p,\mathrm{out}}). \tag{3.vii}$$

The cutoff is one on the inner stencils, and its support, enlarged by those stencils, is inside the outer cube. The difference product rule writes $\Delta_\eta(\chi u)$ as $\chi\Delta_\eta u$ and shifted products of $D\chi, Du$ or $D^2\chi, u$. Their norms give the last two terms. All formulas on nested cubes below use this stencil convention. If a fixed ratio R/η is bounded, the direct estimate by R^{-a} times the field norm on the containing stencils gives the corresponding derivative bound.

For a one-form v, put $(dv)_{\mu\nu} = \partial_\mu v_\nu - \partial_\nu v_\mu$ for $\mu < \nu$. The differences commute and $d^*d + dd^* = \Delta_\eta$ on one-forms. Applying the differentiated inverse to this identity proves

$$\|Dv\|_p \le C_p(\|dv\|_p + \|d^*v\|_p). \tag{3.viii}$$

The identity follows componentwise by canceling the mixed differences in the two terms. With a cutoff it gives the interior estimate with the additional term $C_{p,b}R^{-1}\|v\|_{p,\mathrm{out}}$. The zero-frequency constant causes no term on a torus, as D kills it.

For later use rescale a physical cube of radius R to unit size. At mesh $0 < \eta \le 1$, set $R_1 = (\Delta_\eta + 1)^{-1}$. Young's inequality and eq. 3.iii give

$$\|D^a R_1 f\|_q \le C_{a,p,q}\|f\|_p, \qquad 1 \le p \le q \le \infty, \quad a/2 + 2(1/p - 1/q) < 1. \tag{3.ix}$$

The time integral converges uniformly at zero by the strict inequality and at infinity by the mass. In addition, for finite $p > 4$, $\alpha = 1 - 4/p$, and $0 < |z|_1 \le 1$,

$$\|(\tau_z - 1)DR_1 f\|_\infty \le C_p|z|_1^\alpha \|f\|_p. \tag{3.x}$$

For $\delta = |z|_1$, telescope the heat difference and use the smaller of $C(t+\eta^2)^{-1/2-2/p}$ and $C\delta(t+\eta^2)^{-1-2/p}$ in $L^{p/(p-1)}$. Integrate the first to δ^2 and the second thereafter. Both integrals are $O(\delta^{1-4/p})$. Applying this to $u = R_1 u + R_1 d^* du$ also gives the first-order Morrey estimate for u in terms of its L^p norm and first differences. These estimates apply to interior fields by cutoff.

On a cubic torus of physical side ℓ, with ℓ/η an integer, write G for the mean-zero inverse of Δ_η and $\bar u$ for the mean of u. The corresponding gradient bounds are

$$\|DGf\|_\infty \le C_p \ell^{1-4/p}\|f\|_p, \qquad \|u - \bar u\|_\infty \le C_p \ell^{1-4/p}\|Du\|_p, \qquad p > 4.$$

To verify the first, scale to side one. For $t \le 1$ the image sum in eq. 3.i and its absolute ℓ^1 sum give $\|Dp_t^{\mathrm{per}}\|_{p/(p-1)} \le C(t+\eta^2)^{-1/2-2/p}$. For $t \ge 1$, all nonzero Fourier modes have $\lambda_\eta(k) \ge c|k|^2$ and their sum is $O(e^{-ct})$. The second bound follows from $u - \bar u = Gd^*du$ and the same kernel estimate. On a rectangular torus this proof is uniform when the aspect ratios stay in a fixed compact interval.

Proposition 3.3 (Discrete Schauder multiplier)**.** For $0 < \alpha < 1$, $0 < \eta \le 1$, and any two differences,

$$\|D^2 R_1 f\|_{C^\alpha} \le C_\alpha \|f\|_{C^\alpha}.$$

The norm is the supremum norm plus the seminorm on pairs at distance at most one. The result holds on the full lattice and on every compatible rectangular torus with its shortest periodic distance. On fixed nested cubes with the stencil convention of eq. 3.vii,

$$\|D^2u\|_{C^\alpha(\mathrm{in})} \le C(\|\Delta_\eta u\|_{C^\alpha(\mathrm{out})} + \|u\|_{C^{1,\alpha}(\mathrm{out})}). \tag{3.xi}$$

■

Proof. Set $k = D^2 G_1^\eta$. Its total sum is zero since it is absolutely summable at fixed mesh and each differentiated heat kernel has zero sum. For $r \ge \eta$,

$$\left|\eta^4 \sum_{|z|_1 \le r} k(z)\right| \le C.$$

Indeed split the heat integral at $t = r^2$. For $t < r^2$ use cancellation and bound the sum outside the ball by $C_N(t+\eta^2)^{-1}(1+r/\sqrt{t+\eta^2})^{-N}$. Integrating the bound $C_N r^{-N}(t+\eta^2)^{N/2-1}$ gives C. For $t \geq r^2$ the ball volume times the supremum bound is Cr^4t^{-3}, also integrable with integral C. A ball whose centre is displaced by at most $r/4$ has the same bound because the discrepancy lies in an annulus of radius comparable to r with bounded absolute kernel mass.

Subtract $f(x)$ in the convolution for the supremum bound. Inside radius one sum $C[f]_\alpha|z|_1^\alpha(|z|_1+\eta)^{-4}$ by shells. The remaining tail is integrable. To compare x, x' at distance $\delta \leq 1/16$, subtract $f(x)$ from both convolutions. On $|w-x|_1 \leq 4\delta$, use the increment centred at each kernel centre. The extra constant $f(x')-f(x)$ multiplies a displaced-ball kernel sum, which is bounded as above. This part is $O([f]_\alpha\delta^\alpha)$. On $4\delta < |w-x|_1 \leq 1$, telescoping bounds the difference of kernels by $C\delta|w-x|_1^{-5}$. Shell summation against the increment gives $C_\alpha[f]_\alpha\delta^\alpha$. Beyond radius one the differentiated kernel has an integrable tail and gives $C\delta\|f\|_\infty$. The supremum estimate covers $1/16 < \delta \leq 1$.

Periodic extension to the full lattice has the same supremum and no larger seminorm. The convolution of that extension equals the periodic multiplier by absolute summability. For the local estimate apply the multiplier to $(\Delta_\eta+1)(\chi u)$. The cutoff products used in eq. 3.vii have the stated C^α bounds. Products crossing the support boundary are controlled by the same smooth cutoff differences, with all input stencils still in the outer cube. □

3.3. **Poisson estimates with bounded local data.** For a cube B_r of radius r in the maximum norm, field components are indexed by their bond tails. A source equation on $B_{r-\eta}$ uses the full incoming and outgoing stencils there. Use $\|\cdot\|_{p,B_r}$ with physical counting measure, and use $[\cdot]_{\alpha,B_r}$ with physical distances.

Proposition 3.4 (Interior Poisson estimates)**.** Suppose $r \geq 16\eta$ and $\Delta_\eta u = g + d^*F$ on $B_{r-\eta}$, with u, g, F given on B_r. For $0 < \alpha < 1$, put

$$N = r^{-2}\|u\|_{2,B_r} + r^2\|g\|_{\infty,B_r} + r\|F\|_{\infty,B_r}.$$

On $B_{r/4}$,

$$\|u\|_\infty + r^\alpha[u]_\alpha \leq C_\alpha N. \tag{3.xii}$$

If $F = 0$ there is also the bound

$$r\|Du\|_\infty + r^{1+\alpha}[Du]_\alpha \leq C_\alpha N. \tag{3.xiii}$$

If $F \in C^\alpha(B_r)$, then the right side of eq. 3.xiii is instead $C_\alpha(N + r^{1+\alpha}[F]_{\alpha,B_r})$. The constants depend on the exponent and the fixed cube ratios, uniformly in r and η. ■

Proof. Use a sampled cutoff χ equal to one on $B_{r/2}$ and zero outside $B_{2r/3}$. With $F_\mu^\chi(x) = \chi(x+\eta e_\mu)F_\mu(x)$, the product rule gives the exact full-lattice identity

$$\Delta_\eta(\chi u) = \chi g + d^*F^\chi + \sum_\mu(\partial_\mu\chi)F_\mu + [\Delta_\eta, \chi]u.$$

The support and all shifted product stencils are in B_r. Extend the displayed products by zero and apply G_0^η. In the last term move the commutator onto the kernel by summation by parts. Its adjoint is $-[\Delta_\eta, \chi]$. Every nonzero coefficient of that commutator is in a collar at distance at least cr from the evaluation stencils. For $a = 0, 1, 2$, the commutator applied to $D^aG_0^\eta(x-\cdot)$ is bounded there by Cr^{-4-a}. Its terms have one kernel difference and one cutoff difference or two cutoff differences. It follows that the last two terms, after a output differences, are at most

$$C(r^{-2-a}\|u\|_{2,B_r} + r^{1-a}\|F\|_{\infty,B_r}).$$

The factor r^2 from Cauchy–Schwarz on the collar is included in the first term. These estimates also control the Hölder increments of orders zero and one by telescoping an additional difference along a coordinate path in the evaluation cube.

For the first two terms the relevant kernels are G_0^η and DG_0^η. Their absolute sums over a ball of radius Cr are at most Cr^2 and Cr, respectively. For the difference of two translates of DG_0^η at separation $\delta \le r$, split at distance 2δ from one centre. The near sum is at most $C\delta$ by the first-difference size bound. On the complement telescope and sum the second-difference bound to obtain $C\delta(1+\log(r/\delta))$, increasing the constant for the fixed support radius. This is at most $C_\alpha r^{1-\alpha}\delta^\alpha$. For G_0^η itself the same telescope gives $Cr\delta$. Differences with $\delta > r$ use the supremum bounds. This proves eq. 3.xii and eq. 3.xiii when $F = 0$.

For Hölder F, the remaining output $DG_0^\eta * d^*F^\chi$ uses the same cancellation proof as the massive Schauder multiplier. For clarity, multiply each massless second kernel by a sampled smooth cutoff in its difference variable, equal to one on the evaluation-to-source distances and vanishing beyond a fixed multiple of r. The truncated-sum bound in the proof of Proposition 3.3 used no mass factor, so it holds for the massless kernel. The total sum of the cut-off kernel is bounded by this bound and a shell of bounded absolute kernel mass. After scaling by r, the near and far split of that proof applies verbatim to the convolution with $F^\chi - F^\chi(x)$. The additional term is the bounded total sum times $F^\chi(x)$. The cutoff product has scaled C^α norm at most $C(\|F\|_\infty + r^\alpha [F]_\alpha)$. This proves the third assertion. □

These estimates also apply to a flat unitary connection on a lifted torus cube. Trivialize by parallel transport on the lift and use the componentwise estimate there. Norms are taken on that lift. If every period is at least a fixed fraction of the cube radius, a fixed multiplicity bounds the lifted L^p norm by the norm on the projected set. To periodize a massive kernel for such a connection, insert the commuting unitary holonomies for the deck translations in the image sum. The absolute tail estimates persist by unitarity. A local estimate on an embedded cube makes no smallness assumption on those holonomies.

Lemma 3.5 (Scalar subsolution and covariant modulus)**.** Let $w \ge 0$ and $\Delta_\eta w \le H$ on a lifted ball of radius $2r$ with full stencils where tested, where $H \ge 0$ is constant. Then its value at the centre satisfies

$$w(x) \le C(r^{-2}\|w\|_{2,B_{2r}(x)} + r^2 H). \tag{3.xiv}$$

For r/η bounded the estimate holds by the point-value bound on every region containing x. In particular, for a real unitary connection and a site field u, one may take $w = |u|$ and $H = \|D_U^\mathsf{T} D_U u\|_\infty$ on that ball. ■

Proof. When $r \ge 16\eta$, test the inequality against $\chi G_0^\eta(x - \cdot)$, where $\chi \ge 0$ is one on the inner ball and has its cutoff collar inside the outer ball. Moving the Laplacian gives $w(x)$ and a commutator supported on the collar. The positive test has sum at most Cr^2. The commutator kernel is bounded by Cr^{-4} by the free Green and first-difference bounds. Its L^2 norm on the collar is Cr^{-2}. This proves eq. 3.xiv. If r/η is bounded, use $w(x) \le \eta^{-2}\|w\|_{2,B_{2r}(x)}$.

For $u(x) \ne 0$, take the real inner product of each term $u(x) - \mathrm{Ad}(U_{xy})u(y)$ in the covariant Laplacian with $u(x)/|u(x)|$. Unitarity bounds it below by $|u(x)| - |u(y)|$. Sum over nearest neighbours, with coefficient η^{-2}. At a zero of u, the scalar left-hand side is nonpositive. Thus $\Delta_\eta|u| \le |D_U^\mathsf{T} D_U u|$ at every site with the full stencil. □

3.4. **Scalar block gap and block resolvent.** Fix a physical block side $R = m\eta$ with m a positive integer. Partition a rectangular torus with periods in $R\mathbb{Z}$ into translates of $\{0, \eta, \dots, (m-1)\eta\}^4$. Write q_{ar} for the arithmetic site average on this disjoint partition. This is the flat, one-size version of the site row q. Coarse scalar products have weight R^4 per block. In this subsection the star denotes the Hilbert-space adjoint. All the displayed real operators have the same bilinear transpose.

Proposition 3.6 (Mean, gap, and projection)**.** The adjoint q_{ar}^* embeds a coarse function as a block-constant fine function. Thus $q_{\mathrm{ar}} q_{\mathrm{ar}}^* = 1$ and $P = q_{\mathrm{ar}}^* q_{\mathrm{ar}}$ is an orthogonal projection. If $m \ge 2$,

the Neumann difference d_{N}, using only bonds internal to the individual blocks, satisfies

$$\|(1-P)u\|_2^2 \le \frac{\eta^2}{4\sin^2(\pi/(2m))}\|d_{\mathrm{N}}u\|_2^2 \le \frac{R^2}{4}\|d_{\mathrm{N}}u\|_2^2. \tag{3.xv}$$

For $m = 1$ the left side vanishes. In particular $A = \Delta_\eta + R^{-2}P \ge R^{-2}$ on the torus. Writing $G = A^{-1}$, the coarse Gram matrix and fine projection satisfy

$$q_{\mathrm{ar}}G^2q_{\mathrm{ar}}^* \ge cR^4, \qquad \Pi_{\Delta\ker q_{\mathrm{ar}}} = 1 - Gq_{\mathrm{ar}}^*(q_{\mathrm{ar}}G^2q_{\mathrm{ar}}^*)^{-1}q_{\mathrm{ar}}G, \tag{3.xvi}$$

with a numerical $c > 0$. ■

Proof. Since $R^4m^{-4} = \eta^4$, moving the sum in $\langle q_{\mathrm{ar}}f, u\rangle$ gives exactly the block-constant embedding. The projection identities follow. On a one-dimensional path of m sites, the vectors $\cos(\pi k(j+\frac12)/m)$, $0 \le k < m$, diagonalize the Neumann difference form. Substitution in the second difference gives eigenvalues $4\eta^{-2}\sin^2(\pi k/(2m))$. At the ends, reflect the cosine across the adjacent half-site to remove the missing edge. The eigenvalues are distinct, so the vectors form a basis. Tensoring in four directions adds the four eigenvalues and leaves just the constant vector at zero. The first bound in eq. 3.xv follows. The second uses $\sin(\pi/(2m)) \ge 1/m$. Adding the interblock bonds only increases the form, and $\|u\|_2^2 = \|(1-P)u\|_2^2 + \|q_{\mathrm{ar}}u\|_2^2$, giving the gap.

For the Gram bound choose a right inverse T for q_{ar}. For $m \ge 2$, let Tu on block B be $u(B)$ times

$$\phi(j) = 16\prod_{\nu=1}^{4}\sin^2(\pi(j_\nu+\tfrac12)/m).$$

Each one-dimensional sine square has mean $1/2$ by summing the corresponding roots of unity. Hence $q_{\mathrm{ar}}T = 1$ and $\|T\| \le 16$. The second derivative of a sine square on a continuous interval of length R is bounded by C/R^2. At a block face its endpoint sample and the next interior sample are $O((\eta/R)^2)$, so the second difference across the face is bounded by $CR^{-2}(|u(B)|+|u(B')|)$ for the adjacent block B'. Summing the pointwise bounds with the physical measure gives $\|\Delta_\eta Tu\|_2 \le CR^{-2}\|u\|_2$. For $m = 1$, take $T = 1$ and use $\|\Delta_\eta\| \le 16R^{-2}$. Thus $\|ATu\|_2 \le CR^{-2}\|u\|_2$ in both cases. Now

$$\|u\|_2^2 = \langle q_{\mathrm{ar}}^*u, Tu\rangle = \langle Gq_{\mathrm{ar}}^*u, ATu\rangle \le CR^{-2}\|Gq_{\mathrm{ar}}^*u\|_2\|u\|_2.$$

This is the first part of eq. 3.xvi. The orthogonal complement of $\mathrm{ran}(Gq_{\mathrm{ar}}^*)$ is $\ker(q_{\mathrm{ar}}G)$, which equals $\Delta_\eta\ker q_{\mathrm{ar}}$. Indeed $A = \Delta_\eta$ on $\ker q_{\mathrm{ar}}$, and conversely for $f \in \ker(q_{\mathrm{ar}}G)$ put $u = Gf$. The standard range projection for the injective map Gq_{ar}^* now gives the displayed formula. □

In Euclidean coordinate inner products the unitary conversions are multiplication by η^2 on fine sites and bonds and by R^2 on coarse sites. The conjugated mean has entries m^{-2} on its block, with squared row norm one. Keeping entries m^{-4} in Euclidean coordinates gives squared row norm m^{-4}. Then $q_{\mathrm{ar}}^*q_{\mathrm{ar}}$ fails to fix constants when $m > 1$. These unitaries commute with adjoints, orthogonal projections, and positive or Moore–Penrose inverses by conjugation. The same conjugation applies to a covariance and its constrained covariance, since the constraints act between the corresponding Hilbert spaces. A combinatorial difference is ηd, so its form in eq. 3.xv carries the additional factor η^{-2}.

Proposition 3.7 (Off-diagonal scalar block inverse)**.** Let S be a set of sites of the torus, J_S extension by zero, and $A_S = J_S^*AJ_S$. This includes the whole torus. For nonempty site sets $X, Y \subset S$,

$$\|1_XA_S^{-1}1_Y\|_{2\to2} \le 2R^2e^{-c\,\mathrm{dist}_\eta(X,Y)/R}. \tag{3.xvii}$$

The same estimate for $dJ_SA_S^{-1}$, with an output bond restriction indexed by its tails, has prefactor CR. For the adjoint estimate the prefactor is also CR, and for $dJ_SA_S^{-1}J_S^*d^*$ it is C. The distance is shortest-path distance in the ambient torus, with length η per bond. The constants are independent of m, R, S, and the torus periods. ■

Proof. Let w be real with $|w(x) - w(y)| \le \eta/R$ across each bond, and write $M_t = e^{tw}$. For $|t| \le 1$ we prove

$$\|A_S^{-1/2}(M_t A_S M_t^{-1} - A_S)A_S^{-1/2}\| \le C(|t| + t^2).$$

It suffices to estimate the full-torus form on zero extensions. The compression commutes with each multiplication, and its energy is exactly the energy of the zero extension. For a bond from x to y, put $\delta = w(x) - w(y)$ and $a = e^{t\delta} - 1,\ b = e^{-t\delta} - 1$. The Laplacian conjugation error has form contribution $-\eta^2(a\overline{u_x}v_y + b\overline{u_y}v_x)$. Split the parenthesis as

$$a(\overline{u_x} - \overline{u_y})v_y + b\overline{u_y}(v_x - v_y) + (a+b)\overline{u_y}v_y.$$

Here $|a| + |b| \le C|t|\eta/R$ and $|a+b| \le Ct^2\eta^2/R^2$. Summing and applying Cauchy–Schwarz bounds the error by

$$C|t|R^{-1}(\|du\|_2\|v\|_2 + \|u\|_2\|dv\|_2) + Ct^2R^{-2}\|u\|_2\|v\|_2.$$

For the block term, P has coordinate entries m^{-4} on each block, and the oscillation of w on one block is at most four. The absolute row and column sums of the conjugation error for $R^{-2}P$ are therefore at most $C|t|R^{-2}$. The gap and the gradient energy bound give the asserted relative estimate. This uses the full block mean in the compression, including at blocks cut by the boundary.

Choose a numerical $t_0 > 0$ so the relative error has norm at most $1/2$ for $|t| \le t_0$. Factoring the conjugated operator between $A_S^{1/2}$ on both sides gives the inverse by a Neumann series, with middle factor of norm at most two. This proves $\|M_t A_S^{-1} M_t^{-1}\| \le 2R^2$. On bonds let the multiplier use the tail value of w. Then $M_t d M_t^{-1} = d + \mathcal{E}_t$ with $\|\mathcal{E}_t\|_{2\to 2} \le C|t|/R$, by the same nearest-neighbour bound on w. Thus the conjugated gradient times $J_S A_S^{-1/2}$ has norm at most C, using the energy of the zero extension. The input divergence follows by taking the adjoint of this bound with parameter $-t$. Combining these factors with the middle inverse proves the gradient prefactors.

Finally take $w(x) = \operatorname{dist}_\eta(x, Y)/R$, using the tails for a bond input. The weight is zero on Y and at least $\operatorname{dist}_\eta(X, Y)/R$ on X. Undoing the conjugation proves all the estimates. □

The Dirichlet operator here compresses P itself. Averaging over only the surviving sites would change the operator. The off-diagonal statements are norms on weighted Hilbert spaces. A pointwise estimate for a bounded cell source uses the separate local argument in Lemma 3.5.

3.5. **Local covariant scalar equations.** On a lifted cube of radius comparable to $R \ge \eta$, choose a frame in which $U(x, x + \eta e_\mu) = \exp(i\eta A_\mu(x))$. The frame may be real, with A_μ traceless Hermitian, or complex, with A_μ traceless complex. The coefficient conditions used for finite-p regularity are

$$R\|A\|_\infty \le K, \qquad R^{2-4/q}\|DA\|_q \le K_q \quad (1 < q < \infty). \tag{3.xviii}$$

All differences in these conditions are on the prescribed outer stencils. Only finitely many q are used for each conclusion. For the second-difference Hölder estimate we use the stronger condition, in the same frame,

$$R\|A\|_\infty + R^2\|DA\|_\infty + R^{2+\alpha}[DA]_\alpha \le K_\alpha. \tag{3.xix}$$

On a fixed cube this also controls the scaled C^α norm of A, by telescoping coordinate paths.

For example, suppose the prescribed frame satisfies $d^*A = 0$, $R\|A\|_\infty \le C\kappa$, and each plaquette logarithm satisfies $\|(i\eta^2)^{-1}\log U(\partial p)\| \le \kappa/R^2$. For small κ, multiplying the four exponentials gives

$$(i\eta^2)^{-1}\log U(\partial p) = (dA)(p) + O(\|A\|_\infty^2).$$

The linear term is the oriented sum of the link exponents, and the analytic remainder before division by η^2 is quadratic in those exponents. The interior Hodge estimate eq. 3.viii, applied on a slightly smaller cube, gives $K_q \le C_q\kappa$. This deduction uses only the admitted frame, its Coulomb condition and the curvature bound. For a complex logarithmic perturbation, the corresponding norms in eq. 3.xviii add to those of the real logarithm if the perturbation is specified in the same logarithmic coordinates. No unitary estimate will be used for such a complex coefficient.

With $D_U u(x,\mu) = \eta^{-1}(u(x) - \operatorname{Ad} U_\mu(x) u(x+\eta e_\mu))$, the bilinear transpose satisfies $\Delta_U = D_U^{\mathsf T} D_U$. Its site formula is

$$(\Delta_U u)(x) = \eta^{-2} \sum_\mu (2u(x) - \operatorname{Ad} U_\mu(x) u(x+\eta e_\mu) - \operatorname{Ad} U_\mu(x-\eta e_\mu)^{-1} u(x-\eta e_\mu)). \tag{3.xx}$$

Indeed $(\operatorname{Ad} g)^{\mathsf T} = \operatorname{Ad} g^{-1}$ for the invariant bilinear form. This identity remains valid at complex g.

Lemma 3.8 (Exact lower-order terms)**.** On the rescaled unit cube, with mesh $h = \eta/R$ and potential RA, the difference $\Delta_U - \Delta_h$ is the sum of first differences with bounded coefficients and a multiplication term b with bounded L^q norms. Its bounds are uniform in h, with constants depending only on K, K_q. In the class eq. 3.xix these coefficients are uniformly bounded in C^α. The difference $D_U^{\mathsf T} + d^*$ is a zeroth-order sum with bounded coefficients and one-step shifts. ■

Proof. In unit coordinates, denote the rescaled potential again by A, and put $E_\pm(a) = (e^{\pm ih \operatorname{ad} a} - 1)/h$. In eq. 3.xx, replace each translated u by its central value plus its signed h-difference. The first-difference coefficients are $-E_+(A_\mu(x))$ and $-E_-(A_\mu(x - he_\mu))$ for the forward and the adjoint backward difference. The multiplication coefficient is

$$b(x) = -\sum_\mu h^{-1}(E_+(A_\mu(x)) + E_-(A_\mu(x - he_\mu))).$$

Since $E_+(a) + E_-(a) = O_K(h|a|^2)$, this is a sum of bounded quadratic terms and backward divided differences of $E_+(A)$. The latter are bounded by $C_K|DA|$ by the integral differential of the matrix exponential. In C^α the same identity bounds those terms by products of a smooth function of the two potential values and one difference of A. There is no second difference of A in b. Taking the transpose of the corresponding first-order formula for D_U proves the last assertion. □

Lemma 3.9 (A left change of the connection)**.** Write $U_E(x,\mu) = e^{i\eta E_\mu(x)} U(x,\mu)$. For each direction let $T_\mu u(x) = \operatorname{Ad} U_\mu(x) u(x+\eta e_\mu)$, and put $\widetilde E_\mu(x) = \operatorname{Ad} U_\mu(x-\eta e_\mu)^{-1} E_\mu(x-\eta e_\mu)$. If $\eta(\|E\|_\infty + \|\widetilde E\|_\infty)$ is bounded, the difference $\Delta_{U_E} - \Delta_U$ has no second differences of its argument. The first-difference coefficients relative to the forward and backward transports of U have size $C(|E| + |\widetilde E|)$. Its multiplication coefficients are bounded by

$$C \sum_\mu (\eta^{-1}|E_\mu - \widetilde E_\mu| + |E_\mu|^2 + |\widetilde E_\mu|^2).$$

This is a bilinear identity and norm bound in real or complex frames, on the stencils where both operators are defined. ■

Proof. The shifted operator for U_E is $e^{i\eta \operatorname{ad} E_\mu} T_\mu$, and its inverse is $e^{-i\eta \operatorname{ad} \widetilde E_\mu} T_\mu^{-1}$. Thus the difference of the two positive Laplacians in direction μ is $-\eta^{-2}(M_+ T_\mu + M_- T_\mu^{-1})$, where $M_+ = e^{i\eta \operatorname{ad} E_\mu} - 1$ and $M_- = e^{-i\eta \operatorname{ad} \widetilde E_\mu} - 1$. Write $T_\mu^{\pm 1} u = u + (T_\mu^{\pm 1} - 1)u$. The latter terms are a single transported difference times $M_\pm/\eta$. In the zeroth term $M_+ + M_-$, expand both exponentials to linear order with a bounded quadratic remainder. This gives the asserted coefficient and its precise $\eta^{-1}(E_\mu - \widetilde E_\mu)$ derivative. □

In the same frame, a relative left perturbation also preserves the stated logarithmic coefficient classes. In unit coordinates on a common small exponential chart,

$$(ih)^{-1} \log(e^{ihE} e^{ihA}) = A + E + h\, \mathcal{B}_h(A, E).$$

The last function is analytic and has uniform derivatives of each fixed order on smaller balls. Indeed the logarithm of the product minus its linear term has a zero of order two in h. Division by h^2 extends it at zero. The remainder vanishes when either of the two potentials is zero. Consequently its A-derivative is $O(|E|)$, and its E-derivative is $O(|A|)$. A discrete first difference of this identity uses the integral of these derivatives along the segment joining two neighbouring input pairs. This

gives the finite-p first-difference bounds in terms of $\|DE\|_p + \|E\|_\infty \|DA\|_p$, with the zeroth bound $C\|E\|_\infty$. Products and the same integral formula give the $C^{1,\alpha}$ bound on the logarithm in a $C^{1,\alpha}$ tube. A covariant difference of E in the fixed frame differs from DE by an $O(AE)$ term, with the same scaled C^α estimate in the stronger class.

Proposition 3.10 (Local site bootstraps)**.** Work on nested lifted cubes after scaling their fixed physical radius R to one, and write $h = \eta/R$. The norms in the statement are in these rescaled coordinates, including the first differences and the physical measure h^4. Assume eq. 3.xviii and an outer energy bound

$$\|u\|_{2,\mathrm{out}} + \|D_U u\|_{2,\mathrm{out}} \le B.$$

One may add a zeroth-order operator M to Δ_U, provided that every cutoff step has a reserved outer box containing the full input carrier of M at all source sites in the cutoff support, and $\|Mu\|_{q,\operatorname{supp}\chi} \le K'_q \|u\|_{q,\mathrm{out}}$ on these pairs of stocks for the finite q used below.

If $(\Delta_U + M)u = g$, $4 < p < \infty$, and $0 < \alpha \le 1 - 4/p$, then on the innermost cube

$$\|u\|_\infty + \|Du\|_\infty + [Du]_\alpha \le C(B + \|g\|_{p,\mathrm{out}}). \tag{3.xxi}$$

If $(\Delta_U + M)u = g_0 + D_U^{\mathsf{T}} F$ and $2 < p < \infty$, then

$$\|u\|_p + \|Du\|_p \le C(B + \|g_0\|_{p,\mathrm{out}} + \|F\|_{p,\mathrm{out}}). \tag{3.xxii}$$

For $p > 4$ the left side can also include $\|u\|_\infty$. The constants depend on the exponents, coefficient bounds, and the fixed cutoff and carrier margins. They are independent of h and of the outside domain. After reversing the dilation, $B = R^{-2}\|u\|_{2,\mathrm{out}} + R^{-1}\|D_U u\|_{2,\mathrm{out}}$. The two source norms acquire factors $R^{2-4/p}$ for g, g_0 and $R^{1-4/p}$ for F. The left side uses scaled derivatives, for example $R^{1+\alpha}[Du]_\alpha$ in eq. 3.xxi and $R^{-4/p}\|u\|_p + R^{1-4/p}\|Du\|_p$ in eq. 3.xxii. ■

Proof. The comparison $D_U = -d + O_K(1)$ first bounds the flat outer H^1 norm. A cutoff and the identity $v = R_1 v + R_1 d^* dv$ give $u \in L^3$ by eq. 3.ix, since $\frac{1}{2} + 2(\frac{1}{2} - \frac{1}{3}) < 1$. In the first equation, the new flat zeroth term bu is in L^2 by $b \in L^6$. First-order terms are in L^2. For the cutoff equation $(\Delta_h + 1)v = s$, the cutoff commutator uses only Du, u on the preceding box. With $s \in L^2$ we obtain $D^2 u \in L^2$ from eq. 3.v, $Du \in L^3$ and $u \in L^{12}$ from eq. 3.ix. At the next cutoff, $b \in L^4$ and $u \in L^{12}$ give $bu \in L^3$. The first-order terms are now in L^3, and M reads $u \in L^3$ on its reserved box. An L^3 source yields bounded u and $Du \in L^q$ for every $q < 12$. Take $q = \min(p, 8)$ in this step. If $p > 8$, one more L^8 step gives bounded Du. Thus the final cutoff equation has source in L^p, and eq. 3.x proves eq. 3.xxi.

For a divergence source, each cutoff equation is the sum of a flat divergence of a bounded cutoff times F and an ordinary source. The latter contains the previous commutator terms and only zeroth-order products with F, by Lemma 3.8 and the product rule used for eq. 3.xii. At each step the divergence contribution to the gradient is bounded in L^p by eq. 3.v. The first ordinary source is in L^2 as before, and its gradient is in L^3. If $p \le 3$ this gives eq. 3.xxii already, since the field is in L^3 from the energy step. If $p > 3$, $R_1 d^*$ maps L^p to L^{12} by eq. 3.ix (or first by restriction to a lower exponent if $p > 12$). Thus the field is in L^{12}, including its divergence contribution, and the next ordinary source is in L^3. It gives a bounded field and gradient in every L^q with $q < 12$, sufficient for $q = \min(p, 8)$. If $p > 8$, its next L^8 source gives bounded gradient, while the divergence contribution remains in L^p. For $p > 4$ the divergence contribution to the field is also bounded because $R_1 d^* : L^p \to L^\infty$. This proves the claims without estimating a difference of F. Each invocation reads the preceding box only, with the stated margins. For bounded R/η the desired suprema, differences and seminorms follow directly from the energy on the containing stencils. □

If g_0 is bounded and $F \in C^\alpha$, $0 < \alpha < 1$, the conclusion eq. 3.xxi holds for the divergence equation with the right side replaced by $C(B + \|g_0\|_\infty + \|F\|_{C^\alpha})$ in unit coordinates. To see this, use eq. 3.xxii for some $p > 4/(1 - \alpha)$ on an intermediate box. In the final cutoff equation the

ordinary source, including bu and the first-order terms, is in L^p. Its contribution is controlled by eq. 3.x. For the divergence part use the multiplier in Proposition 3.3 on the cut-off F. The same argument with $F = 0$ gives the corresponding bounded ordinary forcing estimate.

There is also a bounded-data version without an a priori energy bound. For $\Delta_U u = g + D_U^{\mathsf{T}} F$, an outer supremum bound on u gives an interior energy bound by $C(\|u\|_\infty + \|g\|_2 + \|F\|_2)$ in unit coordinates. Indeed write the equation in the flat form of Lemma 3.8 and test against $\chi^2 u$ in the Hermitian pairing. For the flat term, on a bond from x to y, the real part of $(\overline{u_y} - \overline{u_x})(\chi_y^2 u_y - \chi_x^2 u_x)$ equals

$$|\chi_y u_y - \chi_x u_x|^2 - (\chi_y - \chi_x)^2 \operatorname{Re}(\overline{u_y} u_x).$$

This gives the squared gradient of χu up to $C\|u\|_\infty^2$. The first-order and divergence terms are bounded by a small multiple of that gradient square plus $C(\|u\|_\infty^2 + \|F\|_2^2)$, by the product rule and Cauchy–Schwarz. For a shifted first difference, choose the cutoff on the same bond endpoint as the source. The difference from $d(\chi u)$ is a bounded cutoff derivative times u. The zeroth term is bounded by $\|b\|_1 \|u\|_\infty^2$, and the ordinary forcing by Cauchy–Schwarz. This proves the energy bound for real or complex coefficients. It extends to an added M with the stated L^2 carrier bound by testing its output on the complete outer box. After one more cutoff, the bounded force case of eq. 3.xxi yields the gradient and every fixed Hölder exponent below one.

Corollary 3.11 (Second covariant differences)**.** In the class eq. 3.xix, if $\Delta_U u = g$, then on fixed nested unit cubes

$$\|D_U^2 u\|_{C^\alpha,\mathrm{in}} \le C(\|g\|_{C^\alpha,\mathrm{out}} + \|u\|_{C^{1,\alpha},\mathrm{out}}). \tag{3.xxiii}$$

Here a covariant second difference is a composition of forward or backward transported differences on site-based components. The source seminorm includes pairs across adjacent pillar faces whenever the cube crosses those faces. For example, if the source jumps at a face, the first-difference estimate from bounded forcing still applies, while the source seminorm in eq. 3.xxiii then includes the jump. In physical coordinates, set $\|f\|_{C^\alpha_R} = \|f\|_\infty + R^\alpha [f]_\alpha$. Then eq. 3.xxiii reads

$$R^2 \|D_U^2 u\|_{C^\alpha_R,\mathrm{in}} \le C(R^2 \|g\|_{C^\alpha_R,\mathrm{out}} + \|u\|_{\infty,\mathrm{out}} + R\|Du\|_{C^\alpha_R,\mathrm{out}}).$$

■□

Proof. Lemma 3.8 bounds the lower-order terms in C^α, so eq. 3.xi applies to the flat equation. Composing two covariant differences and subtracting the corresponding ordinary ones leaves products of A, DA, u, Du with bounded smooth coefficients, by the same divided-difference identity. Their C^α norms have the stated bound. □

Corollary 3.12 (Checking the strong coefficient condition in a Coulomb frame)**.** Suppose a real Coulomb frame $U = e^{i\eta A}$ is already given on the outer stock, with $d^*A = 0$, $R\|A\|_\infty \le C_0\kappa$, and

$$R^2 \|F\|_\infty \le \kappa, \qquad R^3 \|J_{\mathrm{left}}(U)\|_\infty \le j.$$

Here $F(p) = (i\eta^2)^{-1} \log U(\partial p)$ on the small branch, and J_{left} represents the variation $U \mapsto e^{it\eta B} U$ of the Wilson action $\sum_p (1 - \frac{1}{2} \operatorname{tr} U(\partial p))$ in the physical link pairing. Use complete current stencils throughout this stock. For κ below a fixed ceiling, on a smaller stock and for every $0 < \alpha < 1$,

$$R^2 \|DA\|_\infty + R^{2+\alpha} [DA]_\alpha \le C_\alpha(\kappa + j), \qquad R^3 \|\Delta_\eta A\|_\infty \le C(j + \kappa^2). \tag{3.xxiv}$$

This supplies eq. 3.xix when a Coulomb frame and the actual current are specified. ■□

Proof. Scale to $R = 1$, with mesh $h = \eta/R$, so the potential, curvature and current are multiplied by R, R^2, R^3, respectively. The current scale also follows directly by dilating its defining variation and physical pairing. By the interior flat Hodge estimate, $\|DA\|_p \le C_p \kappa$ on an intermediate stock.

For the pointwise equation let $S_h(F) = (P - P^{-1})/(2ih^2)$ for $P = e^{ih^2 F}$. The exact Wilson current is the covariant curl transpose of $S_h(F)$. In coordinates this reads

$$J_{\mathrm{left},\mu}(x) = h^{-1} \sum_{\nu \neq \mu} (S_{h,\mu\nu}(x) - \operatorname{Ad} U_\nu(x - he_\nu)^{-1} S_{h,\mu\nu}(x - he_\nu)).$$

Indeed varying a positive link gives the plaquette insertion with coefficient ih. Its reversed occurrence gives the negative transported insertion at the other plaquette base. Cyclicity of trace brings both to the varied link tail. In each trace, the traceless insertion pairs with $(P - P^{-1})/2$, since $P + P^{-1}$ is scalar. Conjugation by P leaves $P - P^{-1}$ fixed. Dividing the resulting h^4 contribution by the physical bond weight gives the displayed formula.

Replacing the transport in that formula by the identity costs $C\|A\|_\infty \|F\|_\infty$. The sine remainder satisfies $|S_h(F) - F| \leq Ch^4 |F|^3$, and a first difference of this remainder costs at most $Ch^3 \|F\|_\infty^3$. In $F = dA + E(A)$, the analytic curvature remainder satisfies $|DE(A)| \leq C\|A\|_\infty |DA|$ on a fixed stencil enlargement. This follows by subtracting its values at neighbouring plaquettes and differentiating the analytic quadratic remainder with respect to its four link arguments. Since $d^*A = 0$ and $d^*dA = \Delta_h A$ on full Coulomb stencils, we obtain

$$|\Delta_h A - J_{\mathrm{left}}(U)| \leq C(\kappa \max_{\text{stencil}} |DA| + \kappa^2 + h^3 \kappa^3).$$

Thus $\Delta_h A$ is bounded in any fixed L^p on the intermediate stock. Apply DR_1 to the equation for a cutoff of A, taking $p > 4$. Its source consists of the cut-off $\Delta_h A$ and bounded cutoff products of DA, A. Equation 3.ix gives $\|DA\|_\infty \leq C(\kappa + j)$ on a smaller stock. Insert this bound into the pointwise equation to obtain $\|\Delta_h A\|_\infty \leq C(j + \kappa^2)$. The bounded ordinary forcing estimate eq. 3.xiii gives the stated Hölder bound on the innermost stock. For bounded R/η, direct finite-stencil bounds from the potential supremum give the first-difference and Hölder bounds. The same pointwise identity then gives the claimed Laplacian bound. Undo the dilation to obtain eq. 3.xxiv. □

For real coefficients the corresponding transported Hölder seminorm compares values in a common frame by parallel transport on a fixed shortest coordinate path inside the cube. In the chosen frame, such a transport over distance s differs from the identity by at most $C_K s/R$, by telescoping its factors. The same bound holds in the declared complex class because a product along such a path, and its inverse, have norm bounded by $e^{C_K s/R}$. For first differences use the first-order Morrey estimate on A with some $q > 4/(1 - \alpha)$ and the expression of D_U in terms of d and $E_+(A)$. This gives the transported first-difference bounds from eq. 3.xxi. For second differences use eq. 3.xxiii and its supremum bound. Real site gauges preserve these transported norms.

These local results use full stencils inside a lifted stock. Norms on a periodic lift can be estimated by the corresponding projected norms when the lift has bounded multiplicity, as specified after Proposition 3.4. A Neumann Laplacian on a graph retains only its internal bond terms in its quadratic form, whereas a Dirichlet compression retains the full quadratic form of the zero extension. For a real unitary connection on either graph, the heat semigroup is dominated in norm by the scalar heat semigroup with that same boundary convention. To check this, split the generator into its diagonal degree term and its transported adjacency. The time-ordered expansion with respect to the diagonal term sums over paths in the graph. Each path has a nonnegative scalar weight times a product of unitaries. Taking the norm gives the scalar expansion with the same diagonal term, including the killing term for zero extension. At an unbuffered boundary with bounded R/η, the point-value and present-stencil difference estimates suffice. The interior bootstraps require their stated collars. In particular a complex tube uses the coefficient estimates rather than this unitary domination.

3.6. **Obtaining the local coefficient frame.**

Proposition 3.13 (A small real Coulomb frame)**.** Let R/η be a positive integer. Suppose a real connection is given on a complete lifted cube of radius $8R$, with $\|F_U\|_\infty \le \kappa R^{-2}$ there in the logarithm branch near one. For sufficiently small fixed κ, it admits a real gauge on the cube of radius $6R$ in which $U = e^{i\eta A}$,

$$d^{\mathsf T} A = 0, \qquad R\|A\|_\infty \le C\kappa.$$

The divergence equation holds on complete stencils there. The constant is independent of η/R and exterior data. If the actual Wilson current has $R^3\|j_U\|_\infty \le j$ on the required complete inner stock, the frame also satisfies eq. 3.xxiv on a fixed smaller cube. ■

Proof. Rescale to $R = 1$ and denote the fine spacing by h. In a rooted comb gauge on the original cube, each link holonomy is a product of at most C/h conjugated plaquettes. Unitarity and the defect bound give $\|A\|_\infty \le C\kappa$ for the per-length link logarithm. Expanding each plaquette product gives $F_U = dA + E(A)$, where $|E(A)(p)| \le C\sum_{b\subset\partial p}|A(b)|^2$; the raw coefficients hA are in a fixed small logarithm neighbourhood. Consequently $\|dA\|_\infty \le C\kappa$.

Extend A to a periodic auxiliary lattice of side 32 by multiplying by a fixed smooth cutoff, equal to one on the cube of radius seven and zero before radius eight. Use zero on any link not inner to the original cube. This leaves every complete stencil at radius six unchanged, including the neighbouring tails in its divergence. For fine enough h, each plaquette meeting its nonzero support lies in the original cube, and the difference product rule gives $\|d(\chi A)\|_\infty \le C(\|dA\|_\infty + \|A\|_\infty)$. For the finitely coarse h the direct four-link bound gives the same estimate because h has a fixed positive lower bound. The extended connection therefore has the same small bounds on its potential and physical plaquette curvature up to fixed constants.

On the auxiliary torus use a mean-zero Hermitian site field λ and gauge $e^{i\lambda(x)}$. For its left logarithm write

$$A^\lambda_\mu(x) = (ih)^{-1}\log(e^{i\lambda(x)}e^{ihA_\mu(x)}e^{-i\lambda(x+he_\mu)}) = A_\mu(x) - d_\mu\lambda(x) + N_\mu(A,\lambda)(x).$$

To estimate N, let $l = \lambda(x)$, $r = hA_\mu(x)$ and $s = hd_\mu\lambda(x)$. The analytic logarithm of $e^{il}e^{ir}e^{-i(l+s)}$ vanishes at $r = s = 0$, and its linear part at the origin is $i(r - s)$. After removing that linear part, the remainder is $O((|l| + |r| + |s|)(|r| + |s|))$; its l-derivative is $O(|r| + |s|)$ and its s-derivative is $O(|l| + |r| + |s|)$. These bounds follow by integrating the bounded second derivatives on a fixed smaller matrix ball. Since $|s| \le 2\|\lambda\|_\infty$, division by h leaves a factor A or $d\lambda$ in the remainder estimate. With $\|\lambda\|_* = \|\lambda\|_\infty + \|d\lambda\|_p$, for fixed $p > 8$, this proves

$$\|N(A,\lambda)\|_p \le C(\kappa + \|\lambda\|_*)^2,$$

$$\|N(A,\lambda) - N(A,\widetilde\lambda)\|_p \le C(\kappa + \|\lambda\|_* + \|\widetilde\lambda\|_*)\|\lambda - \widetilde\lambda\|_*.$$

The map $\Delta_h^{-1}d^{\mathsf T}$ into this mean-zero norm is bounded uniformly in h. Its first difference bound is Proposition 3.2. The periodic Poincaré inequality at side 32 bounds its L^p norm by its first difference norm. This Poincaré inequality follows by averaging coordinate paths between pairs of sites. The first-order Sobolev estimate $\|\lambda\|_\infty \le C(\|\lambda\|_p + \|d\lambda\|_p)$ follows from the periodic mean-zero estimate after eq. 3.x, at the fixed side 32. The zero mode is absent because λ has mean zero. Thus

$$\lambda = \Delta_h^{-1}d^{\mathsf T}(A + N(A,\lambda))$$

is a contraction on $\|\lambda\|_* \le C_0\kappa$: choose C_0 larger than twice the linear bound for A, then choose κ so the Lipschitz constant is at most $1/2$ and the quadratic term fits in the remaining half. Its real fixed point satisfies $d^{\mathsf T}A^\lambda = 0$ and $\|A^\lambda\|_p \le C\kappa$. The raw logarithms hA^λ stay in the fixed small branch by $\|\lambda\|_\infty = O(\kappa)$.

Curvature remains $O(\kappa)$ under the real gauge. The same plaquette expansion now gives

$$\|dA^\lambda\|_{p/2} \le C(\kappa + \|A^\lambda\|_p^2).$$

Use the torus Hodge bound eq. 3.viii with zero divergence, and the first-order Sobolev bound at $p/2 > 4$, to obtain $\|A^\lambda\|_\infty \le C\kappa$. Restrict the composition of the two gauges to the original inner cube. Its links agree with the auxiliary connection there. The actual current on complete inner stencils transforms by the unitary colour rotations. The last assertion follows from Corollary 3.12 and rescaling. □

4. The flat classical form and the contour gauge

We first use raw logarithms and counting measures. The fine mesh in the flat comparison is one, the intermediate grid has spacing $n = L^k$, and its curl d_n uses unscaled differences. In the subsequent one-step contour problem we measure length in units of that intermediate spacing. The flat action Hessian is $\|\mathrm{d}a\|_2^2$ for $\langle X, Y\rangle = \frac{1}{2}\operatorname{tr}(XY)$. All flat scalar calculations apply to the three colours separately.

For a positively oriented μ-link let

$$(S_n a)_\mu(y) = n^{-4} \sum_{x \in B_n(y)} \sum_{t=0}^{n-1} a_\mu(x + te_\mu).$$

Write P_n for the raw derivative of the plain k-fold average at the identity, r_n for restriction to its centres, and q_n for the untransported block mean. The flat rows of block averaging give

$$P_n a = S_n a + \mathrm{d}_n \sigma_n(a), \qquad P_n \mathrm{d} f = \mathrm{d}_n r_n f, \qquad S_n \mathrm{d} f = \mathrm{d}_n q_n f \tag{4.i}$$

for a linear site map σ_n. Indeed the first step differs from S_L by the difference of the two averaged contour sums. At each further step the centred block digits and the segment digits partition their longer counterparts. Thus the straight rows compose, and the gradient difference remains a gradient under further straight rows. The two identities on gradients follow by telescoping the segment and the contours, respectively.

Proposition 4.1 (The flat effective action)**.** On a compatible periodic lattice, the raw quadratic value

$$E_n(a) = \min_{P_n v = a} \|\mathrm{d}v\|_2^2$$

is also the minimum subject to $S_n v = a$. It satisfies

$$\|\mathrm{d}_n a\|_2^2 \le E_n(a) \le 256 \|\mathrm{d}_n a\|_2^2. \tag{4.ii}$$

The corresponding translation-invariant matrix has a Fourier symbol holomorphic and bounded on a fixed complex strip, independent of k and the periods. Its kernel and the periodic image sums have uniform exponential decay in intermediate units. The same bounds hold for the real flat unitary connections in their transported coordinates. ■

Proof. Both q_n and r_n are onto. If $P_n v = a$, add a fine gradient with $q_n f = \sigma_n(v)$. If $S_n v = a$, add a fine gradient with $r_n f = -\sigma_n(v)$. By eq. 4.i these changes leave the curl unchanged and give the stated constraints in the other convention.

The coarse curl of $S_n v$ averages fine plaquette sums over n-by-n sheets. Its nonnegative weights have row sum n^2 and counting column sum n^{-2}. For the column sum, first sum over all coarse origins and block sites, which gives each fine translate weight n^{-4}, and then sum the n^2 sheet positions. Weighted Cauchy–Schwarz gives the lower bound in eq. 4.ii.

For the upper bound use the nodal hat $\varphi(t) = (1 - |t|/n)_+$ and interval $\beta(t) = 1_{\{0,\dots,n-1\}}(t)$. Define I_j on coarse j-cochains by the product interpolation with factor n^{-j}, using β in the j oriented coordinate directions and φ in the others. The products are centred at the cochain's starting site, and are periodized on a torus. The identity $\varphi(t+1) - \varphi(t) = n^{-1}[\beta(t+n) - \beta(t)]$ gives $\mathrm{d}I_1 = I_2 \mathrm{d}_n$. The row and column sums of I_2 are n^{-2} and n^2. Hence $\|I_2\|_{2\to 2} \le 1$.

In each coordinate the convolution S_nI_1 has weights

$$a_0 = \frac{3}{4} + \frac{1}{4n^2}, \qquad a_{\pm 1} = \frac{1 - n^{-2}}{8}.$$

In a transverse coordinate the two off-centre coefficients are the sums of the linear tails of the hat. In the segment coordinate they count the overshoots of a translated segment, one triangular sum $1+\cdots+(n-1)/2$ on each side, divided by n^2. Thus S_nI_1 is the same four-factor convolution on every link component. Each factor has Fourier multiplier at least $a_0 - 2a_1 \geq 1/2$. Its inverse commutes with d_n, so $I_1(S_nI_1)^{-1}a$ is a feasible straight lift with curl norm at most $16\|\mathrm{d}_n a\|_2$. This proves the upper bound and nonemptiness. Finite-dimensional orthogonal projection onto the image of the curl gives attainment.

Equivalently the minimizing curvature is the least-norm exact two-cochain with the prescribed sheet-average curl. To verify the converse, lift any such exact cochain as $\mathrm{d}v$. The difference $S_nv - a$ is closed. On the periodic lattice its nonzero Fourier modes are coarse gradients, and its constant mode is a constant link field. These can be removed by a fine gradient and a constant fine link field, respectively. Both operations preserve the fine curl. In a twisted unitary character sector the same argument applies, with a constant mode only in the invariant sector. The sheet averages compose by the same block and segment digit partition. Their kernels on exact fine cochains are therefore nested, and the least-norm curvatures are the orthogonal projections onto the decreasing orthogonal complements of those kernels.

To obtain the symbol, let p be the intermediate momentum and $\zeta_m = (p + 2\pi m)/n$, with $m \in \{-(n-1)/2, \ldots, (n-1)/2\}^4$. Put

$$b_{\nu,m}(p) = \frac{\sin((p_\nu + 2\pi m_\nu)/2)}{n \sin((p_\nu + 2\pi m_\nu)/(2n))},$$
$$f_{\mu,m} = \Big(\prod_\nu b_{\nu,m}^2\Big) b_{\mu,m}^2, \qquad s_m = 4n^2 \sum_\nu \sin^2((p_\nu + 2\pi m_\nu)/(2n)).$$

The removable sinc ratios use their analytic values. With unitary counting Fourier transforms the straight row acts on an alias by the multiplier $n^{-1}e^{i(n-1)\zeta_{m,\mu}/2}(\prod_\nu b_{\nu,m})b_{\mu,m}$. The factor n^{-1} is n from the segment times n^{-2} from the ratio of the square roots of the two site counts. Adding the squared fine divergence to the curl gives the scalar Feynman symbol s_m/n^2 on each alias. Its pushed quadratic covariance is therefore diagonal with entries

$$D_\mu(p) = \sum_m f_{\mu,m}(p)/s_m(p) \quad (p \text{ real}, \ p \neq 0).$$

The squared factors are the continuation of the product of the two oppositely signed filters. The starting-link phase cancels in that product.

Let $t_\mu^\pm = e^{\pm i p_\mu} - 1$, as columns. Removing a fine gradient makes its divergence zero and changes the straight datum only by a coarse gradient. Conversely a fine gradient can restore any such change because q_n is onto. Minimization of the Feynman value over $a + t^+z$ gives the effective symbol

$$\mathcal{E}_n = D^{-1} - \frac{D^{-1}t^+(t^-)^{\mathsf{T}}D^{-1}}{(t^-)^{\mathsf{T}}D^{-1}t^+}, \qquad D = \mathrm{diag}(D_\mu). \tag{4.iii}$$

For a real nonzero p the central alias is positive in each component, and the denominator is strictly positive. The constant mode has zero action by its constant fine lift.

On $|\operatorname{Re} p_\nu| < \pi + \delta$, $|\operatorname{Im} p_\nu| < \delta$ with $\delta > 0$ fixed sufficiently small, the sine denominator gives $|b_{\nu,m}| \leq C(1 + |m_\nu|)^{-1}$. For $m \neq 0$, $\operatorname{Re} s_m \geq c(1 + |m|^2)$: the real sine arguments lie in a fixed subinterval of $(-\pi, \pi)$, and the imaginary squares cost $O(\delta^2)$ after multiplying by n^2. Hence the sum R_μ over noncentral aliases is holomorphic and uniformly bounded by $C\sum_{m\in\mathbb{Z}^4}\prod_\nu(1+|m_\nu|)^{-2}$.

Write $s = s_0$, $f_\mu = f_{\mu,0}$, $B = \prod_\nu b_{\nu,0}^2$, and $H_\mu = (f_\mu + sR_\mu)^{-1}$. Near zero $s = O(|p|^2)$ and $f_\mu = 1 + O(|p|^2)$, with constants independent of n. The exact identity $\sum_\mu t_\mu^- t_\mu^+ / f_\mu = s/B$ gives

$$\sum_\mu t_\mu^- t_\mu^+ H_\mu = sJ, \qquad J = B^{-1} - \sum_\mu t_\mu^- t_\mu^+ R_\mu H_\mu / f_\mu.$$

On a fixed neighbourhood of zero $J = 1 + O(|p|^2)$ is nonzero. Substituting $D^{-1} = sH$, with $H = \operatorname{diag}(H_\mu)$, cancels both powers of s in eq. 4.iii. The extension is

$$\mathcal{E}_n = sH - J^{-1} H t^+ (t^-)^{\mathsf{T}} H.$$

This removes the complex zero cone of s, and gives the stated value at zero. Away from that neighbourhood, the central real $b_{\nu,0}$ are bounded above and away from zero, and s and $\sum_\mu |t_\mu^+|^2$ are bounded below. Thus D and the denominator in eq. 4.iii are bounded away from zero on the remaining real compact set. The uniform alias estimates and Cauchy's bound on a smaller complex neighbourhood keep these lower bounds on a fixed strip there. The expressions agree on overlap by the displayed cancellation. Reindexing aliases proves periodicity. Shifting the Fourier contour in each direction gives an exponential kernel bound; sampling the symbol on a finite torus gives its absolutely convergent image sum.

For a real flat connection, take a parallel frame on the cover. The commuting unitary adjoint holonomies decompose into characters of the deck group. The symbol is unchanged in each character sector and is sampled at shifted real momenta. The images acquire unit-modulus factors, which preserve the same bound. The interpolation and gradient corrections also commute with these deck actions. □

For a real reference V on a lattice of spacing one, write $P_V = Dq_V(0)$, where

$$q_V(a) = i^{-1} \log(\mathcal{M}(e^{ia}V)\mathcal{M}(V)^{-1}).$$

On a retained family of complete L-blocks include all faces incident to those blocks, with their double blocks and all internal contour paths. Extend a field by zero on the remaining links if only retained coordinates are varied. Let $b_V a$ be the derivative of the relative contour logarithm

$$i^{-1} \log[T_{e^{ia}V}(y,x) T_V(y,x)^{-1}]$$

at zero, with one row for each $x \neq y$ in a retained block. At the identity it is the arithmetic contour path sum. The quadratic form $\|b_V a\|_2^2$ is the relative trace-contour Hessian. When the contours of V are one it is the absolute trace-contour Hessian.

Lemma 4.2 (The outer curl and contour inequality)**.** For L fixed and sufficiently small plaquette defect on the complete retained stocks,

$$\|a\|_2^2 \leq C_L(\|\mathrm{d}_V a\|_{2,\mathcal{P}}^2 + \|b_V a\|_2^2 + \|P_V a\|_2^2). \tag{4.iv}$$

Here the supported links are those incident to retained blocks, the plaquette stock $\mathcal{P}$ contains the internal plaquettes of the retained blocks and the face plaquettes comparing neighbouring crossing links, and P_V includes each incident face row. If the field vanishes on a spanning tree inside each retained block, the contour term can be omitted. The constants are independent of the retained family and volume. ■

Proof. At a flat reference use a parallel frame on each rectangular one- or two-block stock. Inside a retained block, a path from its centre to the head of an internal link and a path to its tail followed by that link differ by a sum of elementary plaquette boundaries. To see this, replace successive path steps by coordinate paths to the centre, interchanging adjacent different coordinate steps by their plaquette. The paths have bounded length depending on L; summing these elementary differences and averaging over the prescribed contours gives a bound on the internal link by two contour sums and a finite sum of curls. A tree path can be used instead when its link values vanish. Internal links of an unretained neighbour have zero values.

On a face, a plaquette with one transverse step expresses the difference of adjacent crossing values using its curl and its two internal links. The transverse grid is connected and has L^3 vertices. Thus all crossing values differ from a common one by terms already bounded. Every straight segment in the face row has one crossing and otherwise internal links. Its plain row is their arithmetic average plus the endpoint contour correction. Its total coefficient on the common crossing value is one. The row therefore bounds that common value. Squaring these bounds uses only constants depending on L, and each link or plaquette is used a bounded number of times.

For a real curvature-small field, put each such stock in its comb frame. The Wilson curl, the contour derivative and the plain row differ in that frame from the flat matrices by $O_L(\max|\partial V-1|)$ in the local coefficient norms. This follows by differentiating the admitted finite contour and logarithm formulas on their fixed one-step balls. The change of frame is a unitary rotation at each input or output tail, and it preserves zero tree values. The squared errors summed over the complete stocks are at most $C_L \max|\partial V-1|^2\|a\|_2^2$ by their overlap. Decrease the defect ceiling to absorb them. If a stock wraps, make the estimates on its rectangular cover with the given projected branch tests and descend by deck invariance. □

Corollary 4.3 (Actual contour precisions)**.** Let C_V be the derivative of the actual pivot graph and h_V its pivot-supported right inverse of P_V. Use the ambient Wilson action whose plaquette set contains every internal and face-comparison plaquette in the stock $\mathcal{P}$ of Lemma 4.2, with the varied link field extended by zero. Any selection of free coordinates is compressed after forming the chart precision. At a real reference with sufficiently small defect, the one-step matrix

$$C_V^{\mathsf{T}} H_W(V) C_V + (b_V C_V)^{\mathsf{T}}(b_V C_V) - \left\langle h_V^{\mathsf{T}} J,\, D^2 q_V(0)[C_V\,\cdot, C_V\,\cdot]\right\rangle \tag{4.v}$$

has spectrum in $[c_L, C_L]$ for real independent raw currents J with sufficiently small supremum norm. For the actual Wilson current at an axial reference it is the Hessian of the Wilson and absolute contour actions in the exact fibre chart. At a constrained-critical reference this is the physical Gaussian Hessian. Its inverse is holomorphic on a fixed smaller complex coefficient neighbourhood and has an exponential kernel there.

At the flat field the corresponding next-step precision after k classical averaging levels is $C_1^{\mathsf{T}}\mathcal{E}_n C_1 + (b_1 C_1)^{\mathsf{T}}(b_1 C_1)$, with $n = L^k$ and the intermediate lattice spacing taken as one. Its spectral bounds and the exponential kernel bounds for this precision and its inverse are uniform in k and compatible volume. The flat upper spectral bound may be chosen as CL^8, with a numerical C independent of L and k. ■□

Proof. The pivot graph has free restriction equal to the identity and $P_V C_V = 0$. Its second derivative lies on the pivots and equals $-h_V D^2 q_V(0)[C_V\,\cdot, C_V\,\cdot]$ by two differentiations of the constraint. The chain rule gives the current term in eq. 4.v, including its sign. The absolute contour term has the relative Hessian at the axial reference, and its current is zero there.

The finite one-step pivot, mean and logarithm bounds give $\|C_V u\|_2 \le C_L\|u\|_2$ and operator norm at most $C_L\|J\|_\infty$ for the current term. The Wilson second variation is the transported curl square up to $C_L \max|\partial V-1|\|C_V u\|_2^2$, by expanding the four insertion factors in a plaquette trace. Apply eq. 4.iv and use $\|C_V u\|_2 \ge \|u\|_2$ to absorb the two errors. The same finite coefficient bounds give the upper estimate. The entries are holomorphic on admitted complex tubes. A smaller coefficient increment changes the matrix in norm by less than half the gap, since each row and column reads a fixed number of raw coordinates. Inverting by a Neumann series proves uniform analyticity. Conjugation by $e^{s\,d(\cdot,b)}$ changes the matrix in norm by $O_L(|s|)$ because it has fixed range, so the conjugated inverse stays bounded for a fixed small $|s|$. Taking a single column gives the exponential bound.

For the flat classical form use eq. 4.ii and eq. 4.iv; its curl-null fields include the constant modes, which are controlled by the actual outer rows and contour form. The flat $\mathcal{E}_n$ has uniformly summable exponential kernels by Proposition 4.1. Composing with the fixed local C_1 and b_1 preserves such

kernels on the block lattice. The same conjugation estimate applies using the exponential first moment in place of finite range. For the explicit flat upper bound, the straight raw row S_L has absolute row sum L and column sum L^{-3}. The internal contour mean has row sum $O(L)$ and column sum at most one. Thus $P_L = S_L - \mathrm{d}_L\sigma$ has norm at most $CL^{1/2}$. The pivot right inverse has norm L^3, giving $\|C_1\| \le CL^4$. The contour derivative has row sum $O(L)$ and column sum at most L^4, and it does not read the crossing pivots. Hence $\|b_1C_1\|^2 \le CL^5$. Combine these counts with $\mathcal{E}_n \le 256\mathrm{d}_n^{\mathsf{T}}\mathrm{d}_n$ and the bounded unit-grid curl to obtain CL^8. Periodic image bounds and the flat character decomposition preserve both estimates on tori. □

5. Site averages, restoration and determining row lifts

Write Ξ_s for multiplication by ξ_y on site tops. For a fine bond or plaquette the scale L is the largest assigned corner scale. We use the field energy space X with norm

$$\|v\|_X^2 = \|\nabla_U v^0\|_2^2 + \|v\|_{\mathrm{m}}^2, \qquad \|v\|_{\mathrm{m}} = \|\mathsf{L}^{-1}v\|_2. \tag{5.i}$$

Here v^0 is zero off the active bonds. The comparison norm counts its boundary jumps. If a stencil has two nonzero bond values, the connecting shift is inside the active graph. With only one nonzero value its norm is independent of the arbitrary unitary exterior connection. The form of the Wilson action still uses only inner plaquettes. Let Y be the field-row space with norm

$$\|b\|_Y^2 = \|a_f^{1/2}b\|_{2,r}^2, \qquad a_f(c) = d_c\xi_c^{-2}, \quad 0 < d_* \le d_c \le d^*. \tag{5.ii}$$

Thus Y is uniformly equivalent to the norm $\|b\|_{Y_0}^2 = \sum_c \xi_c^2|b_c|^2$. The dual norm for the physical row pairing is $\|\omega\|_{Y^*} = \|a_f^{-1/2}\omega\|_{2,r}$.

5.1. **Weighted site estimates.** Choose $\delta > 0$ small and fixed, then impose $\rho\delta \ge 1$. On the whole torus set

$$\ell(x) = \inf_z\{\xi(z) + \delta\ \mathrm{dist}_\infty(x,z)\}. \tag{5.iii}$$

It is δ-Lipschitz and $\xi(x)/L \le \ell(x) \le \xi(x)$. In fact if x has level j, a z at level at least $j-1$ has $\xi(z) \ge \xi_j/L$; a z at a smaller level is at distance $> \rho\xi_{j-1}$ by separation. On the bare layer $\ell = \eta$. The values at neighbouring sites and on their fine bond are comparable, and $|D(\ell^{-1})| \le C\delta\ell^{-2}$ for a fine difference. In norms containing a bond scale, ℓ is evaluated at its tail. The metric defined using ℓ instead of ξ lies between d and Ld.

Set $\Delta_U = D_U^{\mathsf{T}}D_U$ on the true graph and $a_s(y) = c_y\xi_y^{-2}$, where $0 < c_* \le c_y \le c^*$. Let $A_s = \Delta_U + q^{\mathsf{T}}a_sq$. For a real connection in eq. 2.xii,

$$\begin{aligned} \|\ell^{-1}u\|_2^2 &\le C(\|D_Uu\|_2^2 + \|\Xi_s^{-1}qu\|_{2,s}^2), \\ \|u\|_2^2 &\le C(\|\ell D_Uu\|_2^2 + \|qu\|_{2,s}^2). \end{aligned} \tag{5.iv}$$

For each nonbare pillar, use a comb frame on its complete surrounding box. Its potential is $O_L(\kappa/\xi_y)$. The q-transports are $I + O_L(\kappa)$ in this frame by the summable block transport comparison. The ordinary block Poincaré inequality follows by writing a difference as a coordinate path and summing over both endpoints. The covariant gradient and the actual mean change its right side by at most $C_L\kappa^2\xi_y^{-2}\|u\|_{2,B_y}^2$ after enlarging its constant. Absorb this term by taking κ small. The bare pillar has the identity mean. Sum the block inequalities to prove the first line of eq. 5.iv, or multiply each by ξ_y^2 before summing to prove the second. Only internal pillar bonds are used. In particular A_s is positive definite.

Lemma 5.1 (Compact lifts of the actual site average)**.** Under the preceding real curvature condition there is a right inverse E_s of q with seeds supported in their pillars. For $r = -1, 0, 1$,

$$\|\ell^{-r}E_sb\|_2 \le C\|\Xi_s^{-r}b\|_{2,s}, \qquad \|\ell^{-r}D_UE_sb\|_2 \le C\|\Xi_s^{-r-1}b\|_{2,s}. \tag{5.v}$$

It can be chosen with the additional bound

$$\|E_s b\|_{V_2} \le C\|\Xi_s^{-2} b\|_{2,s}, \qquad \|u\|_{V_2}^2 = \|\Delta_U u\|_2^2 + \|\ell^{-1} D_U u\|_2^2 + \|\ell^{-2} u\|_2^2. \tag{5.vi}$$

The constants are independent of mesh, depth and volume. ■

Proof. On a complete nonbare box of side $C_L\xi_y$ containing the pillar, choose its comb frame. For a fixed colour b_y let u minimize the covariant gradient energy with boundary value b_y in that frame. The Dirichlet form is positive on zero boundary fields by paths to the boundary, and its equation is $\Delta_U u = 0$ at interior sites. Comparison with the constant field and the small potential give $\|D_U u\|_2 \le C_L\kappa\xi_y|b_y|$. For $u - b_y$ the zero-boundary Poincaré inequality has the bound $\|u - b_y\|_2 \le C\xi_y\|D_U(u - b_y)\|_2$. The ordinary inequality follows by summing along paths parallel to one axis to the boundary. In the comb frame its replacement by D_U costs $C_L\kappa\|u - b_y\|_2$, which is absorbed. Thus $\|u - b_y\|_2 \le C_L\kappa\xi_y^2|b_y|$.

Multiply u by a fixed smooth bump sampled at scale ξ_y, supported strictly inside B_y and with positive mean. For sufficiently fine ratios η/ξ_y, its two-step collar lies in the harmonic region. Its first two differences are bounded by $C\xi_y^{-1}$ and $C\xi_y^{-2}$. The product rule therefore gives $\|\chi u\|_2 \le C\xi_y^2|b_y|$ and $\|D_U(\chi u)\|_2 \le C\xi_y|b_y|$. Moreover

$$\|\Delta_U(\chi u)\|_2 = \|[\Delta_U, \chi]u\|_2 \le C(\xi_y^{-1}\|D_U u\|_2 + \xi_y^{-2}\|u\|_2) \le C|b_y|.$$

For the commutator, expand the two opposite covariant shifts of Δ_U. Terms with a shift difference of u cost a first difference of χ, and the terms proportional to u have its second difference. This uses no derivative of the connection. The transported average of χu is $(m_y I + O_L(\kappa))b_y$, where m_y is the ordinary mean of χ, uniformly bounded away from zero. The error uses the L^2 bound on $u - b_y$ and the actual transports on the pillar. Normalize the seed by the inverse of this matrix and undo the frame.

For the finitely coarse ratios, use a single site seed, normalized by the inverse of its exact site-row coefficient. Its amplitude is bounded by $C|b_y|$ because the number of sites is then bounded. Its i differences cost at most $C\eta^{-i}$ in a fixed collar, which is $O(\xi_y^{-i})$ in these cases. This includes the bare pillar. Distinct pillars are disjoint and their fixed-stencil collars have bounded overlap and comparable scales. Summing proves eq. 5.v and eq. 5.vi. If a box does not embed, construct on a large periodic cover and average the linear lift over simultaneous deck shifts of source and output. The right inverse equation and the norm estimates are preserved and descend with the common sheet-count factor. □

Proposition 5.2 (Normalized site Gram matrix and restoration)**.** Under the preceding real hypotheses put

$$G_s = A_s^{-1}, \quad F = G_s q^{\mathsf{T}} \Xi_s^{-2}, \quad M = F^{\mathsf{T}} F, \quad P = F M^{-1} F^{\mathsf{T}}, \quad R = I - P. \tag{5.vii}$$

There are uniform constants $m_*, m^* > 0$ with $m_* I \le M \le m^* I$. The normalized columns $F, \ell D_U F$ and M^{-1} are bounded after conjugation by $e^{s\phi}$ for $|s| \le s_0$, where ϕ is one-Lipschitz for the metric of ℓ and its top samples are used on top spaces. In particular their normalized cell-L^2 entries decay exponentially. The fine blocks of G_s have the bound $C\xi_y\xi_z e^{-s_0 d(y,z)}$ from B_z to B_y; those of $D_U G_s$ have bound $C\xi_z e^{-s_0 d(y,z)}$.

Define

$$\Lambda = G_s R D_U^{\mathsf{T}}, \qquad S = I - D_U\Lambda. \tag{5.viii}$$

Then

$$\begin{aligned} &\operatorname{ran} R = \Delta_U(\ker q), \qquad qG_sR = 0, \qquad \Delta_U G_s R = R, \\ &RD_U^{\mathsf{T}} S = 0, \qquad QS = Q, \qquad S^2 = S, \\ &\|\Lambda v\|_{V_2} + \|D_U\Lambda v\|_X + \|Sv\|_X \le C\|v\|_X. \end{aligned} \tag{5.ix}$$

On a fixed smaller weighted C^1 complex tube these formulas continue holomorphically, with the same bounds up to fixed constants. R there is the bilinear projection given by eq. 5.vii. ■

Proof. Let $u = Fb$ and $\mathcal{E}_+(u) = \|\ell D_U u\|_2^2 + \|qu\|_{2,s}^2$. Test $A_s u = q^{\mathsf{T}} \Xi_s^{-2} b$ against $\ell^2 u$. On each real bond the exact real-part identity

$$\operatorname{Re}\langle D_U u, D_U(\ell^2 u)\rangle = |D_U(\ell u)|^2 - |D\ell|^2 \operatorname{Re}\langle u_-, \operatorname{Ad} U_b u_+\rangle$$

bounds the summed gradient term below by $\frac{1}{2}\|\ell D_U u\|_2^2 - C\delta^2\|u\|_2^2$. The weighted difference of the actual mean and its constant weight on a pillar satisfies

$$\|\Xi_s^{-2}(q(\ell^2 u) - \ell(y)^2 qu)\|_{2,s} \le C_L \delta \|u\|_2.$$

Indeed the absolute variation of ℓ^2/ξ_y^2 on a pillar is $O_L(\delta)$; the transported average is a contraction on the physical L^2 spaces. Since $\ell(y) \ge \xi_y/L$, the site mass term is bounded below by $c\|qu\|_{2,s}^2 - C_L\delta^2\|u\|_2^2$ by Young's inequality. Also $\|\Xi_s^{-2} q(\ell^2 u)\|_{2,s} \le \|u\|_2$. Choose δ small enough to absorb the errors by eq. 5.iv. The result is

$$c\mathcal{E}_+(Fb) \le \|b\|_{2,s}\|Fb\|_2. \tag{5.x}$$

For the lower bound test the same equation against $E_s \Xi_s^2 b$. Its exact q-value is $\Xi_s^2 b$. The $r = 1$ lift bound controls the gradient term by $C\|\ell D_U F b\|_2 \|b\|_{2,s}$; the mass term is the pairing of qFb with $c_y b$. Thus $\|b\|_{2,s}^2 \le C\mathcal{E}_+(Fb)^{1/2}\|b\|_{2,s}$. Together with eq. 5.x this proves the lower bound, and eq. 5.iv proves the upper bound.

Conjugate the weighted site equation by $e^{s\phi}$. On a fine bond the change of a gradient is at most $C|s|\ell^{-1}$, since a single bond has metric length $O(\eta/\ell)$. A site mean and its transpose change by $C|s|$ in normalized form norms because its carrier has bounded metric diameter. The preceding weighted energy test is therefore changed by $O(|s|)$ times its energy. It bounds the conjugated F and $\ell D_U F$ and gives an $O(|s|)$ difference from the unweighted operators by the resolvent identity. This applies to the opposite weight and hence to the conjugated transpose. Consequently the conjugated M differs in norm by $O(|s|)$ and its inverse is bounded for a fixed sufficiently small s_0. For G_s use the first line of eq. 5.iv to test the conjugated equation in the ordinary energy norm $\|D_U u\|_2 + \|\ell^{-1} u\|_2$. Its right hand side f has dual bound $C\|\ell f\|_2$. Taking ϕ to be distance from the source cell and restricting the outputs gives the two stated blocks, including their length factors.

The range identities follow directly from $F^{\mathsf{T}} A_s = \Xi_s^{-2} q$. If $q\lambda = 0$ then $A_s\lambda = \Delta_U \lambda$ belongs to $\ker F^{\mathsf{T}}$. Conversely $F^{\mathsf{T}} f = 0$ gives $qG_s f = 0$ and $\Delta_U G_s f = f$. Thus $\ker F^{\mathsf{T}} = \operatorname{ran} R$.

We need two energy derivatives on $\ker q$. Set $w = \ell^{-1}$, $f = \Delta_U \lambda$ and $q\lambda = 0$. The first block inequality applied to $w\lambda$ has mean error at most $C_L\delta\|\ell^{-2}\lambda\|_2$. This follows by subtracting $w(y)q\lambda$ in each pillar. It follows that $\|\ell^{-2}\lambda\|_2^2 \le C\|D_U(w\lambda)\|_2^2$ after absorbing the error. Testing the equation against $w^2\lambda$ and using the same bond identity gives

$$\|D_U(w\lambda)\|_2^2 \le \|f\|_2\|\ell^{-2}\lambda\|_2 + C\delta^2\|\ell^{-2}\lambda\|_2^2.$$

A second absorption and the product rule prove $\|\ell^{-2}\lambda\|_2 + \|\ell^{-1} D_U \lambda\|_2 \le C\|f\|_2$.

For the remaining derivative, extend λ by zero outside Ω_0. It vanishes on the bare layer because its actual mean there is its site value. The separation therefore keeps every nonzero two-step stencil of this extension inside the active region. For the unitary covariant shifts, expanding the squared curl and divergence on the full torus yields the component-gradient square up to terms containing a commutator of two shifts. The diagonal terms agree by unitarity. In the mixed terms interchanging the two shifted directions leaves a factor $\operatorname{Ad} U(\partial p) - I$ divided by η^2. The scale bound in eq. 2.xii bounds the sum of these terms by $C_L\kappa\|v\|_{\mathrm{m}}^2$. The curl of $D_U\lambda$ is itself such a commutator applied to λ, with norm at most $C_L\kappa\|\ell^{-2}\lambda\|_2$. Applying this calculation to $v = D_U\lambda$ gives

$$\|\lambda\|_{V_2} + \|\nabla_U D_U \lambda\|_2 \le C\|\Delta_U\lambda\|_2. \tag{5.xi}$$

This calculation uses no exterior curvature, since all terms with nonzero fields or shifted fields lie in the bare separated collar or its interior.

Now $\lambda = \Lambda v$ is in $\ker q$ and $\Delta_U \lambda = R D_U^{\mathsf{T}} v$. The actual divergence is bounded by $C\|v\|_X$: it is the restriction of the divergence of the zero extension, with exterior bond values zero. Thus eq. 5.xi proves the norm bounds. The range identities imply $R D_U^{\mathsf{T}} S = 0$, and S is the identity on that kernel. The exact relation eq. 2.vi gives $Q D_U \Lambda = 0$, hence $QS = Q$.

For a complex shift of size ε the gradient coefficient changes by $C\varepsilon\ell^{-1}$ and the actual normalized site row by $C\varepsilon$. The latter follows by the holomorphic transported-mean comparison and the geometric sum of raw radii over its complete backward blocks. The A_s energy form, the plus-weight tested form, and their conjugated forms therefore have $O(\varepsilon)$ perturbations in their declared energy norms. A relative Neumann series gives G_s, F, and a second one gives M^{-1} and R. In particular $\|R_{\mathbf{U}} - R_U\|_{2\to 2} \le C\varepsilon$. The range identities are algebraic. The covariant Laplacian difference obeys

$$\|(\Delta_{\mathbf{U}} - \Delta_U)\lambda\|_2 \le C\varepsilon(\|\ell^{-1} D_U \lambda\|_2 + \|\ell^{-2}\lambda\|_2).$$

To obtain it, collect opposite shifts before estimating: one connection increment multiplies a first difference of λ, and the derivative of the increment multiplies λ. The weighted C^1 bound controls these by ε/ℓ and ε/ℓ^2, including the product of two increments. On the bare collar the direct finite-stencil bound has those same powers because $\ell = \eta$. For $q_{\mathbf{U}}\lambda = 0$, subtract $E_s q_U \lambda$ to use eq. 5.xi. Its V_2 norm is at most $C\|\Xi_s^{-2}(q_U - q_{\mathbf{U}})\lambda\|_{2,s} \le C\varepsilon\|\ell^{-2}\lambda\|_2$ by eq. 5.vi. The subtracted lift's full gradient-derivative norm is bounded by its V_2 norm up to C: the preceding curl-divergence calculation applies to interior seeds and the finitely coarse collars are controlled by their atomic norm. Absorbing these errors proves eq. 5.xi on the moving kernel, and then the claimed restoration bounds. □

5.2. **Energetic lifts of the field rows.**

Lemma 5.3 (A right inverse of the determining field rows)**.** For the separated no-inward determining geometry and the curvature-small real class, the actual per-length Q has a right inverse $E_Q : Y \to X$ with norm at most C. There is a right inverse $E_L = S E_Q$ into $X_L = \ker(R D_U^{\mathsf{T}})$, with the same uniform bound. The lifts and the row inf-sup estimate extend to a fixed smaller complex tube about a real reference. On this tube the X norm may be taken at the real centre. ■

Proof. At the flat field consider a level $j \ge 1$ pillar with $n = L^j = 2m + 1$ sites per side. A seed for its μ-direction has only that component nonzero. Inside the pillar, at fine index u, take the product $\phi(u_\mu)\prod_{\nu\neq\mu}\chi(u_\nu)$ times its colour, using only bonds with both endpoints in the pillar. Here χ is a nonnegative even bump sampled from a fixed smooth profile of u/n supported in $(-1/4, 1/4)$. For sufficiently large n, set $\beta = n^{-1}\sum_u \chi(u)$ and $s = n^{-1}\sum_u (u/n)^2\chi(u)$. Both are uniformly positive. The straight per-length row weights are n^{-5} times their segment multiplicities. The two rows incident to this pillar have longitudinal counts $u + m + 1$ and $m - u$ at the positive link with tail u. Their response functionals are

$$\ell_+(\phi) = \beta^3 n^{-2}\sum_u (u + m + 1)\phi(u), \qquad \ell_-(\phi) = \beta^3 n^{-2}\sum_u (m - u)\phi(u).$$

The profiles

$$\phi_+(u) = \beta^{-4}\chi(u) + \frac{m}{n}\frac{\beta^{-3}}{s}\frac{u}{n}\chi(u), \qquad \phi_-(u) = \beta^{-4}\chi(u) - \frac{m+1}{n}\frac{\beta^{-3}}{s}\frac{u}{n}\chi(u)$$

have exact responses $(1, 0)$ and $(0, 1)$, by evenness and the two moments. Their first differences are $O(n^{-1})$. For the remaining odd $3 \le n$ below the fixed cutoff, use transverse centre atoms and longitudinal atoms at $u = -1, 0$. The two response columns form $n^{-5}\left(\begin{smallmatrix} m & m+1 \\ m+1 & m \end{smallmatrix}\right)$, of determinant $-n^{-9}$. Its inverse gives the required seeds with constants depending only on the cutoff.

Assign each row to a level-j endpoint, using the interior endpoint for an outward row, and choose the seed with the response on that side. The full regular grid at that level sees exactly the one

assigned row. At the bare level the seed is the individual stored fine link. At most eight seeds are assigned to one pillar. The support volume ξ_j^4 and one inverse length from each energy factor give

$$\|E^0 b\|_X^2 \le C \sum_c \xi_c^2 |b_c|^2$$

for the sum E^0 of the seeds. This includes the bare boundary jump, whose scale is η.

A level-j seed is invisible to lower rows because their carriers miss Ω_j. A row of level $j+2$ or higher has carrier in Ω_{j+1}, so it also misses the seed. The only possible leakage is into the next level. The flat straight rows on full grids satisfy

$$Q_{j+1}^{0,\mathrm{full}} = Q_L^0 Q_j^{0,\mathrm{full}}. \tag{5.xii}$$

The site and segment digits partition their larger ranges, so the weights multiply to $(Ln)^{-5}$. The one-step Q_L^0 has row sum one and counting column sum L^{-4}, the two longitudinal counts adding to L. Its counting L^2 operator norm is at most L^{-2}. The Y_0 norms add a factor L across levels. Therefore $Q^0 E^0 = I + T$ with only adjacent-level upper leakage and $\|T\|_{Y_0 \to Y_0} \le L^{-1}$. No depth factor enters: the level blocks of T have distinct source and target levels. Unit transports in local flat frames preserve these bounds, also with nontrivial flat holonomies.

At a real curvature-small reference, put the same seed profiles in their local comb frames, rotating the colour to its own row tail frame. Use a common real frame on a complete row carrier when computing its action. The gauge transition between two such frames differs from its value at the pillar centre by $C_L\kappa$, by telescoping the bond equation along a path of length $O_L(\xi_j)$; both potentials are $O_L(\kappa/\xi_j)$. The per-length straight row differs from its local flat row by at most $C_L\kappa n^{-4}$ per fine input, by summing the relative one-step distortions $O_L(\kappa L^{i-j})$ over its backward scales. This is the straight-row comparison on the complete carriers. The same normalization bounds row and column sums. Each seed affects only a bounded number of rows at its own and next level, and each such row meets at most C_L seeds. The entrywise errors thus give a $C_L\kappa$ perturbation in Y_0. The leading next-level response is bounded by the scalar Q_L^0 weights, since the fixed colour rotations have norm one. Hence $\|QE^0 - I\|_{Y_0 \to Y_0} \le L^{-1} + C_L\kappa < 1$. The seed energy estimate is unchanged up to a fixed constant: the connection term in its first difference costs at most $C_L\kappa/\xi_j$.

The formula $E_Q = E^0(QE^0)^{-1}$ now proves the result on real fields. For a complex shift, keep the real seeds fixed. On a smaller tube the analytic row coefficients change by $C_L\varepsilon n^{-4}$ per input on the same carriers. This follows by Cauchy's formula applied to the bounded normalized coefficient on the fixed physical perturbation ball. The contraction reserve therefore persists and the lift is holomorphic. The identities eq. 5.ix give $QE_L = I$ with horizontal range. Testing a row covector on $E_L b$, where b attains its dual norm, proves the inf-sup bound C^{-1} in X_L. Short periods are handled on a fixed large cover by averaging the seed lift over simultaneous deck translations before solving QE^0. All the norms and the right inverse equation descend. □

5.3. **Site Green bounds for a bounded cell source.**

Proposition 5.4 (Local norms of the site Green operator)**.** Use the true graph, actual site average and real curvature hypotheses of Proposition 5.2. If f is supported on one pillar B_z and $u = G_s f$, then

$$\xi_y^{-2}\|u\|_{\infty,B_y} + \|\Delta_U u\|_{\infty,B_y} \le Ce^{-md(y,z)}\|f\|_\infty. \tag{5.xiii}$$

In the regular local coefficient class, for $0 < \alpha < 1$ the left side may also include $\xi_y^{-1}\|D_U u\|_{\infty,B_y} + \xi_y^{\alpha-1}[D_U u]_{\alpha,y}^U$. The first-difference stencils and transported Hölder pairs use fixed comparable-scale enlargements of B_y, inside the complete local stock. At a bare boundary use the present stencils. With this local coefficient class the estimates continue to the admitted weighted C^1 complex tube, where the norms use the fixed real frames. The rate is positive with a fixed reserve for the scale ratios used below. ■

Proof. The actual mass term $q^{\mathsf{T}} a_s q u$ at a site of B_w has norm at most $C\xi_w^{-4}\|u\|_{2,B_w}$: the transported average costs ξ_w^{-2} by Cauchy–Schwarz, and a_s supplies ξ_w^{-2}. Apply the real scalar modulus estimate of Lemma 3.5 on an interior ball of radius comparable to ξ_y, with its neighbouring whole pillars retained for this mass. The scales there are comparable by separation. Its bound is $C\xi_y^{-2}\sum_w \|u\|_{2,B_w} + C\xi_y^2\|f\|_{\infty,\mathrm{loc}}$ over a fixed enlarged stock. At a bare boundary the point-value bound directly costs $C\eta^{-2}\|u\|_{2,\mathrm{loc}}$.

The two exponential energy blocks of Proposition 5.2, and $\|f\|_{2,B_z} \le \xi_z^2\|f\|_\infty$, give a bound $C\xi_y^{-1}\xi_z^3 e^{-s_0 d(y,z)}\|f\|_\infty$ for the first term. By eq. 2.ix, the factor $(\xi_z/\xi_y)^3$ is absorbed at a smaller positive rate, using the prescribed separation. A local force contributes only at bounded metric distance from z. The equation then bounds $\Delta_U u$ from the local L^∞ bound on u, proving eq. 5.xiii.

For the differentiated assertion, apply Proposition 3.10 in the declared local frames at radius comparable to ξ_y. The normalized input energy $\xi_y^{-2}\|u\|_{2,\mathrm{loc}} + \xi_y^{-1}\|D_U u\|_{2,\mathrm{loc}}$ has exactly the preceding bound. The mass term is zeroth order there, with L^p norm bounded by $C\xi_y^{-2}\|u\|_{p,\mathrm{outer}}$ on its complete carrier. For any fixed $\alpha < 1$ choose $p > 4/(1-\alpha)$. The result bounds the scaled first differences and their transported Hölder quotient. In the finitely coarse bare collars present-stencil differences and Hölder quotients follow from the point-value bound and $\xi_y = \eta$. For complex coefficients use the continued energy blocks and the complex local bootstrap in the same real frames. The mass and scale estimates are unchanged up to constants because the complex transport condition numbers are uniformly bounded on their complete carriers. The equation again supplies the Laplacian bound.

In particular, for the normalized site column F_z of F in a physical orthonormal top colour basis, the right side in $A_s F_z = q^{\mathsf{T}}\Xi_s^{-2} e_z$ has supremum norm $C\xi_z^{-4}$ and is supported in B_z. Here e_z is a unit top vector with value of magnitude ξ_z^{-2}. Thus

$$\|F_z\|_{\infty,B_y} + \xi_y^\alpha [F_z]^U_{\alpha,y} \le C_\alpha \xi_y^{-2} e^{-m' d(y,z)}$$

in the regular class. This follows from the derivative bound and the bounded metric diameter of the cell, and spends an additional $4\log L/\rho$ to replace $\xi_y^2\xi_z^{-4}$ by ξ_y^{-2}. The estimate is for a column with its physical normalization, rather than for a point-mass fine source. □

Corollary 5.5 (Site Green operator on a cell flux)**.** Use the regular local coefficient class and the actual graph of Proposition 5.4. Let F be a bounded field supported on the bonds incident to a pillar B_z. Put $u = G_s D_U^{\mathsf{T}} F$, with the divergence in the physical variational pairing. For $2 < p < \infty$ and $0 < \alpha < 1$, one has

$$\xi_y^{-1}\|u\|_{\infty,B_y} + \xi_y^{\alpha-1}[u]^U_{\alpha,y} + \xi_y^{-4/p}\|D_U u\|_{p,B_y} \le C_{p,\alpha} e^{-m d(y,z)}\|F\|_\infty. \tag{5.xiv}$$

If F also has the scaled C^α bound on the complete lifted source stock, then

$$\|D_U u\|_{\infty,B_y} + \xi_y^\alpha [D_U u]^U_{\alpha,y} \le C_\alpha e^{-m d(y,z)}(\|F\|_\infty + \xi_z^\alpha [F]^U_{\alpha,z}). \tag{5.xv}$$

Here the source stock includes the zero values of the supported flux on the collars used by any cutoff meeting it. In particular its seminorm includes increments to those zero values. Output seminorms and bare stencils use the same conventions as Proposition 5.4. Both statements hold on the same type of smaller complex coefficient tube, with a fixed positive reserve in the rate. ■□

Proof. The weighted site energy estimate in the proof of Proposition 5.2 applies with the dual source $D_U^{\mathsf{T}} F$. Indeed, testing the divergence against a site field uses one difference of the test. Under exponential conjugation its additional multiplication coefficient is $O(|s|\ell^{-1})$, by the difference product rule across each bond. This is also controlled by the site energy. Taking the distance weight from the source stock gives, on every fixed enlarged output pillar,

$$\xi_y^{-1}\|u\|_{2,\mathrm{loc}} + \|D_U u\|_{2,\mathrm{loc}} \le C e^{-s d(y,z)}\xi_z^2\|F\|_\infty.$$

The physical volume of the complete flux stock accounts for ξ_z^2. Thus the rescaled energy input in Proposition 3.10 is bounded by $C\xi_y(\xi_z/\xi_y)^2 e^{-sd(y,z)}\|F\|_\infty$. The strict scale reserve absorbs the ratio with loss at most $2\log L/\rho$ in the rate.

On a complete stock the equation is $(\Delta_U + q^{\mathsf{T}} a_s q)u = D_U^{\mathsf{T}} F$. The mass is an admitted zeroth-order term for that proposition on each cutoff support, with its complete pillar inputs. The rescaled L^p flux norm on those stocks is at most $C\xi_y\|F\|_\infty$, and it vanishes unless the stock is at bounded metric distance from z. Apply the divergence estimate for the requested p. For the value and its Hölder bound take a finite exponent $p_1 > 4/(1-\alpha)$ and use the first-order Morrey estimate. This proves eq. 5.xiv.

For eq. 5.xv, use the Hölder-flux version of the same bootstrap, with the mass term still estimated as an ordinary finite-p source. When the local flux is nonzero, the two scales are comparable and the rescaled C^α flux norm is at most $C\xi_y(\|F\|_\infty + \xi_z^\alpha [F]^U_{\alpha,z})$. Undoing the output scaling yields the bound. At a bare boundary the corresponding atomic energy estimates cost η^{-2} for a point value and $\eta^{-\alpha}$ for a Hölder quotient. The displayed physical weights cancel these factors. The complex proof uses the continued weighted site energy estimate and the bilinear local divergence equation. Its local estimates use the fixed real comparison frames. □

6. Coercivity on separated determining regions

We use the covariant curl eq. 2.iv on the inner plaquettes, with physical measure η^4.

Let $H_W(U)$ be the Wilson Hessian in the physical left exponential coordinates. For the actual per-length nonlinear straightened map $\mathcal{Q}_U$, put

$$\begin{aligned} B_\omega(v,w) &= \sum_c \xi_c^4 \langle \omega(c), D^2\mathcal{Q}_{U,c}(0)[v,w]\rangle, \\ \gamma_\omega &= \sup_c \xi_c^3 |\omega(c)|. \end{aligned} \tag{6.i}$$

The fixed-input-leg moment of the actual averaging derivative is $C_L\xi_c(\eta/\xi_c)^4$. Pairing this moment with $\xi_c^4|\omega(c)|$ leaves at most $C_L\gamma_\omega\eta^4\xi_c^{-2}$ on each fine input. The carriers have bounded overlap and their scales are comparable to the fine scales they read. By symmetry the same bound holds on either leg. Weighted Schur then gives

$$|B_\omega(v,w)| \le C_L\gamma_\omega \|v\|_{\mathrm{m}}\|w\|_{\mathrm{m}}. \tag{6.ii}$$

The argument applies on the admitted complex chart. The estimate is a coefficient estimate for all input vectors, even though it was obtained on a small analytic ball.

For real curvature-small U the Wilson Hessian satisfies

$$\begin{aligned} |\langle v, H_W(U)w\rangle_f - \langle \mathrm{d}_U v, \mathrm{d}_U w\rangle_f| &\le C_L\kappa\|v\|_X\|w\|_X, \\ \langle v, H_W(U)v\rangle_f &\ge (1 - C_L\kappa)\|\mathrm{d}_U v\|_2^2 - C_L\kappa\|v\|_{\mathrm{m}}^2. \end{aligned} \tag{6.iii}$$

Indeed varying a plaquette inserts four ηv's transported by its successive prefixes. The trace Hessian with the final plaquette matrix replaced by I is the pairing of their sums, since the trace kills the commutator parts. In the difference each term contains the final plaquette defect $O(\kappa\eta^2\mathsf{L}^{-2})$ and two factors η. Their product supplies the physical measure and the mass weight. The prefix curl and d_U differ only by a coefficient $\eta^{-1}(I - \operatorname{Ad} U(\partial p))$ on two of the link values, as follows by multiplying the four prefixes and using the reversed-link rule. Its norm is at most $C_L\kappa\|v\|_{\mathrm{m}}$ because $\eta \le \mathsf{L}$. Cauchy–Schwarz and the bounded number of plaquettes at a link give eq. 6.iii. In particular no boundary-crossing action term has entered this estimate.

6.1. **The flat decomposition.**

Lemma 6.1 (A regular decomposition at a real flat connection)**.** On the true active graph at a real flat connection V, with the separated no-inward geometry and the actual site and field rows, every field has a decomposition

$$v = D_V\lambda + r, \qquad q\lambda = 0, \qquad \|r\|_X \le C(\|\mathrm{d}_V v\|_2 + \|Qv\|_Y).$$

The constant is independent of topology, flat holonomies, depth and compatible volume. ■

Proof. First we relate the outward average to differences of individual pillar values. Two pillars sharing a fine face have equal or adjacent levels. For an equal-level pair the coarse difference is its interior row. For an unequal pair the larger scale is s. In a parallel frame on their two-scale stock, the larger value a and the L^4 smaller values b_w in the whole outer block enter the outward row as $s^{-1}(a-\bar b)$, with $\bar b$ their arithmetic mean. All smaller rows internal to that block are retained. For each face neighbour w, write $a - b_w = (a-\bar b) + (\bar b - b_w)$. The finite L-grid Poincaré inequality estimates the second difference by the internal smaller differences. Conversely choose one face neighbour w_0 and write $a - \bar b = (a - b_{w_0}) + L^{-4}\sum_w (b_{w_0} - b_w)$. The last sum uses the same smaller rows along coordinate paths of length at most $4L$. Thus a route through a bounded number of comparable pillars is controlled by its actual interior and outward differences, together with the internal rows of each full outer block. The length factors on that route differ by bounded powers of L. This proof on lifts retains all faces, even if a periodic pair of pillars has more than one shared face.

Next consider a convex fine box D of side lengths comparable to s. In a parallel frame project the field orthogonally onto gradients of site functions on that box. The remainder r_D is orthogonal to gradients, including the boundary site equations, and has curl equal to the given field's curl. In one dimension with $N+1$ vertices, the scalar path Neumann matrix has eigenvalues $4\eta^{-2}\sin^2(\pi m/[2(N+1)])$, $0 \le m \le N$. This follows by substituting the cosines with half-site offset. The edge matrix has exactly its nonzero spectrum, since the incidence map is onto edges. The cochain Hodge matrix on a box is the sum of those tensor factors, using the edge matrix in the component direction and the scalar matrices in the others. This identity follows by expanding the curl and divergence squares, whose mixed terms cancel. Each component has its nonzero edge frequency. Its quadratic form controls the interior component differences and its lowest eigenvalue is at least cs^{-2}. Therefore

$$s^{-1}\|r_D\|_{2,D} + \|\nabla_V r_D\|_{2,D/2} \le C\|\mathrm{d}_V v\|_{2,D}. \tag{6.iv}$$

The gradient potential can be given zero average on a specified central pillar of side comparable to s by subtracting a parallel constant.

Choose a fixed large $K = K(L)$ and take ρ larger than all the required $C_L K$ collars. A box D_y of radius $K\xi_y$ about a pillar top can meet at most two adjacent assigned levels. Otherwise it would cross an intervening layer of width greater than its radius, by separation. Such boxes and fixed enlargements have bounded overlap, by counting tops of comparable scales. Smooth sampled bumps supported in the doubled pillar cubes and equal to one on their pillars, divided by their sum, give a partition $\sum_y \chi_y = 1$, with first two difference bounds $C_L\xi_y^{-1}$ and $C_L\xi_y^{-2}$. Indeed the sum is between one and the fixed overlap bound, and each term contributing to a stencil has comparable scale.

At the true boundary, first write $v = v_b + w$, where v_b equals v within $H\eta$ of the exterior and vanishes beyond $2H\eta$, with $H > 10K$ fixed. Take $\rho > 4H + C_L$. All its nonzero bonds are bare determining rows, where $Q_0 v = v$, and no higher carrier meets that support. The cutoff difference costs at most $C\eta^{-1}$ there. Thus

$$\|v_b\|_X + \|\mathrm{d}_V w\|_2 + \|Qw\|_Y \le C(\|\mathrm{d}_V v\|_2 + \|Qv\|_Y).$$

A nonbare box is inside the active region by separation. A bare box meeting the exterior lies entirely in the zero collar of w, where its decomposition is set to zero. Hence every nonzero local decomposition uses only the flat connection on an inner rectangular box.

On D_y write $w = D_V\lambda_y + r_y$ with $q_y\lambda_y = 0$ using eq. 6.iv. If two cutoff supports or their two-step collars intersect, their scales are comparable and their centres are within $C_L\sigma$, where $\sigma = \max(\xi_y, \xi_z)$. For K chosen large enough their half-boxes intersect in a neighbourhood containing the central pillars and a coordinate route between them. Only C_L pillars of comparable sizes meet this route. Enlarge it to include the complete outward stocks used in the first paragraph; this fits into the intersection by the choice of K. On this route $\overline{D}q\lambda_z = Q(w - r_z)$ by eq. 2.vi. Its actual site value at z is zero. The route estimate gives

$$|q_y\lambda_z| \le Ce_{yz}, \qquad e_{yz}^2 = \sum_{c \text{ in route}} \xi_c^2|Qw(c)|^2 + \|\mathrm{d}_V w\|_{2,D_y\cup D_z}^2.$$

For the remainder contribution, the per-length row has physical L^2 functional norm at most $C\xi_c^{-2}$, by the product of its coefficient sum and its maximum coefficient divided by η^4. Equation eq. 6.iv bounds $\xi_c|Qr_z(c)|$ by $C\|\mathrm{d}_V w\|_{2,D_z}$. This proves the stated route estimate. If a box meets the exterior, all boxes whose cutoffs overlap its cutoff are bare and remain inside the zero collar of w, by $H > 10K$ and separation. Their potentials and remainders are zero. Every other pair neighbourhood used in the estimate is inner. Poincaré with the y-pillar mean on this comparable-volume intersection now gives

$$\|\lambda_y - \lambda_z\|_2 \le C\sigma^2 e_{yz}, \qquad \|D_V(\lambda_y - \lambda_z)\|_2 \le C\sigma e_{yz}.$$

The gradient bound comes from $r_z - r_y$. For the first bound the mean costs $\sigma^2|q_y\lambda_z|$ in physical L^2. Bounded overlap gives $\sum e_{yz}^2 \le C(\|\mathrm{d}_V w\|_2^2 + \|Qw\|_Y^2)$.

Put $\lambda_* = \sum_y \chi_y\lambda_y$. In $w - D_V\lambda_*$ the derivatives of the cutoff multiply only differences $\lambda_y - \lambda_z$, because their sum is zero. One more difference produces at most $\sigma^{-2}\|\lambda_y - \lambda_z\|_2 + \sigma^{-1}\|D_V(\lambda_y - \lambda_z)\|_2$ for each overlap. The remaining terms cost the two terms in eq. 6.iv. The same estimates control the mass norm. Thus the X bound for the glued remainder follows by bounded overlap. Since $q_y\lambda_y = 0$, the glued mean $q_y\lambda_*$ only reads those same potential differences. The physical L^2 norm of its averaging functional is ξ_y^{-2}, which cancels the factor σ^2. Consequently

$$\sum_y |q_y\lambda_*|^2 \le C(\|\mathrm{d}_V w\|_2^2 + \|Qw\|_Y^2).$$

At a flat field the compact scalar bump lifts of the site averages obey $\|D_V E_s b\|_X^2 \le C\sum_y |b_y|^2$: the first weighted and second ordinary differences have pointwise size $C\xi_y^{-2}|b_y|$ on volume $O(\xi_y^4)$. Bare atomic seeds give the same estimate. Subtract $E_s q\lambda_*$ from the potential and add its gradient and v_b to the remainder. This gives the decomposition.

For short periods apply the construction on a compatible cover on which these finitely sized boxes embed. Deck averaging the potential preserves its zero site means and the norm bound for its remainder. The input and resulting remainder norms have the same sheet-count factor. This uses no global choice of flat frame. □

6.2. **The real and complex form gaps.**

Theorem A (Uniform layered form gaps)**.** On the separated aligned geometry of eq. 2.i, use the real curvature class eq. 2.xii with κ sufficiently small and the positive weights of eq. 5.ii and eq. 5.vii. The actual forms

$$\begin{aligned} K_0 &= H_W(U) + D_U R D_U^{\mathsf{T}} + Q^{\mathsf{T}} a_f Q, \\ K_F &= H_W(U) + D_U D_U^{\mathsf{T}} + Q^{\mathsf{T}} a_f Q \end{aligned} \tag{6.v}$$

satisfy $c\|v\|_X^2 \le \langle v, K_i v\rangle_f \le C\|v\|_X^2$ for $i = 0, F$. The same is true after subtracting the form B_ω from K_0 when γ_ω is sufficiently small. The constants are uniform in mesh, depth, shapes and compatible volume. The corresponding normalized energy inverses are holomorphic and uniformly bounded on a fixed smaller weighted C^1 complex tube, with the actual constraints and projections continued there. ■□

Proof. At a real flat reference V, use Lemma 6.1. Its potential satisfies

$$\Delta_V \lambda = R D_V^{\mathsf{T}}(v - r)$$

by the range identity eq. 5.ix. Equation 5.xi yields $\|D_V\lambda\|_X \le C(\|RD_V^{\mathsf{T}} v\|_2 + \|r\|_X)$. Together with the remainder bound this proves the K_0 lower bound at V. The upper bound follows from the local differences and the physical Schur bound for Q.

Suppose U and a real flat reference V on the same geometry are in a common gauge with $U_b = e^{i\eta A_b} V_b$ and $\varepsilon = \sup_b \mathsf{L}(b)|A_b|$ small. The local comparison of the actual rows and the shifted gradient coefficients gives

$$\begin{aligned} \|\ell(D_U - D_V)u\|_2 &\le C\varepsilon\|u\|_2, \\ \|(q_U - q_V)u\|_{2,s} + \|\Xi_s^{-2}(q_U - q_V)(\ell^2 u)\|_{2,s} &\le C\varepsilon\|u\|_2, \\ \|(Q_U - Q_V)v\|_Y &\le C\varepsilon\|v\|_{\mathrm{m}}. \end{aligned}$$

For the row bounds pass to a flat frame on each complete carrier and use the averaging coefficient estimates with the shift as parameter. An entry of a per-length row changes by $C_L\varepsilon(\eta/\xi_c)^4$, since its relative perturbation at each intermediate scale is $O_L(\varepsilon L^{i-j})$ for a level-j row. The geometric sum and the bounded carrier overlap yield the stated physical norms. The same transported average comparison proves the q bounds.

The site column difference solves

$$A_{s,U}(F_U - F_V)b = (q_U^{\mathsf{T}} - q_V^{\mathsf{T}})\Xi_s^{-2} b - (A_{s,U} - A_{s,V})F_V b.$$

Test it with $\ell^2(F_U - F_V)b$. The left side is the plus-weight energy in the proof of Proposition 5.2. In the gradient difference one D is replaced by an $O(\varepsilon\ell^{-1})$ coefficient; the remaining derivative and the weight are controlled by the plus-weight energies using the product rule. The mean difference uses the two site-row bounds just proved. It follows that $F_U - F_V$ and its ℓ-weighted first derivative are $O(\varepsilon)$ in their physical operator norms. The Gram inverse identity then gives $\|R_U - R_V\|_{2\to 2} \le C\varepsilon$. Also $\|(R_U D_U^{\mathsf{T}} - R_V D_V^{\mathsf{T}})v\|_2 \le C\varepsilon\|v\|_X$, since the adjoint coefficient difference is bounded by $C\varepsilon\|v\|_{\mathrm{m}}$. Finally $\mathrm{d}_U - \mathrm{d}_V$ obeys the same mass bound. Equations 6.iii and eq. 6.v give a form difference $C(\kappa + \varepsilon)\|v\|_X\|w\|_X$. This proves the gap in the common gauge by absorption in the flat form.

To remove the global small-potential condition, localize on the original graph. We give the two estimates needed for localization. Write $P = I - R$. The mass-normalized matrix

$$K_s^{\mathrm{sm}} = \ell D_U P D_U^{\mathsf{T}} \ell = (\ell D_U F) M^{-1} (\ell D_U F)^{\mathsf{T}}$$

has exponential cell-L^2 entries by Proposition 5.2. By taking ρ large with fixed reserve in eq. 2.xi, these entries have summable polynomial moments on the layer metric, in particular their second moment.

Choose a fixed large radius $R_* > 1$, then increase ρ so that balls of radius $C_L R_* \xi_y$ meet at most two adjacent scales. There is a squared partition $\sum_i \chi_i^2 = 1$ with χ_i supported on balls of radius $2R_*\xi_i$, bounded overlap, and

$$\sum_i |D\chi_i(x)|^2 \le C R_*^{-2}\ell(x)^{-2}, \qquad \sum_i |\chi_i(x) - \chi_i(z)|^2 \le C\min\{1, d_\ell(x,z)^2/R_*^2\}.$$

To construct it, select centres in decreasing scale with disjoint balls of radius $R_*\xi_i/(20L)$ until maximal. Their enlarged balls cover by comparability of scales where they meet. Packing gives bounded overlap. Use smooth bumps equal to one on the covering balls and divide by the square

root of their squared sum. Include bare exterior cells for this construction if the active region is proper. Integrating the vector of cutoff derivatives along a metric path proves the second bound. Within each cell the sum of the squared suprema of the changes from its centre is $O(R_*^{-2})$, by the derivative bound and the bounded number of overlapping supports.

The localization error of K_0 is bounded by $CR_*^{-2}\|v\|_{\mathrm{m}}^2$. In fact for a symmetric matrix K its error is the double cutoff $-\frac{1}{2}\sum_i[\chi_i,[\chi_i,K]]$. For K_s^{sm}, on a cell pair expand each cutoff difference into its centre difference and the two within-cell differences. Each term of the double commutator is left and right multiplication by the corresponding scalars or cutoff deviations. By Cauchy–Schwarz in i, its block operator norm is bounded by $CR_*^{-2}(1+d_\ell(y,z))^2$ times the entry norm. Summation uses the second moment, with $\ell^{-1}v$ as input to the normalized smoothing matrix. The remaining fine-range Wilson and divergence matrices have coefficients $O(\eta^{-2})$ in physical array form. Their squared cutoff differences cancel that denominator, since $|D\chi_i|$ costs $CR_*^{-1}\ell^{-1}$. The row mass has uniformly bounded normalized coefficient Schur sums on its bounded metric carriers, so the same double-cutoff estimate applies. The local shift formula for $\|v\|_X^2$ gives the same localization error. A zero-extension jump is exact since it has only one nonzero value.

We also need that changing a connection outside a large patch changes its compressed projector exponentially little. Suppose two admitted real fields agree on the cube of radius $24R_*\xi_i$ in a common gauge, with comparable scales there, and have the same determining geometry. For a cutoff θ with gradient in an intermediate annulus, the exact identity

$$\theta G_{s,1} - G_{s,2}\theta = G_{s,2}(A_{s,2}\theta - \theta A_{s,1})G_{s,1}$$

has middle factor supported in the annulus with the site energy-to-dual-energy bound. Its terms are the gradient commutator and the two actual site-row cutoff differences, since the coefficients agree wherever θ or its required neighbours are nonzero. The exponential conjugated energy estimates of Proposition 5.2 make the relevant tails $O(e^{-cR_*})$ in that norm. Here a path leaving the patch from its core must pay length at least cR_* before leaving its comparable-scale cube. The fine inverse carries its two factors of ξ_i; for F the source Ξ_s^{-2} cancels them. For top input on a middle cube of radius $8R_*\xi_i$ this gives an $O(e^{-cR_*})$ bound on the difference of the two columns in the interior. Their outputs outside a larger middle cube have the same small norm by the column tail estimate. The two Gram matrices compressed to the middle top cube therefore differ in norm by $O(e^{-cR_*})$.

Both compressed Gram matrices retain the uniform gap. The difference between the compressed inverse and the compression of the global inverse, applied to input on the inner cube, is the product of the compressed inverse, the off-diagonal Gram block, and the exterior tail of the global inverse. This follows by restricting $MM^{-1} = I$ to the middle cube. The exterior tail is $O(e^{-cR_*})$ by the weighted inverse bound. Thus inserting $FM^{-1}F^{\mathsf{T}}$, with its analogous column and transpose tails, gives

$$\|1_{\mathrm{core}}(R_1 - R_2)1_{\mathrm{core}}\|_{2\to 2} \le Ce^{-cR_*} \tag{6.vi}$$

on the cube of radius $3R_*\xi_i$. These estimates are on supported vectors and require no sum of errors over the number of fine sites.

For an interior localization patch choose a comb frame on the cube of radius $32R_*\xi_i$, then extend its potential by a scalar cutoff on that cube, equal to one on the agreement cube. Keep the original graph and all its assigned averages. The comb estimate gives potential size $C_L R_*\kappa/\xi_i$ there. The cutoff derivative is $O((R_*\xi_i)^{-1})$. Expanding the plaquette product shows that the new curvature is at most $C_L(\kappa + R_*^2\kappa^2)/\xi_i^2$: the linear plaquette difference comes from the original curvature and the cutoff derivative, and the quadratic product costs $C_L R_*^2\kappa^2/\xi_i^2$. Thus the comparison field is globally in a small potential gauge with the identity on the same graph. For $v_i = \chi_i v$, the inner Wilson and actual row forms agree exactly with this comparison field, by their complete stocks. The divergence is supported in the core, so eq. 6.vi changes its gauge form by at most $Ce^{-cR_*}\|v_i\|_X^2$. We used orthogonality of R to write the form as $\langle D_U^{\mathsf{T}} v_i, RD_U^{\mathsf{T}} v_i\rangle$.

If a bare patch's enlarged comparison cube would meet the exterior, its entire support is in the bare boundary collar by the larger separation. On that support $Qv_i = v_i$, higher carriers miss it, and the row mass is at least $d_*\eta^{-2}\|v_i\|_2^2$. The possible negative Wilson remainder is $C\kappa\eta^{-2}\|v_i\|_2^2$ by eq. 6.iii. The direct bound $\|\nabla_U v_i^0\|_2 \le C\eta^{-1}\|v_i\|_2$ therefore gives the required gap on these patches. Choose R_* so the two localization errors are small relative to the fixed flat gap, then choose the separation for the patches and the rate reserve. Finally reduce κ to admit all comparison fields in the common-gauge gap. Summing the local bounds proves the K_0 lower bound on the real curvature class. The same argument on a compatible cover descends to short periods, since the site and Gram inverses preserve the deck-invariant space.

The upper bounds for both forms follow from their local differences, the row bound and $\|R\| \le 1$. Also $K_F - K_0 = D_U P D_U^{\mathsf{T}}$ is nonnegative at a real reference. Equation 6.ii gives the asserted multiplier perturbation. On the complex tube, the actual R, D, Q comparison from Proposition 5.2 and the analytic averaging bounds give an $O(\varepsilon)$ form difference in X. For the Wilson coefficients, the C^1 weighted shift changes each plaquette by $O(\eta^2\varepsilon\mathsf{L}^{-2})$ and each prefix transport by $O(\eta\varepsilon\mathsf{L}^{-1})$. The calculation for eq. 6.iii gives the same X-form perturbation. A relative Neumann series about the real positive form proves bounded holomorphic inverses. The complex radii are chosen after the real uniform gap. □

Corollary 6.2 (The constrained Lagrangian saddle)**.** Under Theorem A, with γ_ω small, the Lagrangian Hessian $H_L = H_W - B_\omega$ is coercive on $X_L \cap \ker Q$. Given $b \in Y$ and a force f in the dual of X_L, the equations on X_L

$$H_L v - Q^{\mathsf{T}}\mu = f, \qquad Qv = b$$

have a unique solution with $\|v\|_X + \|\mu\|_{Y^*} \le C(\|f\|_{X_L^*} + \|b\|_Y)$. The conclusion concerns the specified inner action and actual row curvature at the given pair. ■□

Proof. On the simultaneous kernel the two positive penalties in K_0 vanish exactly, so the form gap gives the coercivity. Write $v = E_L b + z$ using Lemma 5.3, solve the coercive equation on $z \in X_L \cap \ker Q$, and define $\langle \mu, b'\rangle_r = (H_L v - f)(E_L b')$. Every vector of X_L is the sum of its $E_L Q$ part and a vector of the simultaneous kernel. This gives the full saddle and the estimate. Testing the difference of two solutions on the kernel and then on the lift proves uniqueness. □

6.3. **Ambient compression and the flat tree bound.** Let the real connection be specified also on the complete ambient plaquettes touching the active graph. Compress its Wilson Hessian by zero extension of the field variables. The site operator may use its corresponding Dirichlet compression, or the original graph Laplacian. For the Dirichlet convention include its boundary diagonal in the site form. We retain the same complete determining rows and their physical pairings. Assume the scaled curvature bound also on the crossing plaquettes.

The alternative single-comb averages use the full differential of the relative average in [5, eqs. (89)–(92), (124)], with its matching centred site average

$$q_{\mathrm{comb}}\lambda(y) = L^{-4}\sum_{x\in B_y} \mathrm{Ad}\, U(\Gamma_{yx})\lambda(x),$$

where Γ_{yx} is the chosen comb path from y to x. Here is the field average's one-step definition. For a coarse bond c, let $\Gamma_{c,x}$ run from its initial centre along the comb to x, along the translated long segment to x', and back along the terminal comb. Write $W(c)$ for the central long segment. The relative mean is

$$\overline{W}(c) = \exp\left(L^{-4}\sum_x \log\{W(\Gamma_{c,x})W(c)^{-1}\}\right) W(c).$$

All loop logarithms use their admitted small branches. For a fine tangent a, first solve the tree equations $(a - D\psi)|_{\mathrm{comb}} = 0$ with $q_{\mathrm{comb}}\psi = 0$. Tree integration determines ψ up to a transported

constant on each block, which the last mean condition fixes. Let $\overline{W}_t$ be this mean evaluated on $(e^{i\eta t a}U)^{e^{-it\psi}}$, and put

$$Q_{\mathrm{comb}}a(c) = (iL\eta)^{-1}\,\frac{d}{dt}\log\{\overline{W}_t(c)\overline{W}_0(c)^{-1}\}\Big|_{t=0}.$$

This full differential includes all differentiated transport and endpoint factors. Iterate these actual maps with the complete outward endpoint mean. These are the single-comb maps used in the compressed background operator of [6, pp. 392–394, eqs. (3.13)–(3.19)].

Proposition 6.3 (The compressed zero-current estimates)**.** Under the preceding separated geometry and coefficient bounds, the analogues of A_s, M, K_F, K_0 for the ambient compression have the same uniform normalized real gaps and energy inverse bounds. The row lift and zero-average restoration remain uniformly bounded. These assertions also hold for the full centred single-comb differential and its matching site rows just defined. Its precise point, Laplacian, divergence-source and Hölder output ranges are proved in Corollary 10.4 below, after the local source and field norms have been introduced. ■

Proof. The Dirichlet site form differs from the original one by a nonnegative diagonal supported in the bare boundary collar. Each diagonal coefficient is at most $8\eta^{-2}$: it counts the missing outside neighbours in the gradient form of the zero extension. The real weighted energy tests and the compact lift proof for the site Gram remain valid with this term. It is nonnegative in the tested form and bounded by the bare mass norm, also when paired with the compact lift. It vanishes on $\ker q$, because q is the identity on bare sites. Hence $\Delta_{\mathrm{Dir}}(\ker q) = \Delta_U(\ker q)$; the real orthogonal R is unchanged. The divergence of an active field is the restriction of the ambient divergence of its zero extension, which vanishes at all exterior sites.

A crossing plaquette contributes its transported curl square and a remainder as in eq. 6.iii, using zero field values on its inactive links. Every nonzero field value in these extra terms is in the bare collar by separation. The principal squares are nonnegative. The sum of the remainder absolute values is bounded by $C_L\kappa\eta^{-2}$ times the squared L^2 norm on that collar, whereas the sum of the principal squares is bounded above by the same mass bound without κ. Absorbing the small remainder in Theorem A proves the compressed field gap and the upper bound. This argument retains the crossing terms; it does not subtract them as a small operator.

The single-comb rows agree at a flat field after centring the blocks. In the same local real comb frames, the summable one-step comparison gives $O_L(\kappa)$ differences in both physical row and column norms, with the same $(\eta/\xi_c)^4$ coefficient density for field rows. These are the full finite-path analytic row estimates; see [5, eqs. (139)–(147), pp. 39–40; Proposition 7, p. 43]. For the iteration, write a per-length one-step derivative as $S_i + E_i$ in a common real comb frame. Its absolute row sum is $1 + O_L(\delta_i)$, its column sum is $L^{-4}(1 + O_L(\delta_i))$, and each entry of E_i is $O_L(\delta_i L^{-4})$. The relative potential at that step gives $\delta_i \le C_L\kappa L^{i-j}$ for a row at final level j. Telescope the product, use the later row sums and earlier column sums on either side of each error entry, and sum this geometric sequence. This proves both normalized error sums and the point density $C_L\kappa(\eta/\xi_c)^4$, uniformly in the depth. The common-frame site resolvent and Gram comparison used in Theorem A applies to this change of q at fixed connection, also in the presence of the same bare diagonal. Thus R and the field forms change by $O_L(\kappa)$ in the corresponding normalized norms. The seed lift proof applies with these complete row coefficients. To check the exact gradient identity, put $a = D\lambda$ in the defining tree correction. Then $\lambda - \psi$ is parallel on every comb, and its value at the centre is $q_{\mathrm{comb}}\lambda$ because $q_{\mathrm{comb}}\psi = 0$. Coarse endpoint covariance of the relative mean therefore gives

$$Q_{\mathrm{comb}}D\lambda = \overline{D}_{\mathrm{comb}}q_{\mathrm{comb}}\lambda.$$

At an outward row the complete further endpoint average is included in $\overline{D}_{\mathrm{comb}}$. Iteration preserves this identity and gives the corresponding restoration identities.

The conjugated energy estimates use the same gradient, actual row and finite-range commutators as before; the extra diagonal commutes with the weights and the crossing Hessian uses only bare stencils. Away from the bare boundary the complete local elliptic equations have the same stencils and the same coefficients as in the interior bootstrap. At the boundary the bare energy column gives the point and derivative bounds by the atomic estimates with $\xi_y = \eta$. Applying the interior site and zero-current component estimates with these energy columns gives the stated point and first-difference bounds. The site operator does not acquire a fine Dirichlet boundary in a nonbare layer. □

For clarity the flat strict tree case has a lower bound with an explicit dependence on L. Take in each retained block the tree whose paths from its centre follow a single fixed coordinate order. If the field vanishes on those trees and outside the retained stock, then for its flat straight one-step constraint,

$$S_L a = 0 \quad \Longrightarrow \quad \|a\|_2^2 \le CL^5 \|\mathrm{d}a\|_2^2. \tag{6.vii}$$

Here the raw S_L differs by the factor L from the per-length straight row, which does not affect its kernel. For an internal bond the difference of the two tree paths and that bond bounds its value by a sum of at most CL plaquette curls. Insert the bond step into the fixed coordinate order and move it past the later coordinate segments, forming strips of width one. Each strip has no repeated plaquettes and there are at most four strips. Cauchy–Schwarz and at most $4L^4$ internal bonds give the bound CL^5 on a block.

On a face normal to μ, use the central crossing as reference, with value β. The loop consisting of a tree path to a crossing link, that link, the return tree path, and the central segment has a filling by at most CL^2 plaquettes in the double block. To see the area and multiplicity, split the transverse coordinates into those before and after μ in the order. Move the unit crossing step past the later transverse segments, using width-one strips. The earlier transverse segments then bound rectangles of width L along the central segment. This uses at most six strips or rectangles, each without repeated plaquettes. Consequently each crossing value differs from β by at most $CL\|\mathrm{d}a\|_{2,\text{double block}}$. The same bound holds for a complete translated longitudinal segment: extrude its initial tree path in direction μ by L; each transverse segment sweeps one such rectangle, while the longitudinal segment has zero swept area. The tree values and the internal parts of the central segment vanish. The zero straight average of the translated segments bounds $|\beta|$ by the same error. Squaring and summing over L^3 crossings gives CL^5. Summing over blocks and faces uses fixed overlap. The proof also applies on rectangular lifts and descends to compatible tori.

At a flat reference the mass-removed Schur form on supported top data, in either proper or compressed convention, dominates their coarse curl square with coefficient one. Indeed the minimizing lift has exact row value, its action curl plus the gauge term are the mass-removed form, and the flat flux row is a physical contraction. In the compressed case the additional flat plaquette squares are nonnegative. Combining this observation with eq. 6.vii gives the strict tree gap cL^{-5} on top-supported data in the kernel of their outer straight constraint.

6.4. **The deterministic curvature slices.** Let $\mathcal{R}_j$ be the level-j sheet average on exact fine two-cochains, including $\mathcal{R}_0 = I$. In counting measure let W_j be the orthogonal complement of $\ker \mathcal{R}_j$ inside the exact cochains and P_j its orthogonal projection. The sheet composition gives $W_{j+1} \subset W_j$. Thus $P_j P_k = P_k P_j = P_k$ for $j \le k$, and every exact cochain F satisfies

$$F = P_K F + \sum_{j=1}^{K} (P_{j-1} - P_j) F. \tag{6.viii}$$

The terms are mutually orthogonal because they belong to W_K and $W_{j-1} \cap W_j^{\perp}$, respectively. These are identities of deterministic finite operators. The linear background current also has its row form.

Since P_jF is in the range of the transpose of $\mathcal{R}_j$ restricted to exact cochains, it has the form $P_{\mathrm{ex}}\mathcal{R}_j^{\mathsf{T}}\phi$. The identity $\mathrm{d}_jS_{L^j} = \mathcal{R}_j\mathrm{d}$ then gives

$$\mathrm{d}^{\mathsf{T}}P_jF = S_{L^j}^{\mathsf{T}}\mathrm{d}_j^{\mathsf{T}}\phi.$$

Here P_{ex} is the projection onto exact cochains, and $\mathrm{d}^{\mathsf{T}}P_{\mathrm{ex}} = \mathrm{d}^{\mathsf{T}}$. Thus the minimizing background has a row multiplier even away from the zero datum.

Proposition 6.4 (Kernel of a flat curvature slice)**.** For $1 \le j \le K$ at the flat periodic reference, the counting-measure kernel of $P_{j-1} - P_j$ obeys

$$|(P_{j-1} - P_j)(p,q)| \le C_L n^{-4} e^{-c_L d_{j-1}(p,q)}, \qquad n = L^{j-1},$$

where the fine mesh is $1/n$ in level-$(j-1)$ units and d_{j-1} is periodic distance in those units. The bound is uniform in j and compatible volume. ∎

Proof. Let $H_n a$ be the unique least-norm fine curvature with sheet-average equal to the curl of the intermediate datum a. Then $H_n^{\mathsf{T}}H_n = \mathcal{E}_n$ in counting coordinates, by Proposition 4.1. We first show that H_n has a kernel bounded by $Cn^{-2}e^{-cd_{j-1}}$.

Use the notation of eq. 4.iii. Define $B_m = \prod_\nu b_{\nu,m}$ and the diagonal matrix

$$(\mathcal{D}_m)_{\mu\mu} = e^{-i(n-1)\zeta_{m,\mu}/2}B_m b_{\mu,m}.$$

Write

$$t^{\mathrm{f}}_{m,\mu} = e^{i\zeta_{m,\mu}} - 1.$$

The Feynman minimizer of the gauge-adjusted datum has alias $n\,\mathcal{D}_m\mathcal{E}_n a/s_m$. Indeed its fine covariance multiplier is n^2/s_m, the transpose filter is $\mathcal{D}_m/n$, and the minimizing coarse force is $\mathcal{E}_n a$ by eq. 4.iii. Correcting the datum by a fine gradient leaves its curl unchanged. At a fine offset r in the intermediate block the curvature symbol is therefore

$$\frac{1}{n}\sum_m e^{i\zeta_m\cdot r}\frac{t^{\mathrm{f}}_m \wedge (\mathcal{D}_m\mathcal{E}_n a)}{s_m}. \tag{6.ix}$$

The additional factor n^{-2} converting fine alias coefficients to the point value has already been included.

For the noncentral aliases, the uniform strip bounds from Proposition 4.1 give $|t^{\mathrm{f}}_m| \le C(1+|m|)/n$ and $|s_m| \ge c(1+|m|^2)$. The phase factors are bounded on that strip because $|r_\nu| \le n/2$. Thus the noncentral sum is bounded by $Cn^{-2}\|a\|$ times

$$\sum_{m\in\mathbb{Z}^4}(1+|m|)^{-1}\prod_\nu(1+|m_\nu|)^{-1}.$$

This converges by dyadic shells: on the shell of radius 2^l the first factor costs 2^{-l}, and the four truncated harmonic sums give at most $C(1+l)^4$.

For the central alias use $\mathcal{E}_n = sH - J^{-1}Ht^+(t^-)^{\mathsf{T}}H$ from the origin cancellation proof, where $s = s_0$. The term $\mathcal{D}_0Ht^+$ has leading part $\mathcal{D}_0\,\mathrm{diag}(f_{\mu,0}^{-1})t^+ = nB_0^{-1}t^{\mathrm{f}}_0$. This follows by cancelling $\sin(p_\mu/2)$ against $b_{\mu,0}$, including its phase. Its wedge with t^{f}_0 is zero. The remaining part is divisible by s, since $H - \mathrm{diag}(f_{\mu,0}^{-1})$ is divisible by s. Hence the quotient in eq. 6.ix is holomorphic and $O(n^{-1})$ there. Away from zero, s is bounded away from zero on a smaller strip and the same bound follows directly. The sum is periodic by alias reindexing. The shifted contour and the periodic image sum prove the bound for H_n.

The slice subspace $W_{j-1} \cap W_j^{\perp}$ is the range of H_n on intermediate link data a with $S_La = 0$ and zero link values on the prescribed internal coordinate trees. Indeed any element of this slice is H_na and has $\mathrm{d}_LS_La = 0$ by sheet composition. A gradient on the intermediate grid and a constant link field can remove this closed straight datum without changing $\mathrm{d}a$. Next subtract the site potential obtained by integrating along each internal tree. Choose its additive block constants so its block

average is zero. Its gradient then leaves $S_L a = 0$ and makes all tree links vanish. The reverse inclusion follows from the straight flux identity.

Let C_{tr} parametrize this hard-tree and straight-constraint subspace by its remaining nonpivot coordinates. The flat crossing pivot coefficient is L^{-3} and it is the only pivot in its row, so C_{tr} is bounded and local with constants depending only on L. Its free restriction is the identity. By eq. 6.vii and eq. 4.ii, the matrix $A_{\mathrm{tr}} = C_{\mathrm{tr}}^{\mathsf{T}} \mathcal{E}_n C_{\mathrm{tr}}$ has a uniform positive gap. Its inverse has an exponential kernel by the same weighted conjugation argument as in Corollary 4.3. The orthogonal projection onto the slice is exactly

$$H_n C_{\mathrm{tr}} A_{\mathrm{tr}}^{-1} C_{\mathrm{tr}}^{\mathsf{T}} H_n^{\mathsf{T}}.$$

Both exterior factors give n^{-2}, and their product is summed only over the fixed-width intermediate block cells. Multiplying the summable exponential kernels at a smaller rate therefore gives $C_L n^{-4} e^{-c_L d_{j-1}}$. □

6.5. **The full complex saddle on fixed data spaces.** Fix a real reference U_0 of Theorem A and use its energy norm X. Let $Z_0 = \operatorname{ran} R_0$ with its physical site L^2 pairing. At a complex perturbation $U_A = e^{i\eta A} U_0$ in the admitted weighted C^1 tube, define

$$\mathcal{C}_A v = (Q_A v, \Gamma_A v), \qquad \Gamma_A = R_0 R_A D_A^{\mathsf{T}} : X \longrightarrow Z_0.$$

Only the target is fixed by R_0. The site and field operators are the actual continued ones.

Proposition 6.5 (The moving-slice saddle)**.** For sufficiently small fixed tube radius and $\gamma_\omega = \sup_c \xi_c^3 |\omega_c|$, let $H_{A,\omega} = H_W(U_A) - C_\omega(U_A)$, with C_ω the operator of the form eq. 6.i. The independent multiplier ω may be complex. For $f \in X^*$, $b \in Y$ and $z \in Z_0$, the system

$$\begin{aligned} H_{A,\omega} v - Q_A^{\mathsf{T}} \mu - D_A R_A R_0 \nu &= f, \\ Q_A v = b, \qquad \Gamma_A v &= z \end{aligned}$$

has a unique solution in $X \times Y^* \times Z_0$, with norm at most $C(\|f\|_{X^*} + \|b\|_Y + \|z\|_2)$. The inverse on this fixed product is holomorphic. At $z = 0$, the field and physical row multiplier are exactly the solution of the constrained saddle on the actual moving slice $\ker(R_A D_A^{\mathsf{T}})$. ■

Proof. At the real centre the combined constraint has exact lift

$$(b, z) \longmapsto E_{L,0} b + D_0 G_{s,0} z.$$

The first term has zero projected divergence. The second has zero Q_0-value by eq. 2.vi and $q_0 G_{s,0} R_0 = 0$; its projected divergence is z. The restoration and field lift bounds give the uniform energy norm. On its simultaneous kernel the real Wilson form is coercive by Theorem A, because both positive penalties vanish. The full reference saddle is therefore invertible: lift the two data, solve on the coercive kernel, and evaluate the residual functional on the same lift to recover the two multipliers.

The row, gradient and normalized projector estimates give $\|\mathcal{C}_A - \mathcal{C}_0\| \le C\|A\|_{\mathrm{tube}}$ in the fixed spaces. Bilinear symmetry of R_A, R_0 gives the displayed transpose of Γ_A. The Wilson comparison and the second-leg moment give

$$\|H_{A,\omega} - H_W(U_0)\|_{X \to X^*} \le C(\|A\|_{\mathrm{tube}} + \gamma_\omega).$$

For the Wilson term the prefix variation costs $O(\eta \|A\|_{\mathrm{tube}}/\mathsf{L})$ and the curvature variation costs $O(\eta^2 \|A\|_{\mathrm{tube}}/\mathsf{L}^2)$. The first uses the undifferentiated potential, the second its weighted first difference. Their insertion into the actual plaquette form gives the stated energy comparison. A relative Neumann series about the reference saddle gives the estimate and holomorphy.

For $w \in \operatorname{ran} R_A$, $R_0 w = 0$ implies $w = (R_A - R_0) w = 0$ by the small norm difference. Hence $\ker \Gamma_A = \ker(R_A D_A^{\mathsf{T}})$. Restriction of the first equation to this kernel eliminates ν. Conversely Γ_A is onto, since $\mathcal{C}_A$ has a right inverse obtained by perturbing the displayed lift. Thus a residual functional vanishing on its kernel belongs to its transpose range, with a unique gauge multiplier.

If f was supplied only on the slice, any norm-preserving extension to X gives the same field and physical multiplier. The fixed-space holomorphy claim uses fixed source extensions. □

7. Projected Green operators and coarse response

All inverses in this section act on the prescribed finite fine-field or site space. We use the physical bilinear transpose T, and use Hermitian norms on their complexifications. Write Ξ_s and Ξ_f for multiplication by the assigned lengths on the site and field rows, respectively. At an outward field row this is the row length, even when the other endpoint has a smaller site length. Let a_s and a_f be the positive diagonal row masses, comparable to Ξ_s^{-2} and Ξ_f^{-2}. Put

$$X_y(u) = \|\ell^{-1}u\|_{2,B_y} + \|\nabla_U u\|_{2,B_y}.$$

The row space remains Y with its physical dual Y^* from eq. 5.ii. Here B_y is the fine cell of an assigned top, with a fixed derivative collar when a difference is measured, and ℓ is the comparison length of eq. 5.iii. The norms X_y sum in squares, up to bounded overlap, to the field energy norm X. At the bare physical boundary use the atomic difference convention for X. We work with the separated domains, complete rows and admitted real backgrounds of theorem A. Their complex tubes use the same spaces and fixed reference norms.

For a site function u and a site source h, define

$$\|u\|_{\mathcal{Y}_s} = \|u\|_2 + \|\ell Du\|_2, \qquad \|u\|_{\mathcal{Z}_s} = \|\ell^{-2}u\|_2 + \|\ell^{-1}Du\|_2 + \|\nabla_U Du\|_2, \qquad \|h\|_{\mathcal{Y}_s^\flat} = \sup_{v\neq 0} \frac{|\langle h, \ell^2 v\rangle|}{\|v\|_{\mathcal{Y}_s}}.$$

The local versions use the complete cell collars and bare atomic stencils.

Lemma 7.1 (Balanced site bounds and positive elimination)**.** Keep $A_s, G_s, F, M, P, R, \Lambda, S$ of proposition 5.2. In the distance weights $w = e^{s\psi}$ of that proposition, G_s satisfies $\|wG_sw^{-1}h\|_{\mathcal{Y}_s} \le C\|h\|_{\mathcal{Y}_s^\flat}$ for arbitrary site sources. For the inner Wilson Hessian H_W put

$$K_F = H_W + DD^T + Q^T a_f Q, \quad G_F = K_F^{-1}, \qquad K_0 = K_F - DPD^T.$$

Then $\Theta = M - (DF)^T G_F DF$ has a uniform positive upper and lower bound and

$$G_0 = K_0^{-1} = G_F + G_F DF\Theta^{-1}(DF)^T G_F. \tag{7.i}$$

The identities and inverse bounds extend holomorphically to a common smaller complex tube. ■

Proof. For an arbitrary source h, test $A_s u = h$ against $\ell^2 u$. The left side is the balanced test of eq. 5.x: its leading terms are comparable to $\|\ell Du\|_2^2 + \|qu\|_{\mathrm{top}}^2$. The differences of the weight cost $C_L \operatorname{Lip}(\ell)\|u\|_{\mathcal{Y}_s}^2$ by the product and complete-row estimates there. The pillar Poincaré bound absorbs them. On the right the pairing is bounded by $\|h\|_{\mathcal{Y}_s^\flat}\|u\|_{\mathcal{Y}_s}$ by definition, which proves the balanced inverse bound. Conjugation by w adds a relative error $C_L|s|$ from first differences and the $O_L(|s|)$ complete-row oscillations. This proves the weighted bound on arbitrary sources.

The layered gap gives $K_0 \ge \delta K_F > 0$ with a fixed $\delta > 0$. In real orthonormal physical coordinates use $C = G_F^{1/2} DFM^{-1/2}$. Conjugating the preceding inequality gives $I - CC^T \ge \delta I$. The singular values of C then give $I - C^T C \ge \delta I$ and consequently $\delta M \le \Theta \le M$. Multiplying the right side of eq. 7.i by $K_F - DFM^{-1}(DF)^T$ cancels the two terms containing Θ^{-1}, by the definition of Θ. This proves the formula.

For the tube, the local differences and finite row moments give relative form perturbations of size $C\varepsilon$ for A_s and K_F and a perturbation of the same size in the balanced site bound. In each covariant first difference the change is multiplication by $O(\varepsilon/\ell)$; for a row the weighted sums of the changed coefficients are $O(\varepsilon)$ in normalized row units. The Wilson form perturbation is controlled by its principal difference and curvature multiplication terms. These bounds and the real gaps give the site and field inverses by the Neumann formula. They give F in the balanced norm and Θ in the normalized row norm, since DF, from the site-top space with the physical L^2 pairing in eq. 2.ii

to X^*, is bounded by $\|F\|_2$. Thus M^{-1} and Θ^{-1} continue by the same argument. The displayed identities are algebraic with bilinear transpose. □

We record the weighted restoration estimate used with the current. In the same distance weights as above,

$$\|\ell^{-2} w\Lambda w^{-1} v\|_2 + \|wD\Lambda w^{-1} v\|_X + \|wSw^{-1} v\|_X \le C\|v\|_X. \tag{7.ii}$$

For this purpose, the balanced estimate shows that wFw_{top}^{-1} and its conjugate with $-s$ differ from F by $O(|s|)$ in the top-to-fine L^2 norm. Expand wMw^{-1} using these two factors, in the top weight. Its inverse is then a uniform small perturbation of M^{-1}. Thus wRw^{-1} is bounded on fine L^2. Set $\lambda = \Lambda w^{-1} v$. It has $q\lambda = 0$ and $\|w\Delta\lambda\|_2 \le C\|v\|_X$, by the first-difference product rule for D^T. Apply the zero-average pillar estimate with $h = \ell^{-1} w$. The row error of $q(h\lambda)$, since $q\lambda = 0$, is bounded by $C(\operatorname{Lip}\ell + |s|)\|\ell^{-2} w\lambda\|_2$ in the grade-two norm. Testing $\Delta\lambda$ against $h^2\lambda$ and absorbing the corresponding first-difference error bounds $\|\ell^{-2} w\lambda\|_2 + \|\ell^{-1} wD\lambda\|_2$. The covariant curl of $D\lambda$ is a curvature times λ, while its divergence is $\Delta\lambda$. The component-gradient estimate applied to $wD\lambda$ therefore bounds the remaining part of X: both weight commutators in curl and divergence are $O(|s|\ell^{-1} wD\lambda)$. The zero-average bare collar keeps the nonzero stencils inside the action region. This proves eq. 7.ii with only first differences of the weight. The same estimate holds on the smaller complex tube by perturbing the weighted operator and using the same zero-row equation and fixed real norms.

Proposition 7.2 (Current completion)**.** For a fine current J, write

$$(W[J]\lambda)(b) = \tfrac{i}{2}[\lambda(b_-) + \mathrm{Ad}(U_b)\lambda(b_+),\, J(b)].$$

The contraction of the second jet of the finite relative gauge by J is

$$B[J] = W[J]\Lambda + \Lambda^T W[J]^T - \tfrac{1}{2}\Lambda^T (D^T W[J] + W[J]^T D)\Lambda. \tag{7.iii}$$

For the actual Wilson current j_U,

$$H_W - B[j_U] = S^T H_W S.$$

The bound $\|wB[J]w^{-1}\|_{X\to X^*} \le C\gamma_{\mathrm{t}}$ holds with $\gamma_{\mathrm{t}} = \sup_{b \text{ touching } \Omega_1} \ell(b)^3 |J(b)|$. The empty touching stock gives $\gamma_{\mathrm{t}} = 0$. ■

Proof. For u set $\ell_u = (\Lambda u)(b_-)$ and $r_u = \mathrm{Ad}(U_b)(\Lambda u)(b_+)$. Apply the finite gauge $e^{-i\Lambda u}$ to $e^{i\eta u}U$. Its Hermitian bond logarithm relative to U, divided by η, is $(i\eta)^{-1}\log(e^{-i\ell_u} e^{i\eta u} e^{ir_u})$. Its first derivative is Su, and its second differential on u, v is

$$\tfrac{i}{2}([u, \ell_v + r_v] + [v, \ell_u + r_u]) - \tfrac{i}{2\eta}([\ell_u, r_v] + [\ell_v, r_u]).$$

Indeed the degree-two part of the logarithm of the product is half the sum of commutators in the written order. Trace invariance identifies the first term with the first two terms of eq. 7.iii. For the second, use $D\Lambda u = (\ell_u - r_u)/\eta$ and

$$[\ell_u - r_u, \ell_v + r_v] + [\ell_v - r_v, \ell_u + r_u] = 2([\ell_u, r_v] + [\ell_v, r_u]).$$

This gives the last term with its factor $1/2$. The chain rule for the gauge-invariant inner Wilson action gives the identity at j_U.

In a bilinear pairing move each displayed D^T and Λ^T onto its test vector. Each of the resulting terms pairs $W[J]\lambda$ against a field with X-norm controlled by eq. 7.ii. Pointwise endpoint comparison and the current grade give $\|\ell W[J]\lambda\|_2 \le C\gamma_{\mathrm{t}}\|\ell^{-2}\lambda\|_2$ on the touching stock. Indeed both site factors $\Lambda u, \Lambda v$ vanish at bare sites by the bare singleton rows of q. A bond with neither endpoint in Ω_1 therefore contributes zero. In the weighted pairing the endpoint weight ratio is bounded by a fixed constant; apply eq. 7.ii with both signs. No difference is applied to J. □

Let C_ω denote the Hessian of the actual per-length straightened constraint, contracted with the physical row multiplier ω. If its quadratic Taylor term is $C^{(2)}$, the convention is $\langle u, C_\omega u\rangle = 2\langle \omega, C^{(2)}(u)\rangle_{\mathrm{rows}}$. In the raw coordinates the same term is $\Xi_f C^{(2)}$, and the transpose of a raw lift in counting row coordinates is $\Xi_f^3 H^T$. Their counting pairing therefore equals the physical pairing just written. The two-leg row bounds give

$$\|\ell C_\omega u\|_2 \le C\Big(\sup_{c\ \text{nonbare}} \xi_c^3 |\omega(c)|\Big)\|\ell^{-1}u\|_2. \tag{7.iv}$$

In fact the second derivative in one leg has a normalized fixed-leg sum on the full carrier. In the transpose contraction the fine factor η^{-4} cancels that leg's density $(\eta/\xi_c)^4$ against ξ_c^4. The remaining per-length factor has size ξ_c when the two arguments are unscaled field values, so a multiplier of grade three leaves a field mass of grade two. Bounded row incidence and comparable assigned lengths give both absolute sums of the resulting $\ell C_\omega \ell$ matrix. This proves eq. 7.iv by the Schur inequality, and also proves its weighted version because the row diameter in the layer metric is bounded. The actual bare mean is linear and has zero second jet.

Theorem B (Normalized Schur matrices and coupled response)**.** On the admitted real and complex coefficient domains, take the touching current sufficiently small and define, in this order,

$$K_J = K_0 - B[J], \qquad G_J = K_J^{-1}, \qquad T_J = a_f^{1/2} Q G_J Q^T a_f^{1/2},$$
$$H_J = G_J Q^T a_f^{1/2} T_J^{-1} a_f^{1/2}, \qquad \omega = H_J^T J, \qquad K_{1,J} = K_J - S^T C_\omega S.$$

Write L^2_r for the physical row Hilbert space of eq. 2.ii. The second variant has $G_{1,J}, T_{1,J}, H_{1,J}$ defined from $K_{1,J}$ by the same inverse, row matrix and lift formulas. For either variant write K, G, T_f, H for these four objects and put $V_f = QGQ^T$. All weighted norms

$$G : X^* \to X, \quad T_f^{\pm 1} : L^2_r \to L^2_r, \quad H : Y \to X$$

are uniform in mesh, depth and volume. $QH = I$, and H is the K-energy-minimizing lift at real data. In counting orthonormal row coordinates Ξ_f^2, the entries of T_f^{-1} are $O(e^{-md(c,d)})$. The entries of V_f^{-1} as a counting array are $O(\xi_c^{-3}\xi_d e^{-md(c,d)})$, and as a physical row kernel are $O(\xi_c^{-3}\xi_d^{-3} e^{-md(c,d)})$.

For sources $f \in X^*$, $b \in Y$, the solution of $Ku - Q^T\mu = f,\ Qu = b$ is

$$u = (I - HQ)Gf + Hb, \qquad \mu = V_f^{-1}(b - QGf). \tag{7.v}$$

The X and Y^* output norms are bounded by $C(\|f\|_{X^*} + \|b\|_Y)$. Removing $Q^T a_f Q$ changes the multiplier to $\mu - a_f b$. The corresponding datum-to-multiplier map is $V_f^{-1} - a_f$. These formulas are holomorphic and have exponential cell-to-cell bounds in the stated norms. ■□

Proof. First consider K_0. Conjugating its local parts by $w = e^{s\psi}$ changes the weak form by $O(|s|)$ in $X \to X^*$, by the difference product rule and complete-row oscillation bound. For DPD^T use the weighted L^2 estimate of P in the proof of eq. 7.ii. It is $O(|s|)$ close to P, and the two adjacent first differences are bounded $L^2 \to X^*$ and $X \to L^2$, with commutators of size $O(|s|)$. Inverting the conjugated K_0 is thus a uniform perturbation of its real gap. The first current subtraction is small by proposition 7.2.

For the normalized row matrix at zero current the upper bound follows from the energy trace bound of $a_f^{1/2}Q$. For a physical row vector β use the exact field lift $E_Q a_f^{-1/2}\beta$ of lemma 5.3, whose X-norm is at most $C\|\beta\|_{\mathrm{rows}}$. Cauchy–Schwarz in the real K_0 energy bounds

$$\|\beta\|^2_{\mathrm{rows}} = \langle Q^T a_f^{1/2}\beta, E_Q a_f^{-1/2}\beta\rangle \le C\langle \beta, T_0\beta\rangle^{1/2}\|\beta\|_{\mathrm{rows}}.$$

This proves the positive gap. Conjugating $a_f^{1/2}Q$ with fine and row weights introduces an $O(|s|)$ normalized difference. The normalized T_J, conjugated by the row weight, is $O(|s| + \gamma_{\mathrm{t}} + \varepsilon)$ from the real reference T_0, where ε is the tube radius. Thus its inverse is uniform. The lift formula and

$QH_J = I$ now give a uniformly weighted H_J. The first lift readout estimate proved immediately below gives a nonbare ω of size $C\gamma_{\mathrm{t}}$. Use eq. 7.iv and eq. 7.ii for the second subtraction, then apply the identical trace and perturbation argument to obtain $T_{1,J}^{-1}$ and $H_{1,J}$.

Choose ψ to be the distance to the input cell, truncated above at the output distance. The weighted bounds give the exponential cell estimates. Conversion from the normalized counting array to V_f^{-1} multiplies the c, d entry by $\xi_c^{-2} a_f(c)^{1/2} a_f(d)^{1/2} \xi_d^2$, and conversion to a physical kernel divides by ξ_d^4. This proves both exponents of the physical lengths. Substitution proves eq. 7.v. At real data Ku for $u = Hb$ is in $\operatorname{ran} Q^T$ and hence K-orthogonal to $\ker Q$; strict positivity proves the minimizing assertion. The multiplier shift follows by substituting $Qu = b$ into the removed term. Every analytic inverse above is a Neumann perturbation in fixed reference norms. □

Corollary 7.3 (The two physical current readouts)**.** For $i = 0, 1$, put $H_{0,J} = H_J$ and $\omega_i = H_{i,J}^T J$. For either variant

$$\sup_c \xi_c^3 |\omega_i(c)| \le C\gamma, \qquad \sup_{c \text{ nonbare}} \xi_c^3 |\omega_i(c)| \le C\gamma_{\mathrm{t}}, \qquad \gamma = \sup_b \ell(b)^3 |J(b)|.$$

The first readout alone enters $K_{1,J}$. Both variants depend on J through its restriction to the touching stock. The unrestricted readout has the exact additional bare component obtained by identifying each nontouching current bond with its bare row. ■□

Proof. The proof for the first variant uses only its weighted lift already obtained before constructing $K_{1,J}$. For a unit colour e, let $b^{(c)}(c) = \xi_c^{-1} e$, zero elsewhere. Its Y-norm is bounded above and below by fixed constants. The identity

$$\xi_c^3 \langle e, \omega_i(c) \rangle = \langle H_{i,J} b^{(c)}, J \rangle$$

uses the exact physical row volume. On B_y, $\|\ell J\|_2 \le C\gamma$ by the cell volume bound. The weighted lift estimate bounds $\|\ell^{-1} H_{i,J} b^{(c)}\|_{2,B_y}$ by $Ce^{-md(y,c)}$. Summing the pairings uses the metric growth with rate strictly smaller than m. Hence the constant is independent of the number of cells. A datum supported on nonbare rows lifts to a field vanishing on the bare rows since $QH_{i,J} = I$ and the bare row is the fine coordinate. A current outside the touching stock therefore contributes only to its identical bare row, with both measures η^4. It drops out of C_ω. Apply the estimate to the restricted current for the nonbare bound, first for H_J and then for the second variant. □

For clarity about the independent-current extensions, put $\delta J = J - j_U$. The preceding identities give, for either variant,

$$KDG_sR = DR - B[\delta J] DG_sR, \qquad GDR = DG_sR + GB[\delta J] DG_sR. \tag{7.vi}$$

Indeed $SDG_sR = 0$, $QDG_sR = 0$, and $RD^T DG_sR = R$. Transposing and composing with the row factors of the lift gives its slice defect. In particular it vanishes when $J = j_U$ on the touching stock. If a physical pair additionally satisfies $j_U = Q^T \omega_{\mathrm{phys}}$, the first lift gives $H_J^T j_U = \omega_{\mathrm{phys}}$. We next verify the Lagrangian Ward identity for this pair.

Write $\mathfrak{q}(a)$ for the actual per-length nonlinear constraint in the reference chart and $\Gamma_a \lambda$ for the gauge tangent in that chart. Differentiating the exact average and node-factor covariance gives

$$D\mathfrak{q}(a)\Gamma_a \lambda = \overline{D}(a)\tau_a \lambda, \qquad D\mathfrak{q}(0) = Q, \quad \Gamma_0 = D, \quad \tau_0 = q, \quad \overline{D}(0) = \overline{D}_{\mathrm{ext}}.$$

Here the two moving coarse endpoint factors are included in $\overline{D}(a)$. At an outward row the further one-step mean of the finer endpoint factors enters that operator. Thus its value at zero is the extended row difference on own-top values. The Wilson Ward identity $\langle j_U, D\lambda \rangle = 0$, the equality $j_U = Q^T \omega_{\mathrm{phys}}$, and onto q give $\overline{D}_{\mathrm{ext}}^T \omega_{\mathrm{phys}} = 0$.

For $q\lambda = 0$, differentiate the displayed covariance at zero in direction v and pair with ω_{phys}. The term differentiating $\overline{D}(a)$ vanishes because $\tau_0 \lambda = 0$; the term differentiating τ_a vanishes by the

coarse transpose identity. What remains is

$$\langle v, C_{\omega_{\mathrm{phys}}} D\lambda\rangle + \langle j_U, \Gamma_0'[v]\lambda\rangle = 0.$$

The differentiated action Ward identity gives the same equation with H_W in place of $C_{\omega_{\mathrm{phys}}}$. Subtract to obtain the full identity $(H_W - C_{\omega_{\mathrm{phys}}})D\lambda = 0$ on $\ker q$, with arbitrary test v. Since the range of $I - S$ lies in $D\ker q$ and the form is symmetric, $S^T(H_W - C_{\omega_{\mathrm{phys}}})S = H_W - C_{\omega_{\mathrm{phys}}}$. On that pair this gives

$$K_{1,J} = H_W - C_{\omega_{\mathrm{phys}}} + DRD^T + Q^T a_f Q.$$

This identification uses the stated critical-pair identity as input. The construction of the two Green operators and their independent-current bounds needs only the admitted coefficient data.

The constrained Green operator is

$$\mathfrak{G} = G_{1,J} - G_{1,J}DRD^T G_{1,J} - H_{1,J}QG_{1,J}.$$

On shell eq. 7.vi gives $QG_{1,J}DR = 0$ and $RD^T G_{1,J}DR = R$. Thus $\mathfrak{G}f$ belongs to $\ker Q \cap \ker RD^T$. On this common slice, pairing $K_{1,J}\mathfrak{G}f$ with a test field gives exactly the pairing with f, since both subtractions are in the respective transpose ranges. This is the inverse on the stated slice. The same formula defines the independent-current extension. The first lift on shell is the stationary field for the Wilson quadratic form on $Qu = b$, $RD^T u = 0$: its penalized equation has multiplier $V_f^{-1}b$, the gauge term vanishes, and S acts as the identity on both tests and solutions. The stationary value is $\frac{1}{2}\langle b, (V_f^{-1} - a_f)b\rangle_{\mathrm{rows}}$.

7.1. **A local augmented system with independent sources.** The site spaces $\mathcal{Y}_s, \mathcal{Z}_s, \mathcal{Y}_s^\flat$ are those in the proof of lemma 7.1. A site divergence $h_0 + D^T h_1$ has $\mathcal{Y}_s^\flat$-norm at most $C(\|\ell^2 h_0\|_2 + \|\ell h_1\|_2)$. This follows by applying D to $\ell^2 v$ in the test pairing. In particular the norms below admit divergence data with no pointwise differences of their coefficients.

Proposition 7.4 (Unfolded response)**.** Use the same real or complex admission as theorem B, and set $W = W[J^{\mathrm{t}}]$, where J^{t} is the restriction to the touching stock. In the second variant use the first lift's nonbare ω from that theorem. For $\epsilon = 0, 1$ put

$$e_\omega = \epsilon C_\omega(A - D\lambda), \qquad h_J = W^T A - \tfrac{1}{2}(D^T W\lambda + W^T D\lambda) - D^T e_\omega.$$

The finite system with arbitrary sources

$$\begin{aligned} K_F A - Dv - W\lambda - e_\omega - Dt &= f_0, \\ A_s\lambda + v - D^T A &= g, \qquad q\lambda = c, \\ A_s v - q^T \Xi_s^{-2}\beta &= h, \\ A_s t + q^T \Xi_s^{-2}\beta' - h_J &= h', \\ A_s\zeta - t &= r, \qquad q\zeta = d_0 \end{aligned} \tag{7.vii}$$

has a unique solution, with bound

$$\begin{aligned} &\|A\|_X + \|\lambda\|_{\mathcal{Z}_s} + \|\zeta\|_{\mathcal{Z}_s} + \|v\|_{\mathcal{Y}_s} + \|t\|_{\mathcal{Y}_s} + \|\beta\|_{\mathrm{top}} + \|\beta'\|_{\mathrm{top}} \\ &\quad \le C(\|f_0\|_{X^*} + \|g\|_2 + \|r\|_2 + \|h\|_{\mathcal{Y}_s^\flat} + \|h'\|_{\mathcal{Y}_s^\flat} + \|\Xi_s^{-2}c\|_{\mathrm{top}} + \|\Xi_s^{-2}d_0\|_{\mathrm{top}}). \end{aligned}$$

Adding $-Q^T\mu$ in the first equation and $QA = b$ yields the analogous bound with $\|\mu\|_{Y^*}$ and $\|b\|_Y$ added to the two sides. At zero auxiliary sources the field inverse is G_J or $G_{1,J}$, respectively. These statements concern the actual system on its prescribed full finite domain. ∎

Proof. Fix A. Write $f = D^T A + g$ and $v_0 = G_s h$. The second and third equations have the unique solution

$$\beta = M^{-1}\{F^T(f - v_0) - \Xi_s^{-2}c\}, \qquad v = v_0 + F\beta, \qquad \lambda = G_s(f - v).$$

The constraint follows from $qG_sF = \Xi_s^2 M$. The balanced site inverse and M gap bound v, β in their stated norms. On a complex tube use the right lift

$$E_s(U) = E_s(U_0)\{q_U E_s(U_0)\}^{-1}.$$

The matrix in braces is $I + O(\varepsilon)$ in the top norm $\|\Xi_s^{-2} \cdot \|_{2,s}$, by the actual row perturbation and the compact real lift estimate. Its Neumann inverse gives $q_U E_s(U) = I$ and the same $\mathcal{Z}_s$ bound. Subtract this site lift $E_s c$ from λ and use the zero-average second-difference estimate with source

$$\Delta(\lambda - E_s c) = f - v - q^T a_s c - \Delta E_s c.$$

The actual row norm and the bound on E_s show this source is L^2-bounded by the declared norms. This gives λ in $\mathcal{Z}_s$. For a given h_J the last two equations have the unique solution

$$t_0 = G_s(h' + h_J), \quad \beta' = M^{-1}\{F^T(t_0 + r) - \Xi_s^{-2} d_0\}, \quad t = t_0 - F\beta', \quad \zeta = G_s(t + r).$$

Subtracting $E_s d_0$ now gives the same $\mathcal{Z}_s$ estimate for ζ.

At zero auxiliary sources $v = PD^T A$, $\lambda = \Lambda A$ and $t = RG_s h_J$. Since $DRG_s = \Lambda^T$, the terms $-W\lambda - Dt - e_\omega$ in the field equation equal $-B[J]A - S^T e_\omega$, with $e_\omega = \epsilon C_\omega S A$. This proves exact elimination to K_J or $K_{1,J}$. For nonzero auxiliary sources subtract these zero-auxiliary solutions at the same A. The v, λ differences are bounded by $\|g\|_2 + \|h\|_{\mathcal{Y}_s^\flat} + \|\Xi_s^{-2} c\|_{\mathrm{top}}$ from the preceding formulas. The resulting change in h_J is bounded in $\mathcal{Y}_s^\flat$ by those same norms. Indeed $\|\ell W\lambda\|_2 \le C\gamma_{\mathrm{t}} \|\ell^{-2}\lambda\|_2$, and testing $W^T A$ against $\ell^2 v$ gives $C\gamma_{\mathrm{t}} \|A\|_X \|v\|_{\mathcal{Y}_s}$. Replace A by $D\lambda$ for the third term in h_J, and use the divergence estimate and eq. 7.iv for the other two terms. Thus the errors $Dv, Dt, W\lambda, e_\omega$ have X^*-norm at most the auxiliary source bound. The actual field inverse, or eq. 7.v with the added constraints, bounds A; the site equations then give all other estimates. This proof uses the same site algebra and relative perturbation estimates on the smaller complex tube. In particular the finite-carrier equations do not require invertibility of their compressions with new artificial boundary conditions. □

8. The proper site inverse in a strong norm

The site gauge parameter requires a second derivative in a Hölder norm. This conclusion uses a regular frame on each complete interior box. An actual-current bound in an admitted Coulomb frame is one sufficient way to verify this input.

Let U_0 be a real proper reference on the original determining graph. On the nested complete interior boxes it obeys the small curvature bound and admits real frames satisfying eq. 3.xix, uniformly after rescaling by the assigned length. Alternatively, it admits the Coulomb frames and actual-current bounds of Corollary 3.12, with enough additional outer boxes for that corollary. Proposition 3.13 constructs these real Coulomb frames when its enlarged curvature stocks are available. Let q_0 be its own-pillar site average, and evaluate the operators G_s, F, M, R of Proposition 5.2 at this reference. Write R_0 for its projector R. The site-gap and normalized-Gram estimates supply these inverses, as in Proposition 5.2 and Lemma 7.1. Their exponential column bounds use the physical orthonormal top basis. The two pairings in F^{T} must therefore be retained.

Fix $0 < \alpha < 1$. On each nonbare pillar, take the nested complete boxes specified for the strong interior estimates. Hölder increments compare values after the fixed short coordinate transport inside the same box. They include increments across adjacent pillar faces. Define

$$\|\rho\|_{\alpha,2} = \sup_y \{\xi_y^2 \|\rho\|_{\infty,B_y} + \xi_y^{2+\alpha}[\rho]_{\alpha,y}^{U_0}\}, \tag{8.i}$$

$$\|\lambda\|_{Y^{2,\alpha}} = \sup_y \{\|\lambda\|_{\infty,B_y} + \xi_y \|D_0\lambda\|_{\infty,B_y} + \xi_y^2 \|D_0^2\lambda\|_{\infty,B_y} + \xi_y^{2+\alpha}[D_0^2\lambda]_{\alpha,y}^{U_0}\}. \tag{8.ii}$$

At a bare pillar $\xi_y = \eta$, use the fixed finite collars of present stencils. The condition $q_0\lambda = 0$ then fixes the actual bare-site value to zero. No smoothness through a rough exterior boundary is required.

Proposition 8.1 (Strong proper site inverse)**.** Under the preceding real curvature and admitted-frame hypotheses, with the fixed separation large enough for the column summations,

$$\|R_0\rho\|_{\alpha,2} \le C_\alpha\|\rho\|_{\alpha,2}, \qquad \|G_sR_0\rho\|_{Y^{2,\alpha}} \le C_\alpha\|\rho\|_{\alpha,2}. \tag{8.iii}$$

For $\rho \in \operatorname{ran} R_0$, the field $\lambda = G_s\rho$ is the unique solution on the actual graph of

$$q_0\lambda = 0, \qquad R_0D_0^{\mathsf{T}}D_0\lambda = \rho. \tag{8.iv}$$

■

Proof. Proposition 5.4 gives the normalized site-column estimates

$$|F_z(x)| \le C\xi_y^{-2}e^{-md(y,z)}, \qquad [F_z]^{U_0}_{\alpha,y} \le C_\alpha\xi_y^{-2-\alpha}e^{-md(y,z)}.$$

These are estimates for the physical orthonormal column indexed by z. A source of unit norm in eq. 8.i has physical L^2 norm at most a constant on each pillar. Pairing it with F_z and summing the exponential cell bounds therefore gives uniformly bounded top coefficients. The normalized Gram inverse preserves this bound by its two-sided exponential sums. Summing the output columns proves the first estimate in eq. 8.iii for $R_0 = I - FM^{-1}F^{\mathsf{T}}$. For the Hölder seminorm, take the increment of each output column before summing. The source is not differentiated.

Set $u = G_sR_0\rho$. Since $F^{\mathsf{T}} = \Xi_s^{-2}q_0G_s$ and $F^{\mathsf{T}}R_0 = 0$, we have $q_0u = 0$. The defining resolvent equation then gives

$$D_0^{\mathsf{T}}D_0u = R_0\rho. \tag{8.v}$$

The exponential energy blocks of G_s imply

$$\xi_y^{-2}\|u\|_{2,B_y} + \xi_y^{-1}\|D_0u\|_{2,B_y^+} \le C\|\rho\|_{\alpha,2}.$$

Indeed a resolvent block has factor $\xi_y\xi_ze^{-md(y,z)}$, and the source has bounded cell L^2 norm. The remaining scale ratio is summable by eq. 2.xi. The derivative block is treated in the same way.

Rescale a nonbare interior box by ξ_y. The bounded-force bootstrap in Proposition 3.10 gives a $C^{1,\alpha}$ bound on a smaller box: choose a finite $p > 4/(1-\alpha)$ and use eq. 8.v. Its right-hand side has the required C^α bound by the first estimate already proved. The admitted-frame hypothesis supplies the uniformly bounded $C^{1,\alpha}$ reference potential on the larger box; in the Coulomb-frame alternative this is Corollary 3.12. Corollary 3.11 on one further nested box gives the second derivatives and their Hölder seminorm. Rescaling gives exactly the powers of ξ_y in eq. 8.ii.

At a bare cell, the physical atomic L^2 estimate bounds the finitely many values in its fixed collar. A difference of order r costs at most $C\eta^{-r}$, and a Hölder quotient costs at most $C\eta^{-\alpha}$. The weights in eq. 8.ii cancel those factors. This proves the same estimate on present bare stencils, with $q_0u = 0$ retaining the actual zero values.

For uniqueness, suppose $q_0\lambda = 0$ and $R_0D_0^{\mathsf{T}}D_0\lambda = 0$. Since R_0 is the orthogonal projection onto the complement of $\operatorname{ran} F$, write $D_0^{\mathsf{T}}D_0\lambda = Fa$. The vanishing site average gives $\lambda = G_sFa$. Applying q_0 yields

$$0 = q_0G_sFa = \Xi_s^2Ma.$$

The normalized Gram is invertible, so $a = 0$ and $\lambda = 0$. Existence follows from eq. 8.v. □

9. Connected families for the finite inverses

A family coefficient has a finite connected set of tagged cubes, together with the endpoint, mesh, row and domain data read on that set. A tag includes its level. A cube on the exterior is allowed as a geometry tag, with the bare scale. Besides the fine connection, a coefficient reads the complete plaquette and row stocks, the row weights and the zero-extension and free-coordinate masks on its carrier. In particular a row with some endpoints outside a patch retains its actual coefficients.

For a cell B_y, use the force norm $\mathcal{F}_{2,y}(f) = \xi_y\|f\|_{2,B_y}$ and the energy norm X_y of the preceding section, with the same conventions for scalar energy. For spaces involving derivatives choose a

smooth square partition subordinate to fixed enlargements of the assigned cells. One such partition is obtained by taking sampled smooth bumps equal to one on each cell, supported in fixed enlargements, and dividing each by the square root of the sum of their squares. Separation bounds the number of overlaps and the ratios of scales at every overlap. The difference of order a of the bump is $O(\xi_y^{-a})$ on its support. Thus the product rule gives bounded multiplication on the local energy spaces, their duals and the second-difference spaces. On L^p and row spaces one can use their actual cell partition. Write ϕ_y for these localization multipliers. The identity used at an intermediate slot is $\sum_y \phi_y^2 = I$.

For a coefficient from z to y, let $\sigma(C)$ be a fixed additive constant plus the connected length of its tagged carrier. Include both canonical endpoint tags. The additive constant pays for the fixed short join between carriers at a common localization slot. In particular $\sigma(C \vee C') \le \sigma(C) + \sigma(C')$. Use the norm

$$\|\mathbf{T}\|_{s,\vartheta} = \max\left\{ \sup_y \sum_{z,C} e^{sd(y,z)+\vartheta\sigma(C)} \|\phi_y T_C \phi_z\|,\ \sup_z \sum_{y,C} e^{sd(y,z)+\vartheta\sigma(C)} \|\phi_y T_C \phi_z\| \right\}. \tag{9.i}$$

Here each coefficient norm includes the supremum over that coefficient's admitted local field and current data, taken before either endpoint sum. The geometry and declared read set are fixed during that supremum. The operator norms use the two declared normalized local spaces, including their physical measures. Supports and any derivative collars of ϕ_y, ϕ_z are included in the filing. For each product insert the square partition and retain the joined carrier for every intermediate slot. The two triangle inequalities and the separate absolute sums prove the product bound. One may keep the sum of the intermediate spatial distances in the weight of each word instead of its endpoint distance. This gives the same product estimate and bounds decay in the length of a connecting path through the successive slots. The norm of a summed matrix alone would give no such bound.

Lemma 9.1 (Local positive Green families)**.** Fix a finite carrier rate ϑ. On the separated hierarchy, with a sufficiently large fixed separation, G_s and G_F have exact normally convergent connected families from local force to energy and from local energy dual to energy. The bounds in eq. 9.i hold for a fixed $s > 0$, uniformly in mesh, depth and compatible volume. For G_s they also hold from unweighted L^2 to $\mathcal{Z}_s$, and from $\mathcal{Y}_s^\flat$ to $\mathcal{Y}_s$, in the physical local norms. Every term uses the prescribed positive operator compressed by zero extension. The same families are holomorphic on a common smaller weighted C^1 tube around the admitted real reference. ■

Proof. We give the construction for $\mathcal{A} = A_s$ or K_F. On a set D_0 of actual variables let E_{D_0} be zero extension. The positive form $E_{D_0}^T \mathcal{A} E_{D_0}$ inherits the global lower bound in the energy of the zero extension. For a crossing row, the row value is evaluated on that extension. There is no inf-sup claim for a newly renormalized truncated row. Multiplication by $e^{s\psi}$, with ψ a bounded one-Lipschitz function of the layer metric, preserves this support. Its conjugation of the restricted form changes the energy-dual bound by $C|s|$. For the difference part this follows from

$$D(wu)(b) - w(b_-)Du(b) = \eta^{-1}\big(w(b_-) - w(b_+)\big)\,\mathrm{Ad}(U_b)u(b_+), \qquad w = e^{s\psi},$$

with the same estimate for every translated component difference. For a row subtract its scalar weight at the top. Its oscillation in relative units is $O(|s|)$, and the normalized row and column bounds control the two row factors. In the Wilson form the covariant curl has the same product rule, and the curvature coefficients have a finite shift and size $O(\xi_y^{-2})$. These observations prove the conjugation estimate also on the domain of zero extensions. The same estimates for the changed coefficients, with size $O(\varepsilon)$ in the tube, give the complex inverse by a Neumann argument. Distance weights now give columns bounded by $Ce^{-s_0 d(y,z)}$ from $\mathcal{F}_{2,z}$, or from the local energy dual, to X_y. For the latter source one tests the compressed variational equation against the weighted solution. Cutoff multiplication of a functional is defined by multiplying its test vector.

We next choose cutoffs using only the local geometry. Take a large m and use the comparison length ℓ of eq. 5.iii, putting $r_x = m\ell(x)$. Take $m\delta$ small and $\rho\delta$ sufficiently large depending on L, with the ambient bare scale outside Ω_1. One can choose the common slope after the largest cutoff radius parameter used in the finitely many constructions. The estimates preceding the cutoff choice are uniform under such a decrease of the slope and increase of the separation. To verify the read collar, a point at level at most two below the level of x is separated from x by a complete intervening layer. Its distance term alone exceeds $\xi(x)$ if $\rho\delta$ is large. A point of the same or higher level cannot improve the zero-distance candidate. Hence all decisions for $\ell(x)$ are made inside a ball of radius $C_L\ell(x)/\delta$, including decisions about exterior labels. This collar is included even if it exceeds the support of the cutoff. Smooth difference estimates for ℓ itself are unnecessary in this construction.

Perform the center counting and all local patch constructions on one compatible cover on which the enlarged balls and their decision collars embed. In particular a short fundamental period does not truncate the ambient fine centers in a cutoff ball. The periodic-sector calculation below returns the resulting operators and their family estimates to the original period.

Choose a fixed smooth nonnegative bump φ, supported at radius $1/2$ and equal to one at radius $1/4$, invariant under the hypercubic group. At every ambient fine center x put

$$p_x = \eta^4 r_x^{-4}, \qquad N(v) = \sum_x p_x \varphi((v-x)/r_x)^2, \qquad \chi_x(v) = N(v)^{-1/2}\varphi((v-x)/r_x).$$

For bond slots sample this formula at their geometric midpoints. If $|x-v| \le a r_x$ for a fixed a, then $|r_x - r_v| \le am\delta r_x$. Thus the radii are comparable there. Counting fine centers in the smaller and larger radius balls gives $c \le N(v) \le C$ and $\sum_{x: v\in B(x,ar_x)} p_x \le C$. These constants are uniform because $r_x \ge m\eta$. Differentiating the sampled formula takes differences only in v, and each derivative of the bump costs r_x^{-1}. The lower bound for N and the product rule give $|D^a\chi_x| \le C_a r_x^{-a}$ for $a \le 2$, and in fact for any fixed a used below. We have the exact identity $\sum_x p_x\chi_x^2 = I$.

Let D_x be the actual variables in $B(x, 2r_x)$, and $I_x = E_{D_x}(E_{D_x}^T \mathcal{A} E_{D_x})^{-1} E_{D_x}^T$. Include its full finite interaction collar. The support of χ_x and each stencil reading it lie strictly inside the artificial boundary. The separation and local scale comparison ensure that the whole enlarged patch has at most two adjacent scales and that the only rough physical boundary is in the bare layer. The exact commutator satisfies

$$\|[\mathcal{A}, \chi_x]u\|_{\mathcal{F}_{2,y}} \le \frac{C}{m}(\|\nabla_U u\|_{2,B_y^+} + \xi_y^{-1}\|u\|_{2,B_y^+}). \tag{9.ii}$$

For the interior covariant Laplacian, expand two differences by the product rule. The terms have either a first cutoff difference and a first difference of u, or two cutoff differences and u. The finite curvature terms give a single cutoff difference of relative size $O(1/m)$. In the row mass, insert the cutoff value at each row location between the two row factors. On the complete row support the difference from this value is $O(1/m)$; the two normalized absolute sums control the resulting L^2 operator. At a physical bare boundary, the present matrix entries are $O(\eta^{-2})$ and connect only finitely many neighboring bare-scale coordinates. Their cutoff differences are $O(1/m)$, so the weighted force bound follows from the bare mass norm directly. No missing plaquette is filled in. The commutator reads u only on the cutoff stock and its stencil collar, away from the artificial boundary.

Set

$$\mathcal{M} = \sum_x p_x \chi_x I_x \chi_x, \qquad \mathcal{T} = \sum_x p_x [\mathcal{A}, \chi_x] I_x \chi_x.$$

Zero extension and the interior support of χ_x give $\mathcal{A}\mathcal{M} = I + \mathcal{T}$ exactly. The local inverse column and eq. 9.ii give exponential cell bounds $Ce^{-s_1 d(y,z)}$ for the first summand and $Cm^{-1}e^{-s_1 d(y,z)}$ for the second, each multiplied by p_x. For a fixed y the sum of contributing p_x is bounded, by the overlap estimate on the enlarged patch. The same holds with z fixed. Sum the remaining

exponential with the strict growth reserve of lemmas 2.1 and 2.2. In particular the number of fine centers or the volume of the patch does not enter the bound.

File each summand on the patch, its complete row and stencil collars, the normalization and length decision collars, and its endpoint tags. Cubes of side $M_0\xi_j$ at the adjacent scales, with $M_0 \geq C_L/\delta$, cover this set with at most B_L tags after bounded joins. Increase separation relative to LM_0. The constant B_L is independent of the radius parameter m. It follows that $\|\mathbf{T}\|_{s,\vartheta} \leq (C/m)e^{\vartheta B_L}$, and $\|\mathbf{M}\|_{s,\vartheta} \leq Ce^{\vartheta B_L}$. Choose m so the former is less than $1/2$. This is consistent with first taking δ small relative to m^{-1} and then increasing M_0, ρ: the local inverse columns and overlap constants are independent of m, δ, M_0, ρ once the fixed slope and rate conditions hold. The exact family is the expansion

$$\mathcal{A}^{-1} = \mathcal{M} \sum_{n\geq 0} (-\mathcal{T})^n. \tag{9.iii}$$

The rightmost error may act on a local energy-dual source by the variational column estimate, while every subsequent error acts on the force space. The zero-error term uses the variational column directly. This gives the energy-dual assertion without changing the contraction.

For a site L^2 source, the interior equation on the output cutoff is $\Delta u = f - q^T a_s q u$, plus first and second cutoff terms. The normalized row sums bound the last term in local L^2 by $C\xi_y^{-1}$ times local scalar energy. Apply the local component estimate

$$\|\nabla_U Du\|_{2,B_y} \leq C\{\|\Delta u\|_{2,B_y^+} + \xi_y^{-1}\|Du\|_{2,B_y^+} + \xi_y^{-2}\|u\|_{2,B_y^+}\}.$$

This follows by applying the covariant curl-divergence identity to a smooth cutoff of Du: its curl is curvature times u, and the cutoff derivatives have size C/ξ_y. Use the atomic estimate at a true bare stencil. Consequently the leftmost inverse with cutoff in eq. 9.iii maps $\mathcal{F}_{2,z}$ to $\mathcal{Z}_{s,y}$ with an additional ξ_y^{-1}. The unweighted input L^2 contributes ξ_z, and the intervening normalized factors keep only the ratio ξ_z/ξ_y. Pay it once by the strict scale reserve. For $\mathcal{Y}_s^\flat \to \mathcal{Y}_s$, scalar energy on B_y is comparable to $\xi_y^{-1}\mathcal{Y}_{s,y}$ and its dual to $\xi_y^{-1}\mathcal{Y}_{s,y}^\flat$. This gives the ratio ξ_y/ξ_z, paid in the same way. These proofs use only the displayed second-difference estimates and the fixed real coercive references. They extend to the common tube using the same form and local coefficient bounds. □

9.1. **Coarse inversion by a grouped short power.** The matrices to be inverted next act on top or row slots with a bounded number of counting orthonormal coordinates per assigned cell. The number of fine variables in the reads can be arbitrarily large. Normalize the top coordinate by $J_s = \Xi_s^2$ and the field row coordinate by $J_f = \Xi_f^2$. In particular

$$\widehat{F} = FJ_s^{-1} = G_s q^T \Xi_s^{-4}, \qquad \widehat{M} = \widehat{F}^T\widehat{F}, \qquad \widehat{\Theta} = \widehat{M} - (D\widehat{F})^T G_F D\widehat{F}.$$

The transpose here uses counting coordinates at the indicated normalized slot and the physical coordinates at every fine slot. The density of $q^T\Xi_s^{-4}$ on the complete own pillar of a unit counting top is $O(\xi_y^{-4})$. Its $\mathcal{Y}_{s,y}^\flat$-norm is bounded, since the ℓ^2 in the test and the physical volume factor in L^2 each contribute ξ_y^2. The positive site family therefore gives $\widehat{F} : \ell^2_{\mathrm{top}} \to \mathcal{Y}_s$ and $D\widehat{F} : \ell^2_{\mathrm{top}} \to \mathcal{F}_2$, with the same bounds for their physical transposes into the corresponding dual slots. Taking their products and using $G_F : \mathcal{F}_2 \to X$ gives the exact input families for $\widehat{M}, \widehat{\Theta}$. Both have positive uniform real gaps by lemma 7.1.

Lemma 9.2 (Inversion of a positive coarse family)**.** Let $A = A^T$ have an exact connected family on such a coarse coordinate graph, with a real gap $aI \leq A \leq b_0 I$ and a common complex tube. Suppose its coefficients admit the norm eq. 9.i at some positive spatial rate and at a carrier rate with a fixed reserve over the desired output rate. The bound uses local coefficient suprema inside the sums. For a short power combined entry, assume that every admitted real datum on its declared read has a real extension with the same uniform gap and input family bound. The extension may use another compatible volume or hierarchy if it preserves all the geometry, phases, masks,

weights, and averaging reads of the entry. It need not preserve a periodic identification of the read on the lifted lattice. A complex datum is admitted on a smaller tube about a real datum with such an extension, using the cutoff continuation in the proof. The matrices used below have the real extensions described after this lemma. For sufficiently separated geometry, with fixed aligned reblocking, A^{-1} has an exact family at the output rate and a positive spatial rate. The construction also yields grouped powers for local inverse square roots and logarithms. The bounds are uniform in depth and compatible volume. ■

Proof. Write $H = I - A/b$ with $b > 2b_0$, and symmetrize each coefficient by T. On a periodic quotient first average the unmarked input family over its full deck orbit, carrying the tile phases with the tags. Carrier length is preserved by that translation. The short-carrier cutoff below therefore commutes with deck averaging, and its sum S, as well as H, preserves the periodic sector before powers are formed. This preserves the family bound because both row and column sums are used. In a provisional tiling of side $M_0\xi_j$, split the family by its full length, writing

$$S = \sum_{\sigma_0(C)\le R_0} H_C, \qquad E = H - S.$$

Use R_0 large. The tail norm at an intermediate carrier rate is at most Ke^{-cR_0} by the reserve at the input rate. At the real reference H lies between $I/2$ and $(1 - a/b)I$; for a sufficiently small tail $0 \le S \le q_*I$ for a fixed $q_* < 1$. The tail and these estimates hold for the full graph, including compressions already specified as the original input domain of A.

An old tag of level j has metric diameter at most C_LM_0 on the indicated adjacent layers. Its bounded joins have that same size bound. Thus every coefficient of S has its whole read set within $C_LM_0(R_0 + 1)$ metric distance of either endpoint. For $1 \le j \le n$, combine the entries of S^j as matrices before taking their absolute values. Their range is at most $B_j = C_LM_0j(R_0 + 1)$, and a combined entry reads only the two radius-B_j endpoint balls. This follows by following any short word through its successive intermediate cells and including the full carrier at each step. The balls include the geometric decisions and all boundary masks needed to know which short coefficients enter the sum. Truncation by endpoint separation would not give this read set.

Choose separation so $2B_n$ is shorter than a full intervening layer crossing. Within these balls the metric and one of two adjacent lattice units are comparable, so each ball has at most $C_L(1 + B_j)^4$ coarse coordinates, with a fixed factor for the cell dimension. At a common real pair, the sum of squares of the block entries in one row or column of S^j is bounded by Cq_*^{2j}, by the Hilbert norm of that matrix. Cauchy–Schwarz bounds its absolute row and column sums by $C_L(1 + B_j)^2q_*^j$. If the supremum over local data is taken separately for each entry, use the individual bound q_*^j and count all possible entries. This gives the bound $C_L(1 + B_j)^4q_*^j$, with the same fixed real spectral reserve at any admitted real extension. The distinction matters because the extremizing data of different entries can differ.

Take the individual estimate on the lifted entry before summing its periodic images. An averaged phase is formed by translating the filing rule and its field arguments together. Each phase is an exact family for the same operator on the extended data, with the same bounds. The completed field therefore need not be deck invariant. At periodic data the averaged short matrix is deck equivariant, as required for the subsequent projection. Summing over input lifts at one fixed output lift then gives the same bound for the projected row, and the transpose argument gives the column bound.

For use on local complex domains, take a collar at the scale of the assigned cells around the full read set. Sum the squares of the smooth localization profiles whose supports meet the read set to obtain a cutoff equal to one there and vanishing before the outer edge of the collar. Its first differences are $O(1/\xi)$ by bounded overlap and comparable scales. Multiply the local relative C^1 field by this cutoff and extend by zero, relative to the fixed real reference. The full weighted C^1 norm costs a fixed factor by the product rule. Cutoff extension of a current uses only its amplitude

bound. The collar and the local tube divided by this fixed factor ensure that the data extend to the admitted global tube. This construction uses the fixed underlying hierarchy and does not change geometry outside the read set. On a weighted $C^{1,\alpha}$ domain, use the same smooth cutoff with $|D^2\chi| \le C\xi^{-2}$. Its scaled first-difference Hölder norm is bounded by interpolation between $|D\chi| \le C\xi^{-1}$ and this second-difference bound. The discrete product rule then bounds the scaled $C^{1,\alpha}$ norm of χA by a fixed multiple of that of A, including pairs crossing the collar where the cutoff vanishes. Thus the extension also preserves the stronger domain used for the fourth-order inverse. On it the complex short matrix obeys $\|S\|_{2\to 2} \le q' < 1$ by reducing the original tube. Its individual complex entry bounds therefore use q' in place of q_*, even when the local data are varied separately. More precisely, after fixing n, R_0 the difference of each short power from its value at the real reference is holomorphic in its local field, and its entry variation is at most $C_j\varepsilon$ in a smaller radius ε. This follows by telescoping $S^j - S_*^j$, or by the one-complex-variable Cauchy estimate on each line in the local tube. The polynomial entry count above gives a uniform row and column bound Δ_j^+ for this supremum. The tube can be chosen so $\Delta_n^+ \le (q')^n$ and $\sum_{j<n} \Delta_j^+ \le (q')^n$. There is no factor from the number of fine coefficient variables.

The long part is the exact family

$$H^j - S^j = \sum_{i=0}^{j-1} S^i E H^{j-1-i}.$$

Each term keeps its joined carrier and has total family norm at most $CjK^j e^{-cR_0}$ by the product estimate. Choose $R_0 = Dn$ for fixed D with $cD > 2\log(1+K) + 2$. Next choose aligned parent tiles of side $M_c\xi_i$, M_c a multiple of M_0, with $M_c \ge C_L(B_n + 1)$. File each combined short entry on the full two balls. It then has at most a fixed number of new tags. For a long carrier traverse its old connected tree in at most twice its length. Divide the traversal into pieces of at most M_c/M_0 edges. Each piece, with its old collars, meets a bounded number of parent cubes on at most two adjacent levels if $\rho \gg LM_c$. Hence the new tag count is at most

$$C_L(1 + (M_0/M_c)\sigma_0(C)). \tag{9.iv}$$

Endpoint tags, level tags and short joins are included in this estimate. The parent mapping does not enlarge the family norm beyond a fixed factor once its carrier weight is paid from the reserved old rate. Likewise, for any fixed parameter radius $\mathcal{S} > 2$, choose M_c so the factor $\mathcal{S}^{\#C}$ of a carrier is paid by the reserve in eq. 9.iv. The factor for a short combined entry is then bounded independently of n.

In the new filing of $H^n = S^n + (H^n - S^n)$, the norm of the short part is at most $C_L(1+B_n)^4(q')^n$, up to the fixed tag and parameter factors. The long part is bounded by $CnK^n e^{-cDn}$. For the short spatial weight take $s \le \log 2/(B_n + 1)$, still smaller than the input spatial rate. Choose n large enough that the combined family has norm less than $1/2$; then choose M_c, ρ as above and impose the strict spatial reserve for all scale ratios used in the construction. The bound remains valid under that increase of separation. The constants used to choose n are from the initial positive construction and the real gaps, and do not depend on the final parent tile. Finally choose the smaller common complex tube for the finitely many short powers. If no local coefficient suprema are needed, the quadratic power of $1 + B_n$ from the common-pair row estimate suffices, but the preceding stronger construction will be used below.

Use the same filing for H^j, $j < n$, and then

$$A^{-1} = b^{-1} \left(\sum_{j<n} H^j \right) \sum_{v \ge 0} (H^n)^v. \tag{9.v}$$

At complete data this is an ordinary matrix identity. With the joined carriers the geometric series converges absolutely in the family norm. All local coefficients are holomorphic, since the positive

primitive inverses and the finite short products are holomorphic on their declared read sets, and the series is normally convergent. The same grouping can be used for any scalar power series in H with bounded coefficients, in particular the square root and logarithm constructions below. □

For the actual coarse matrices the short-read extension requires only a real energy gap. Enlarge the two endpoint balls and all declared decision and stencil collars to an axis-parallel cube of radius $R\xi_y$ and, where needed, one further fixed enlargement. Here R is fixed after the short word length and the old carrier cutoff. Increase the separation so that the enlarged cube contains at most two adjacent assigned scales. Include every complete plaquette and every projected branch stock in that collar. Its radius increases the short read by a fixed factor and costs a bounded number of parent tags.

If the enlarged cube is interior, keep the hierarchy, all row weights, and all selection masks on a compatible lift where the cube embeds. Use the fine coordinate comb to choose a unitary frame on the cube. Each transformed link is the product of unitarily transported plaquette holonomies in a strip between two comb paths, with at most $CR\xi_y/\eta$ factors. Thus its principal potential a, defined by $U = e^{i\eta a}$, satisfies $\|a\|_\infty \le C_L R\kappa/\xi_y$ on the cube. Reserve an inner collar containing the later relative complex cutoff and all its derivative stencils. Take a scalar cutoff χ equal to one on this entire collar, including every read link, and vanishing in the outer collar before the cube boundary, with first differences bounded by $C/(R\xi_y)$. Extend by $e^{i\eta\chi a}$ in this frame and by the identity beyond its support. For a complete plaquette p in the cutoff region, the product expansion gives

$$\eta^{-2}|\log U_{\mathrm{ext}}(\partial p)| \le C(\eta^{-2}|\log U(\partial p)| + \|D\chi\|_\infty\|a\|_\infty + \|a\|_\infty^2) \le C_L(\kappa + R^2\kappa^2)\xi_y^{-2}.$$

Indeed in the linear sum of the four oriented link potentials one may subtract a common plaquette value of χ; the remainders are bounded by one link length times the cutoff derivative. Both plaquette product remainders are quadratic in the link angles. All stencils meeting the cutoff are complete within the enlarged cube. Undo the comb frame throughout the inner collar, extending that unitary site gauge arbitrarily on the lift. Curvature is unchanged by this last operation. The small-curvature recurrence on complete backward averaging stocks admits the real branches, including the projected branch tests. Copy the independent current on the read and set it to zero elsewhere, which preserves its weighted amplitude bound. Reducing the local curvature ceiling after fixing R puts this extension in the already proved real energy-gap class.

If the enlarged cube meets the physical exterior, separation makes its entire active part bare, including the decision collars. Take a larger level-zero aligned box within the supplied collar and restrict the original active set to it, with no higher regions. Retain the original action-boundary convention and copy the data on all the resulting stencils. For an ambient compression keep the complete ambient crossing stencils in the enlarged collar. Every retained curvature test was a test in that collar. The new box boundary is beyond the entire read and its relative complex cutoff and derivative collar. Inside the read the layer length stays η, the site and bond rows are the identity, and the physical boundary and every cutoff and mask decision are unchanged. Hence the short coefficients are those of this admitted real bare problem. Proper outer coefficients have their enlarged short reads in the interior top stock by the assumed outer separation.

For successive coarse inversions make these choices in their dependency order: first the site matrices, then the row matrix, and then the outer precision. At each stage choose its finite read radius and reduce the local real curvature ceiling enough that its completion lies in every earlier input family class. Choose nested complex tube radii in the same order. Earlier gap and family bounds remain valid on the smaller classes. The maximum of this finite list of read radii fixes the required separation. None of these choices depends on the mesh or the number of averaging levels.

9.2. **Actual current families and the reconstruction slots.**

Theorem C (Connected coefficients for the named propagators)**.** Fix a finite carrier rate and choose the aligned tiles, separation and smaller common complex coefficient tube of the preceding lemmas. For a sufficiently small touching current the maps $P, R, \Lambda, S, G_0, G_J, G_{1,J}$, both normalized Schur inverses and lifts, the two multiplier readouts and the coupled response have exact connected families. In the norms of eq. 9.i they are bounded in their stated energy, balanced site, force and row slots. The same holds for $\mathfrak{G}$, for the zero-current constrained response $(I - H_0Q)G_0$ with $H_0 = G_0Q^T(QG_0Q^T)^{-1}$, and for the mass-removed row response. These assertions include local X^*-sources and local L^2-force sources. ■□

Proof. Obtain $\widehat{M}^{-1}$ and $\widehat{\Theta}^{-1}$ at a stronger intermediate carrier rate by lemma 9.2. The site and field positive families used in their input are first obtained at a still stronger rate. Use aligned parent tiles throughout. Compose to get $P = \widehat{F}\widehat{M}^{-1}\widehat{F}^T : L^2 \to \mathcal{Y}_s$ and $R = I - P$ on $\mathcal{Y}_s, L^2$. The map $\Lambda = G_sRD^T$ takes X to $\mathcal{Z}_s$; then $D\Lambda : X \to X$. The first part of this composition uses $D^T : X \to L^2$. At a force input f, use instead $\|D^Tf\|_{\mathcal{Y}_s^\flat} \le C\|\ell f\|_2$, already proved by the test product rule. Thus $DRG_s = \Lambda^T : \mathcal{Y}_s^\flat \to \mathcal{F}_2$ and $S^T = I - DRG_sD^T$ is bounded on $\mathcal{F}_2$ in the family norm. These bounds arise from the actual site family, including its balanced $\mathcal{Y}_s^\flat \to \mathcal{Y}_s$ bound. The other orientation, $S : X \to X$, follows from the $\mathcal{Z}_s$ output. Formula eq. 7.i gives G_0 from $\mathcal{F}_2$ or X^* to X, with $G_F : X^* \to X$ also used in its rightmost factor.

Before taking local coefficients, write the exact current completion using $W = W[J^{\mathrm{t}}]$. Indeed every occurrence of W in eq. 7.iii has a site factor Λ, or its transpose, and $q\Lambda = 0$ makes that site field zero on bare singletons. Hence replacing J by J^{t} leaves the complete $B[J]$ unchanged. The multiplication $W[J^{\mathrm{t}}]$ takes arbitrary $\mathcal{Z}_s$ inputs to $\mathcal{F}_2$ with norm $C\gamma_{\mathrm{t}}$. Its transpose takes X to $\mathcal{Y}_s^\flat$ by the weighted test pairing. Equation 7.iii, with every W replaced by $W[J^{\mathrm{t}}]$ and its D^TW term interpreted as a divergence source, therefore gives a family for $B[J] : X \to \mathcal{F}_2$ with norm $C_\vartheta\gamma_{\mathrm{t}}$. Every such coefficient reads the local current bond. Hence $G_J = G_0\sum_{n\ge0}(B[J]G_0)^n$ converges in the family norm after decreasing the fixed touching-current ceiling. At an X^* input the rightmost G_0 acts on that slot and every subsequent loop acts on the force slot. The first difference $G_J - G_0$ has the additional factor γ_{t}.

Normalize $\mathcal{Q}_n = J_f a_f^{1/2} Q$. The actual injection $\mathcal{Q}_n^T e_c$ from a unit counting datum has density $O(\xi_c^{-3})$ on its full row carrier, so has bounded $\mathcal{F}_2$ norm and a bounded local family. The transpose takes X to counting rows by the row bounds. Thus $\mathcal{T}_0 = \mathcal{Q}_nG_0\mathcal{Q}_n^T$ has the exact family and positive real gap proved in theorem B. Apply lemma 9.2 once more at the desired rate. The current difference $\mathcal{T}_J - \mathcal{T}_0$ has norm $C_\vartheta\gamma_{\mathrm{t}}$, so an ordinary family Neumann series gives $\mathcal{T}_J^{-1}$, and

$$H_J = G_J\mathcal{Q}_n^T\mathcal{T}_J^{-1}J_f a_f^{1/2}.$$

This formula keeps the row weight on the input.

For the multiplier first use the exact split of corollary 7.3: pair the lift with J^{t}, and add the explicit bare-row readout of the nontouching current. Test each term of the lift family on the datum $b^{(c)}$ and split its pairing with J^{t} by cells. A coefficient for $\xi_c^3(H_J^TJ)(c)$ has the carrier of the lift, joined to the actual current cell B_y, with root c. Its absolute size is bounded by the normalized column norm times $C\gamma_{\mathrm{t}}$, since $\|\ell J^{\mathrm{t}}\|_{2,B_y} \le C\gamma_{\mathrm{t}}$. Sum first over all such current cells and carriers at the fixed root. The weighted column bound gives a uniformly summable rooted family. The added bare-row readout is local and vanishes at nonbare roots. Thus the touching-current estimate holds coefficient by coefficient before the root sum. The two-leg estimate eq. 7.iv now gives a family for C_ω with norm $C_\vartheta\gamma_{\mathrm{t}}$ from X to $\mathcal{F}_2$, since its actual two legs and their row root read only a bounded star. Consequently $S^TC_\omega S$ has that same bound. Inverting the second variant by the corresponding force-space Neumann series gives $G_{1,J}$. For its row inverse use the difference from

$\mathcal{T}_0$ and the same normalized row formula. The same rooted pairing gives the second readout. The first readout was used in constructing this second operator.

All remaining claimed families are finite products of these families. For $\mathfrak{G}$ use $D^T : X \to L^2$, R on L^2 and $D : L^2 \to X^*$, as well as the lift. For the datum-to-multiplier response the physical angle is $\Xi_f b$, and the physical current amplitude is $\Xi_f^3 \mu$. Their family identity is

$$\Xi_f^3\{(QG_iQ^T)^{-1} - a_f\}\Xi_f^{-1} = (\Xi_f a_f^{1/2})(\mathcal{T}_i^{-1} - I)(\Xi_f a_f^{1/2}), \qquad i = J,\ (1, J). \tag{9.vi}$$

This follows by writing $T_f = J_f^{-1}\mathcal{T}_i J_f$. It retains the row mass subtraction. The constrained response and the remaining blocks of eq. 7.v follow by inserting their actual local row maps. This completes the claimed constructions. □

Corollary 9.3 (Comparison of determining data)**.** Use one filing rule, including its aligned tile phase, on a common lift of two admitted geometries. Suppose the two systems agree on a given open read region, including all fine links, independent currents, original inner plaquettes, determining rows and their weights, domain masks, and geometry labels there. The coefficients with full carriers in that common region agree. The difference of the localized complete operators on matching endpoint slots is bounded by the absolute sums of the families whose carriers leave it. Write d_i for the layer metric of system i and r_i for the d_i-distance from the output slot to the complement of the common read region, after subtracting the fixed collar allowances. The difference is bounded by

$$C \sum_{i=1}^{2} e^{-s d_i(y,z) - c(r_i)_+ / M_c}.$$

Here $s, c > 0$ are reserved rates. If the common region contains a radius-r ball measured before first exit, then $r_i \geq r$ in both systems, since their geometry agrees along every path before that exit. One may then use $2Ce^{-s\min(d_1,d_2)(y,z) - cr/M_c}$. A single metric suffices when the two hierarchies agree globally. ■□

Proof. The read rule for a primitive coefficient first decides its comparison lengths and normalization on its full geometry collar. It then uses only the actual compressed finite matrix, its cutoff and endpoint data. The row trace, difference, local current and constraint coefficients have the same literal property. In a short combined entry use the two endpoint balls enlarged by the full data collar. They can be fixed in Euclidean coordinates using radius $C_L B_j \xi_y$ at the y endpoint and $C_L B_j \xi_z$ at the other. Every metric path of length at most B_j stays at the two indicated adjacent scales by separation, so these Euclidean balls contain all short reads. This fixes their construction without needing data at a remote alternative path. Joined products and the rooted current pairing preserve equality when all those reads agree. Thus the terms match individually, including whether they are present. All families converge absolutely, so their common terms cancel in the difference. A connected carrier leaving the common radius has at least cr/M_c tags, up to the fixed endpoint and collar allowances, because each tag and short join has metric diameter at most $C_L M_c$. Reserve part of the carrier exponent for this bound separately in each hierarchy, using its own metric and first-exit distance. Add the two remaining absolute sums. The same argument with the input slot gives the corresponding column comparison. □

The families can be used on periodic quotients without cutting a literal path at a periodic face. First construct all primitive and coarse stages on one common compatible lift large enough for the finite patches and aligned tile choices. The real positive compressions and all local coefficients are defined there. The complete operators on that lift commute with deck shifts. For example, choose isometric pullback of base coordinates into the deck-invariant subspace, with the normalization by the square root of the number of sheets. Each complete deck-equivariant matrix intertwines this injection, as does its bilinear transpose. An individual local coefficient need not preserve the periodic sector; its full deck orbit does. Inverses and products of the complete equivariant matrices

preserve that sector by uniqueness. Average the families over simultaneous deck shifts, translating the filed tags and phases as well, before projection. Convexity preserves both absolute sums. A projected row sum is bounded by the lifted row sum over all input lifts at one fixed output lift. Deck covariance gives the corresponding bound for columns. The projected tag length and distances can only decrease, while the full lifted reads remain declared. There is no factor from the number of sheets.

One may use a common filing on all compatible periods at fixed levels. Fix the aligned tag phases on the universal lattice and use their finite uniform average under translations by the top mesh, modulo the final tile width. Include the phase as part of each coefficient's label. For any finite collection of words the lifted coefficients are independent of period once all their reads embed. The family bounds, including the old carrier tail, dominate the projection sums over longer words. Thus on any period the same rule is obtained by projection of the lifted coefficients, with winding reads retained. The finite-cover sums agree with the complete operator by the preceding periodic-sector identity; absolute convergence gives the equality as the lift increases. This argument also gives literal term agreement on common lifts. A period so small that a chosen support wraps is treated by projection, with its winding and periodic identifications in the coefficient. It is not treated as an embedded copy of that support.

In real gauge changes the coefficient on each output and input slot is conjugated by its induced color map. For example a reversal of a fine bond sends its left logarithmic increment to $-\operatorname{Ad}(U_b^{-1})A(b)$; on a determining row the same expression uses the actual averaged row link. For a site the color map has no reversal sign. These maps preserve the declared bilinear pairings because $\operatorname{Ad}(g)^T = \operatorname{Ad}(g^{-1})$ and each isometry preserves the slot measure. All primitive cutoffs are scalar and each term reads the actual complete rows needed for these endpoint factors. Covariance therefore holds at admitted images, with the tags and their phases carried along. A single fixed phase gives termwise isometry covariance for isometries preserving that phase. The complete sums transform under every admitted isometry of the graph and domains. Complex covariance is this bilinear identity in the appropriate charts.

Corollary 9.4 (Independent carrier parameters)**.** After a sufficiently large fixed aligned reblocking, assign a complex parameter s_Δ to each parameter cube and multiply a joined carrier coefficient by the product of the parameters of its distinct cubes. On every prescribed fixed polydisc $|s_\Delta| < \mathcal{S}$, the resulting family converges normally, with the same physical local spaces and a reserved rate. At all ones its sum is the original operator. Setting parameters outside a wall to zero retains precisely the terms filed in the wall, in addition to any explicitly held guard. Separate wall components have no terms connecting them after their guards and complete read sets are included and intersecting components are merged. ■□

Proof. Apply eq. 9.iv to each old connected carrier. Increase the parameter tile width until the cost per old tag,

$$(\log \mathcal{S}) C_L M_0 / M_c,$$

fits in the old carrier reserve. The constant term costs only $\mathcal{S}^{C_L}$. This yields the absolute row and column bounds with the coefficientwise suprema on every smaller field tube. Normal convergence follows by the Weierstrass criterion there. For a product of operators the parameter monomial of the joined coefficient uses its union of cubes. Repeated visits to a cube do not create new factors in this convention. The product of the absolute majorants for the constituents bounds this marking. Evaluation at all ones preserves operator products. A surviving coefficient at a zero wall parameter has its entire filed support in the allowed wall and held guard. Its connected carrier proves the splitting. Corollary 9.3 proves equality on matched wall reads. □

On a short period, descend the unmarked families first, including their full lifted read collars. Choose the parameter partition on the base torus afterward. At a fixed level measure lengths in

that level's grid units and write $W = M_c$. For an axis with $N \geq 3W$ sites, use $\lfloor N/W \rfloor$ cyclic intervals and distribute the remainder so their lengths lie between W and $2W$. If $N < 3W$, use $\min(2, N)$ balanced cyclic intervals. Products of these intervals are the parameter boxes; on an exactly divisible long period they may be cubes. A ball of radius CW meets at most $\lceil 2C \rceil + 3$ intervals in a long axis and at most two in a short one. Apply this count to the traversal pieces in the proof of eq. 9.iv. Separation leaves at most two adjacent levels in each piece, so the same bound holds with a changed C_L, uniformly in rectangular periods. In particular, on one level with all four axes short there are at most sixteen parameter boxes. Mark a distinct base box once, regardless of the number of lifted visits that project to it. The parameter partition and phase are included among the matched data for wall comparison. The matrix-function extension below uses its separately stated ordinary-product convention.

The ordinary uniform estimates, such as theorem B, use a coefficient and current ceiling determined by the real form gaps and the geometric growth reserve, before introducing any walk tile. The stronger families at a prescribed carrier rate use a finite list of tile choices and may shrink the complex tube and current ceiling after those choices. The real curvature ceiling for independently varied local reference data may also be reduced after the short-read radius is fixed, as in the real extension proof above. The ordinary summed estimates keep their previously chosen ceilings. All localized bounds and sums then remain independent of mesh, depth and compatible volume. A pointwise entry bound in an unnormalized coordinate can still contain its physical mesh powers. Such an entrywise bound, and a whole-volume bound used only to verify convergence, do not enter any of the weighted local bounds or rates.

10. Strong local Green and restoration estimates

Use the fixed real comparison frames on complete nonbare boxes, and the bare atomic convention at true exterior stencils. Take finitely many nested boxes at each nonbare cell, with enough room for the row carriers and the successive interior cutoffs. Their radii are fixed multiples of ξ_y and lie in at most two adjacent layers by separation. The coefficient premises for the finite-p estimates are bounded scaled potentials and scaled first differences in these frames. The estimates with two Hölder derivatives below require a scaled $C^{1,\alpha}$ potential bound on the real reference and the relative perturbation. One can use an admitted strong frame or an admitted Coulomb frame with the actual current hypotheses of corollary 3.12. The independent current in the propagator is a separate variable.

For a field force and field output use

$$\|f\|_{\mathcal{F}_{p,y}} = \xi_y^{3-4/p} \|f\|_{p,B_y^+}, \qquad \|u\|_{\mathcal{A}_{\alpha,y}} = \xi_y \|u\|_\infty + \xi_y^2 \|\nabla_U u\|_\infty + \xi_y^{2+\alpha} [\nabla_U u]_\alpha^U.$$

The Hölder comparisons use short coordinate transports on complete local boxes. On the complex tube they are equivalent to the same comparisons in the fixed reference frames. A site function with amplitude of grade two has scaled first-order norm

$$\|v\|_{\mathcal{V}_{p,y}} = \xi_y^{2-4/p} \|v\|_p + \xi_y^{3-4/p} \|Dv\|_p.$$

A site source $D^T f + h$ is measured by the infimum over representations of $\|f\|_{\mathcal{F}_{p,y}} + \xi_y^{4-4/p} \|h\|_p$, with the complete source collar. Its D^T is variational. A bounded site source has grade-four amplitude $\sup_y \xi_y^4 \|h\|_{\infty, B_y^+}$.

Lemma 10.1 (The interior Wilson stencil)**.** On a complete nonbare stencil, with the Wilson quadratic normalization for which its flat Hessian is the curl energy, the actual operator $H_W + DD^T$ equals the component covariant Laplacian plus a finite-shift kernel Z_U with absolute row and column bounds $C\kappa\xi_y^{-2}$ on a scale-ξ_y box. The identity and bounds hold on the admitted complex tube with the corresponding curvature ceiling. ∎

Proof. For a plaquette, transport the four tangent insertions to its initial vertex and divide their oriented sum by η. Squaring this linear curl gives the Hessian at unit plaquette holonomy. The exact Wilson insertion formula gives the difference as a sum of products of two transported insertions paired with $U(\partial p) - I$, with coefficients $O(\eta^{-2})$ after division by the fine volume. It is therefore a finite shift of size $C\kappa\xi_y^{-2}$. Replacing the four path transports in this curl by the standard covariant forward curl changes a coefficient by $O(\eta^{-1}|U(\partial p) - I|)$: the only alternative routes around the square differ by the plaquette holonomy. Multiplying the two curl matrices gives mixed coefficients $O(\eta^{-2}|U(\partial p) - I|)$, since each curl entry has size C/η and each stencil has a bounded number of entries. This step takes no difference of the curvature.

For the standard curl write $T_\mu = \mathrm{Ad}(U_\mu)\tau_\mu$, where τ_μ shifts by one forward step. Its curl square plus the gradient-divergence square has diagonal $\eta^{-2}\sum_\nu(2 - T_\nu - T_\nu^{-1})$. An off-diagonal entry is a signed difference of the two ordered shifts $T_\mu T_\nu^{-1}$ and $T_\nu^{-1}T_\mu$, divided by η^2. Those two routes differ by one elementary plaquette and hence give a kernel of the same curvature size. Bounded incidence gives the column bounds as well. All transposes here use transport inverses, an algebraic identity also for complex links. □

Lemma 10.2 (Strong columns of the positive primitives)**.** For $0 < \alpha < 1$, finite $p > 4/(1 - \alpha)$, and a fixed carrier rate, $G_F : \mathcal{F}_p \to \mathcal{A}_\alpha$ has a connected family in the norm eq. 9.i. For every fixed $2 < p < \infty$, G_s has a family from the variational site source just defined to $\mathcal{V}_p$. From bounded grade-four site sources it has a family to

$$\xi_y^2\|u\|_{\infty,B_y^+} + \xi_y^3\|Du\|_{\infty,B_y^+} + \xi_y^{3+\alpha}[Du]_{\alpha,y}^U.$$

For site columns $\widehat{F}_z = G_s q^T \Xi_s^{-4} e_z$ this gives the point and Hölder bounds with factor $e^{-sd(y,z)}$ and the physical scales in the displayed norm. Constants are uniform in mesh, depth and compatible volume and hold on a common smaller complex coefficient tube. ■

Proof. Start with a positive local compression from lemma 9.1. The normalized $\mathcal{F}_p$ source embeds into its L^2 force slot by the cell volume bound. Its local energy column has the exponential rate s_0. On an interior box the equation is the component covariant Laplacian by lemma 10.1, with lower-order curvature and row terms. In a fixed comparison frame subtract the ordinary lattice Laplacian. Pairing the forward and inverse shifts gives first-order coefficients $O(\|a\|_\infty)$ and zeroth-order coefficients $O(\|\nabla a\|_\infty + \|a\|_\infty^2)$ in rescaled units. For example the zeroth term from one direction is the sum of the forward $(\mathrm{Ad}\, e^{i\eta a} - I)/\eta$ and its backward inverse counterpart, divided by η; their linear terms leave one first difference of a. The curvature term of lemma 10.1 is a bounded finite-shift multiplication.

The row mass has bounded point forcing on each box. Indeed the complete normalized row has coefficient density $O((\eta/\xi_c)^4)$, and hence $Qu(c)$ is bounded by $C\xi_c^{-2}\|u\|_{2,\mathrm{supp}\, Q_c}$. The physical transpose of Q on one row has bounded point density and $a_f(c) \asymp \xi_c^{-2}$. The result is a point force of grade three controlled by the cell energy. The same argument for $q^T a_s q$ gives grade four from the balanced site energy.

In the unit-scale rescaled box the field H^1 estimate gives the lower-order first differences in L^2. The local finite-p estimates give $W^{2,2}$, and strict lattice Sobolev embedding gives gradient L^3. A second application at exponent 3 gives $W^{2,3}$, gradient L^8 and bounded field. For $p > 8$ use one further application at 8 before the final one at p. For $p \le 8$ the final p application is already valid. Every lower-order term is estimated on the next larger complete box. The strict Morrey estimate gives $\alpha < 1 - 4/p$, with the stated gradient and potential grades after rescaling. These are the finite-p and covariant interior estimates of propositions 3.2 and 3.10. The original energy and source terms keep their exponential column bounds since each box enlargement costs a bounded metric distance. The bounded-site-source case uses the same bootstrap componentwise. A true bare stencil uses its finite number of coordinates and $\xi_y = \eta$, so the scaled value, first difference and Hölder quotient follow from its L^2 value bound directly.

For a site divergence source use the balanced energy column. In a comparison frame the connection lower-order terms may be written as divergence of a bounded coefficient times u, plus bounded multiplication of u. For a forward first difference this is the identity

$$a(x)\frac{u(x+\eta)-u(x)}{\eta}=\frac{a(x)u(x+\eta)-a(x-\eta)u(x)}{\eta}-\frac{a(x)-a(x-\eta)}{\eta}u(x)$$

in one coordinate direction. The covariant divergence differs from the flat one by another bounded multiplication of the specified source. Smooth cutoff multiplication preserves the divergence representation by the same product rule. Starting with H^1 gives $u\in L^3$; the local divergence estimate at 3 gives $W^{1,3}$ and hence $u\in L^8$. If the desired p is smaller, stop at p. Otherwise apply the estimate at 8 and use the resulting bounded u before the final finite exponent. This gives $\mathcal{V}_p$ with the same exponential rate after the fixed scale-ratio reserve. It differentiates no input f.

Use these stronger local columns in the slow parametrix. Each interior commutator has either $D\chi_x\,\nabla_U u$, $D^2\chi_x\,u$, a bounded curvature coefficient times a cutoff difference, or a complete-row cutoff difference as in eq. 9.ii. The same bounds hold in $\mathcal{F}_p$, in the site source class, and in the bounded site force class respectively. In the site divergence case the commutator is even a grade-four L^p source from $\mathcal{V}_p$. All these local error norms are at most $C_{p,\alpha}/m$ after rescaling. The square partition and weighted center sum from the positive walk proof yield a contraction on each input force space once m is fixed sufficiently large. Thus eq. 9.iii gives exact strong families. For $\widehat{F}_z$ the input density is grade four, supported on the complete own pillar, so the bounded-site-source conclusion gives its columns. The stated coefficient perturbation preserves every estimate on the smaller tube. The cutoff and collar choices are independent of the fine number of variables. □

Two useful physical conversions follow. In the unnormalized site array the inverse of $qG_s^2q^T$ has size $C\xi_y^{-4}e^{-sd(y,z)}$: it equals $\Xi_s^{-2}M^{-1}\Xi_s^{-2}$ and the change from normalized counting coordinates contributes $\xi_y^{-2}\xi_z^2$. Also the counting fine array of DPD^T, at fine bonds based in B_y, B_z, has size

$$C\,(\eta/\xi_z)^4\xi_y^{-2}e^{-sd(y,z)}. \tag{10.i}$$

In fact its physical kernel is the composition of the point columns $D\widehat{F}$, the normalized $\widehat{M}^{-1}$ and their transpose columns. Each end contributes its grade-three amplitude and the intermediate counting sums are exponential, giving $C\xi_y^{-3}\xi_z^{-3}$ times the exponential. Multiply by η^4 for the counting array and use the reserved scale ratio ξ_z/ξ_y. With P, DP, PD^T use the same argument with grades two or three according to the differentiated ends. In particular these are smoothed projection kernels, with no diagonal Coulomb singularity.

10.1. **A fourth-order construction for strong restoration.** The formula G_sR gains two Hölder derivatives on a Hölder source, but its second-order positive pieces can have an averaging-mass jump at a pillar face. To obtain the same gain for each local coefficient, use the actual fourth-order positive matrix

$$q^*=q^T\Xi_s^{-4},\qquad A_4=\Delta^2+q^*q,\qquad G_4=A_4^{-1}.$$

Here q^* pairs the raw top values with counting measure. The site form norm is $V_2=\|\Delta u\|_2+\|\ell^{-1}Du\|_2+\|\ell^{-2}u\|_2$. It controls $\nabla_U Du$ on complete stencils by the curl-divergence estimate, with the bare atomic convention elsewhere. The real form $\|\Delta u\|_2^2+\|qu\|_{\mathrm{count}}^2$ is equivalent to V_2^2. For the lower bound subtract E_squ. The remainder has zero averages and its zero-average estimate is bounded by its L^2 Laplacian. The lift has V_2-norm at most $C\|qu\|_{\mathrm{count}}$. The upper bound follows from the physical row bound at grade zero and the weight ℓ^{-2} in the fine L^2 norm. The form equivalence carries over by a small relative perturbation of $\Delta: V_2\to L^2$ and $q: V_2\to \ell^2_{\mathrm{count}}$, including their zero-extension restrictions.

For this paragraph use the stronger coefficient hypothesis stated at the start of the section. A fourth-order source has a representation

$$\rho = \Delta f + D^T g + h \tag{10.ii}$$

with f in grade-two C^α, g of bounded grade three and h of bounded grade four. The norm is the infimum over representations of the sum of these normalized norms on the appropriate complete boxes. Its local energy-dual bound follows by testing $\Delta f, D^T g, h$ against $\Delta u, Du, u$ in the physical pairing. The normalized cell L^2 sizes of f, g, h are $O(1), O(\xi_y^{-1}), O(\xi_y^{-2})$. This source class is preserved by smooth localization. For instance

$$[\Delta, \chi]v = D^T\big((D_{\mathrm{flat}}\chi)(v_- + \mathrm{Ad}(U_b)v_+)\big) - (\Delta_{\mathrm{flat}}\chi)v$$

on the actual graph, with the same present scalar stencils in Δ_{flat}. At the tail the divergence term supplies $(\chi_- - \chi_+)(v_- + \mathrm{Ad}(U_b)v_+)/\eta^2$; its v_- term is canceled by $(\Delta_{\mathrm{flat}}\chi)v$. The head gives the transported opposite contribution. This proves the identity, including for crossing edges on the zero extension. Thus commuting χ across Δf differentiates only χ in the new divergence and bounded point sources. If the reference connection is changed within the tube, expand the difference of its two Laplacians as first-order shifted terms and a zeroth term. The first-order terms have the divergence representation from the preceding proof. Their coefficients and first differences are $O(\varepsilon/\ell)$ and $O(\varepsilon/\ell^2)$, respectively. Consequently the norms for eq. 10.ii at the varying and fixed bases are equivalent without differentiating f or g.

Lemma 10.3 (Strong restoration family)**.** On a common weighted $C^{1,\alpha}$ tube, $G_s R$ has an exact connected family from grade-two C^α to the $Y^{2,\alpha}$ space of eq. 8.ii. Accordingly Λ has such a family from $\mathcal{A}_\alpha$ to $Y^{2,\alpha}$ and S from $\mathcal{A}_\alpha$ to $\mathcal{A}_\alpha$. All norms are in eq. 9.i with fixed s, ϑ after the separation and tube choices. ■

Proof. For a positive compression of A_4, begin with its exponential V_2 energy column. Here the weight needs two differences. Smooth the truncated distance to the source cell by the same squared local partition as in the positive walk, with a fixed radius multiple of ℓ. Over the profiles meeting y, the center distances differ by a bounded amount. Since their weights sum to one, subtract one center value before differentiating the weighted sum. The resulting ψ differs from the truncated distance by $O(1)$ and obeys $|D\psi| \le C\ell^{-1}$ and $|D^2\psi| \le C\ell^{-2}$. The second-difference product rule gives

$$\|e^{-s\psi}\Delta(e^{s\psi}u) - \Delta u\|_2 \le C|s|(\|\ell^{-1}Du\|_2 + \|\ell^{-2}u\|_2).$$

The full-row relative weight oscillation is $O(|s|)$, so the conjugated A_4 form differs by $O(|s|)$ in $V_2 \to V_2^*$. Its positivity or small complex perturbation therefore gives the exponential energy column from the source in eq. 10.ii. Both norms are dimensionless in four dimensions.

If u is that local solution, put $v = \Delta u - f$. On the interior cutoff stock the two equations are exactly

$$\Delta v = D^T g + h - q^* q u, \qquad \Delta u = f + v.$$

Here q^*qu is bounded of grade four by the actual complete row and the local energy bound. The first equation and a squared cutoff test bound v in local H^1 by its local L^2 norm, g and the bounded source, using the real reference in a small complex tube. The divergence bootstrap of the preceding proof then gives v in $W^{1,p}$ for any fixed finite p. Take $p > 4/(1-\alpha)$ to obtain a grade-two C^α bound. Apply the scalar point-force bootstrap and the covariant Schauder estimate to the second equation, in the scaled $C^{1,\alpha}$ reference class. This gives u in $Y^{2,\alpha}$. The discontinuous averaging density appeared only in the first equation. Nested interior cutoffs retain the exponential cell estimate, and at the true bare scale the energy column alone gives the rescaled finite differences. No output norm is used across the artificial compression boundary.

The exact fourth-order cutoff identity is

$$[\Delta^2, \chi]u = \Delta[\Delta, \chi]u + [\Delta, \chi]\Delta u.$$

For strong u, the first term is Δ of a grade-two C^α function bounded by C/m times the strong norm. Use the ordinary two-difference product rule and the first three cutoff differences, so that both its point and Hölder bounds have this factor. In the second term apply the displayed divergence form for $[\Delta,\chi]$ to the bounded Δu, using only its values. This gives a grade-three divergence coefficient and a grade-four bounded term, again with factor C/m. The complete-row commutator of q^*q gives a bounded grade-four term with the same factor. At a true bare boundary the atomic formulas use the original finite matrix and each coefficient has its appropriate bare power. Thus the slow positive parametrix is a contraction on the source class eq. 10.ii. Its exact inverse has the strong local family $G_4 : \rho \mapsto Y^{2,\alpha}$.

For the short-power real extensions of M_4 only its energy family is needed. The same fourth-order compression and cutoff expansion just used yields that family on the real curvature class. In fact the two-difference product rule bounds $\|[\Delta,\chi]u\|_2 \le (C/m)\|u\|_{V_2}$. In the first term of $[\Delta^2,\chi]$ test the outer Δ against the test function. In the second move the skew-adjoint commutator to the test function. This gives $\|[\Delta^2,\chi]\|_{V_2\to V_2^*} \le C/m$. The complete-row cutoff difference gives the same bound for $[q^*q,\chi]$. Use the preceding exponentially weighted energy columns and the same center sums to obtain the convergent walk on $V_2^* \to V_2$, increasing m if needed for both norms. These are the same filed matrix coefficients. The stronger regularity hypothesis is needed on the local collars where their strong output norms are estimated. The rest of a real completion for the short-power spectral bound needs only the energy estimates.

The counting coarse matrix $M_4 = qG_4q^*$ is positive with a uniform gap. For a counting datum β, dualize against $E_s\beta$, use $qE_s\beta = \beta$ and $\|E_s\beta\|_{V_2} \le C\|\beta\|_{\text{count}}$. This gives the lower bound by Cauchy–Schwarz in the real A_4 form. The upper bound follows from the row trace. Its actual family comes from G_4 and the local row maps, with q^* giving a bounded source of grade four and q tracing a grade-zero site output. Apply lemma 9.2. The exact identity is

$$G_sR = (G_4 - G_4q^*M_4^{-1}qG_4)\Delta. \tag{10.iii}$$

At real coefficients the right side applied to f is the unique zero-average minimizer of $\|\Delta u - f\|_2^2$. Its normal equation gives $\Delta u = Rf$ by $\Delta(\ker q) = \operatorname{ran} R$. This is exactly G_sRf. The identity continues bilinearly. Composition of the local families in eq. 10.iii uses $f \mapsto \Delta f$ as a source in eq. 10.ii, without differentiating the input norm. Finally D^T takes $\mathcal{A}_\alpha$ to grade-two C^α on complete boxes, and D takes $Y^{2,\alpha}$ to $\mathcal{A}_\alpha$. They are finite stencils with the stated grade bounds. This proves the two restoration families and their complete identities. □

10.2. **The two strong named propagators.**

Theorem D (Strong current and row response families)**.** Fix $0 < \alpha < 1$, finite $p > 4/(1-\alpha)$ and a finite carrier rate. Work on the weighted C^1 coefficient class of lemma 10.2. After the fixed geometry and tube choices, and a sufficiently small touching current for these choices, $G_0, G_J, G_{1,J}$ have exact connected families $\mathcal{F}_p \to \mathcal{A}_\alpha$. They also have families from bounded field divergence sources $\nabla_U^T F$, with F of grade two, to the point field norm of grade one. If F has the scaled grade-two C^α norm, the output is $\mathcal{A}_\alpha$. In particular one can use $p = 8,\ \alpha = 1/4$ for the first-order regular-pair families.

For either current variant, the coupled response and lift have the same strong families with normalized datum $\Xi_f b$, normalized multiplier $\Xi_f^3\mu$, and field force norm $\mathcal{F}_p$. Thus for sources and data in a single cell or row, each of these normalized output norms has the exponential factor of its distance to the source. The two current readouts and the mass-removed coarse operator retain the families of theorem C. The constrained Green operators are obtained by their stated algebraic subtractions, including the $\mathfrak{G}$ gauge term. ■□

Proof. Use the exact touching-current split before expanding coefficients, as in theorem C. First use the two scalar families of lemma 10.2 and the normalized site Gram inverse. The point column

$\widehat{F}$ has site amplitude of grade two and $D\widehat{F}$ has amplitude of grade three. When $\widehat{F}^T$ pairs a grade-two field with one such column, the physical volume contributes ξ_y^4, canceling the two grade-two factors. Hölder's inequality gives the same cancellation for the normalized $\mathcal{V}_p$ input. Consequently $R = I - \widehat{F}\widehat{M}^{-1}\widehat{F}^T$ is bounded as a family on $\mathcal{V}_p$ and on the grade-two site space with bounded first difference of grade three. The exact adjoints then satisfy

$$\begin{aligned} \Lambda^T &= DRG_s : \{\text{bounded site grade four}\} \to \{\text{bounded field grade three}\}, \\ S^T &= I - DRG_sD^T : \mathcal{F}_p \to \mathcal{F}_p. \end{aligned} \tag{10.iv}$$

The second map uses the finite-p divergence estimate. In particular $\Lambda^T D^T = I - S^T$ applies to a bounded field force through its finite-p norm.

The positive vector family and eq. 7.i give $G_0 : \mathcal{F}_p \to \mathcal{A}_\alpha$. Indeed $D\widehat{F}$ gives bounded field grade-three forcing, and its transpose pairs that grade with grade-one fields on physical cells, so the volume factors again cancel. The intermediate $\widehat{\Theta}^{-1}$ acts in normalized counting coordinates. For $u \in \mathcal{A}_\alpha$ put $\lambda = \Lambda u$. Its input $D^T u$ is bounded of grade two, and the smoothed columns of P show that R preserves this slot. Apply the bounded site point-force family to $RD^T u$, changing the normalized input from grade four to grade two. The extra factor $(\xi_z/\xi_y)^2$ for the local column is absorbed by the scale reserve. This gives λ of bounded grade zero and $D\lambda$ of bounded grade one. These are the site outputs used in the following current form:

$$B[J]u = W[J^{\mathrm{t}}]\lambda + \Lambda^T\big(W[J^{\mathrm{t}}]^T u - \tfrac{1}{2}W[J^{\mathrm{t}}]^T D\lambda\big) - \tfrac{1}{2}\Lambda^T D^T W[J^{\mathrm{t}}]\lambda.$$

The first term is bounded field grade three. In the last term $W[J^{\mathrm{t}}]\lambda$ has that same grade and enters the finite-p divergence slot of eq. 10.iv. Each coefficient reads its current bond and the complete background at all ends. The result is a family from $\mathcal{A}_\alpha$ to $\mathcal{F}_p$ of norm at most $C_{p,\alpha,\vartheta}\gamma_{\mathrm{t}}$. Thus the first Neumann series of theorem C converges in these strong slots.

For the second variant, the first lift gives the same rooted multiplier family as before, since its strong estimate includes the cell pairing required in corollary 7.3. Both absolute normalized sums of the two-leg Hessian are bounded by eq. 7.iv and its pointwise row proof. In particular C_ω takes a bounded grade-one field to a bounded grade-three force. Its two legs and the multiplier retain their joined data carrier. The right factor $Su = u - D\lambda$ is bounded of grade one by the preceding site estimate. On the left use S^T in the finite-p slot eq. 10.iv. Thus $S^T C_\omega S$ has norm $C_{p,\alpha,\vartheta}\gamma_{\mathrm{t}}$ from $\mathcal{A}_\alpha$ to $\mathcal{F}_p$. The second Neumann series converges after the corresponding fixed decrease of the ceiling. The stronger $C^{1,\alpha}$ domain in lemma 10.3 is used for the separate second-derivative site conclusion.

Here are the divergence input slots. Use the normalized field space $X_{1,p,y} = \xi_y^{1-4/p}\|u\|_p + \xi_y^{2-4/p}\|\nabla_U u\|_p$. For $G_F\nabla_U^T F$, the same variational energy start and local divergence bootstrap as for G_s give $X_{1,p}$ with F of grade-two L^p. The equation is componentwise by lemma 10.1, and the additional row force is pointwise bounded by the energy. The slow commutator of G_F now maps $X_{1,p}$ to $\mathcal{F}_p$; only the rightmost factor of the positive walk needs the divergence source estimate. A grade-two C^α source F gives an $\mathcal{A}_\alpha$ output at that rightmost factor and in the no-error term, by the interior divergence Schauder estimate. In that estimate put the connection's lower terms and the finite curvature kernel in finite L^p forcing, using the $W^{1,p}$ bound just obtained. The rest is ordinary divergence of a C^α field. The corresponding statement at the bare boundary follows by finite summation on the atomic stencil.

For the current corrections on $X_{1,p}$, the actual $D^T u$ is grade-two L^p, and R is bounded on that site slot by the same column pairing. The positive site point-forcing bootstrap gives two L^p differences of $\Lambda u = G_s RD^T u$. This is the site estimate proved above with a grade-two L^p point source rather than grade four, and with output grade zero. The extra scale ratio between the two normalizations is paid once in the spatial reserve. Morrey gives bounded λ and $D\lambda$. Therefore the displayed $B[J]$ formula and the C_ω bound hold also from $X_{1,p}$ to $\mathcal{F}_p$. For $S^T C_\omega S$ use the pointwise grade-one output of Su and the same finite-p adjoint. This gives the divergence-input statements

by using the divergence source at the rightmost inverse factor. All subsequent loops use $\mathcal{F}_p \to \mathcal{A}_\alpha$. The row trace and transpose have bounded strong cell norms by their complete row moments. The normalized row inverse and eq. 7.v therefore give the strong lift and saddle.

For $G_{1,J}DRD^TG_{1,J}$, the first $G_{1,J}$ takes a point force to $\mathcal{A}_\alpha$. Its D^T output is grade-two C^α because the comparison connection is Lipschitz. The smoothed columns of P have bounded first differences, so R preserves this Hölder slot. The remaining D is a field divergence of this site datum. Explicitly, for T_μ as in lemma 10.1, $(I - T_\mu)z = (T_\mu^{-1} - I)(T_\mu z)$. Thus the tensor with component $\delta_{\mu\nu}T_\mu z$ gives Dz under the declared forward-difference transpose on full stencils. At a true bare boundary retain the finite remainder $Dz - \nabla_U^T F$ on its actual stencil. It is a bounded grade-three point force, so the ordinary bare force estimate controls it. The $G_{1,J}$ divergence estimate gives the desired output. With a bounded divergence source at the rightmost factor, use $X_{1,p}$ there and the grade-two L^p slot in the middle. This proves the two corresponding estimates for $\mathfrak{G}$. The same column pairings give the pointwise force family $DPD^T : \mathcal{A}_\alpha \to \mathcal{F}_\infty$ and the raw lift transpose bound. □

Corollary 10.4 (Output bounds for the compressed inverses)**.** Use the actual or the full single-comb rows and the ambient coefficient stock of proposition 6.3. Assume the regular local frame bounds of proposition 3.10. Let G_{c} be the inverse of the compressed A_s, or the inverse of the compressed zero-current K_0. In the former case ∇ denotes the site difference and L_{c} the scalar site Laplacian on the prescribed graph or Dirichlet compression. In the latter case ∇ denotes the component difference and L_{c} the compressed component covariant Laplacian.

For bounded f supported in the cell B_z, $u = G_{\mathrm{c}}f$ satisfies

$$\begin{aligned}\xi_y\|u\|_{\infty,B_y} + \xi_y^2\|\nabla u\|_{\infty,B_y} + \xi_y^{2+\alpha}[\nabla u]_{\alpha,y} + \xi_y^3\|L_{\mathrm{c}}u\|_{\infty,B_y}\\ \le C_\alpha e^{-sd(y,z)}\xi_z^3\|f\|_\infty\end{aligned} \tag{10.v}$$

for $0 < \alpha < 1$. For $u = G_{\mathrm{c}}\nabla^T F$ with F supported in the complete source collar of B_z, one has $\xi_y\|u\|_{\infty,B_y} \le Ce^{-sd(y,z)}\xi_z^2\|F\|_\infty$. If F is C^α on that full stock, including its support boundary, the first three terms of eq. 10.v are bounded by

$$C_\alpha e^{-sd(y,z)}(\xi_z^2\|F\|_\infty + \xi_z^{2+\alpha}[F]_{\alpha,z}).$$

Use complete interior stencils and the prescribed bare boundary convention. These bounds continue to a fixed smaller weighted C^1 tube on the same ambient stock, including the crossing plaquettes, around the real reference, with norms measured in its comparison frames.

The normalized compressed row matrix $a_f^{1/2}QG_{0,\mathrm{c}}Q^Ta_f^{1/2}$ has a uniform positive real gap. Its inverse and the exact lift

$$H_{\mathrm{c}} = G_{0,\mathrm{c}}Q^T(QG_{0,\mathrm{c}}Q^T)^{-1}$$

have exponential local bounds, with holomorphic continuation on the same smaller tube. For a datum supported on a row c, the value, first difference and gradient Hölder terms of $H_{\mathrm{c}}b$ in the normalized output above are bounded by $C_\alpha e^{-sd(y,c)}|\xi_c b(c)|$. ■□

Proof. The weighted energy form estimate for either compressed positive inverse gives a cell energy bound of order $e^{-s_0 d(y,z)}\xi_z^3\|f\|_\infty$. In fact the source has energy-dual norm at most $C\xi_z\|f\|_{2,B_z}$. For a divergence source the pairing moves its difference onto the test field, so its energy-dual norm is bounded instead by $C\|F\|_{2,B_z^+} \le C\xi_z^2\|F\|_\infty$. This uses the full prescribed stencil in the source collar. The crossing plaquette forms are local and occur only in the bare stock. The extra scalar Dirichlet diagonal commutes with the spatial weights. For the projected field form use the same site-column bound for R as in the real gap proof. The conjugated perturbation estimates are uniform for the real and complex coefficient classes in the statement.

On complete interior boxes the site equation is

$$L_{\mathrm{c}}u = f - q^Ta_squ.$$

For the field equation the component stencil gives

$$L_c u = f + DPD^T u - Q^T a_f Q u - Z_U u.$$

Both row masses have the stated bounded point forcing from the energy and the full normalized row densities. For $DPD^T u$ use $D\widehat{F}\widehat{M}^{-1}(D\widehat{F})^T u$. Its point columns $D\widehat{F}$ follow from the site bootstrap with the bounded injection $q^T \Xi_s^{-4} e_z$, with the same scaling and complete rows as in lemma 10.2. Their transpose pairs with the local field energy by the physical cell volume, giving a point force of grade three. The finite curvature shifts in Z_U are bounded on the local L^p spaces. In unit cell coordinates the site and vector local bootstraps therefore give the first three point-output bounds. Inserting these bounds in the two equations gives the Laplacian bound. At a true bare boundary $\xi_y = \eta$ and the same conclusions follow from the local energy and the finitely many stencil entries in those units.

For divergence input, retain the divergence in the local equations and apply the finite-p first-order bootstrap. This gives the point field for $p > 4$, including with the bounded projection and row terms just estimated. If F is Hölder, put the other lower-order terms in finite L^p on a smaller cutoff stock. The flat resolvent then bounds their gradient Hölder norm by eq. 3.x, and the divergence term uses the two-difference multiplier of proposition 3.3. The difference between the flat and covariant divergences is bounded multiplication by the connection potential. This proves the claimed derivative range and uses only the fixed scaled first-difference bounds on the complex coefficients. The bare divergence estimate again uses finite stencils. Applying these local arguments to the weighted columns costs only fixed collar widths in the metric. If the site column is read in its grade-two output and grade-four source conventions, conversion to the scales in eq. 10.v costs ξ_z/ξ_y; reserve this single ratio using eq. 2.xi. The normalized single-comb rows have the same finite moment and perturbation bounds by proposition 6.3, so all these computations apply to that full differential as well.

For the row matrix, dualize against the uniformly bounded exact row lift in the compressed energy form, as in Theorem B. This proves the normalized positive gap. The just-proved weighted $G_{0,c}$ bound and the complete row bounds give an $O(s)$ change of the normalized row matrix under exponential conjugation. Its Neumann inverse therefore has exponential decay. The physical row-transpose injection of one normalized datum has bounded grade-three point density. Compose its compressed point-source estimate with that row inverse to obtain the stated lift bounds. The same relative perturbation proves their complex continuation. □

Bounds independent of localization choices. The ordinary point and gradient estimates for the two inverse operators can use the current ceiling of the weighted energy theorem before any walk tile is chosen. For a cell source, the weighted energy inverse bounds the field in X_y, and eq. 7.ii bounds its restored site parameter in $\mathcal{Z}_{s,y}$, with the same exponential localization after a scale reserve. On unit-scale boxes, strict Sobolev embedding gives the field and $D\lambda$ in L^3 and λ in L^{12}. The first current correction is consequently in field L^8: $W[J^t]\lambda$ is in L^{12}, the two site point sources $W[J^t]^T u$ and $W[J^t]^T D\lambda$ are in L^3, and the last source is a divergence with coefficient in L^{12}. A site L^3 point source gives a derivative in L^8 by two finite-p differences and strict embedding. The site divergence source gives the same derivative bound by the variational estimate. Projection by R preserves these bounds by its smoothed columns. The second correction is initially in field L^3 by both absolute sums of its row Hessian and the finite-p bound of S^T.

In the actual field equation, $DPD^T u$ is a bounded point force by the point columns of $D\widehat{F}$ paired with the cell energy. The other terms are the two current corrections just bounded and the given force. Thus the local $W^{2,3}$ vector estimate gives bounded field and gradient in L^8. The site equation for Λu now gives bounded $\lambda, D\lambda$, since its right side $RD^T u$ is in L^8 in scale units. Both current corrections consequently belong to every required finite L^p, by eq. 10.iv and the two-leg point bound. The final $W^{2,p}$ estimate gives the asserted point and gradient Hölder bounds for

$G_J, G_{1,J}$ on $\mathcal{F}_p$. Each estimate is local on fixed boxes and retains the exponential cell weight. For a divergence input, keep it in variational form throughout. The first vector bootstrap gives $W^{1,3}$, then $W^{1,8}$ and bounded field, from the same initial current bounds. This yields the same final finite-p derivative bound for bounded divergence data, and the divergence Schauder step for Hölder data. The ordinary point estimate for the component Laplacian at zero current follows directly from its equation with the bounded row, curvature and DPD^T sources. At a nonzero independent current these arguments give the finite-p forcing bound instead. They require no derivative of that current.

11. Removing the row mass in the outer precision

Use the separated inner hierarchy in top units. Retain a union of complete L-blocks of the next lattice, separated from Ω_k^c by the fixed required collar. Let X_{out} be the level-k bonds touching those blocks, and let ι extend fields on X_{out} by zero to the independent determining row space. All incident double blocks, outer pivots, contours and curl plaquettes used below lie in the interior top stock, with their complete fine backward carriers in Ω_k. These conditions hold for the proper fluctuation geometry with sufficiently large fixed separation. The outer pairing is counting measure, since the supporting top rows have length one. Inner row pairings remain physical.

Form the ordered current inverses and lifts of Theorem B. Write

$$\Sigma_1 = (QG_{1,J}Q^{\mathsf{T}})^{-1}, \qquad \omega_0 = H_J^{\mathsf{T}} J, \qquad \omega_1 = H_{1,J}^{\mathsf{T}} J, \qquad a_{\mathrm{top}} = \iota^{\mathsf{T}} a_f \iota.$$

The first multiplier enters $K_{1,J}$. The second will enter the outer curvature term in the proper extension. On physical stationary pairs they agree, as proved below.

Lemma 11.1 (The top curl after row-mass subtraction)**.** At a real reference in the curvature-small class with a sufficiently small admitted independent current, put $V = \mathcal{M}^k U$ on the complete outer stock. For a supported on X_{out}, extend it by zero to the curl plaquettes $\mathcal{P}$ of Lemma 4.2. Then

$$\begin{aligned} &\langle a, (\iota^{\mathsf{T}}\Sigma_1\iota - a_{\mathrm{top}})a\rangle \geq c\|\mathrm{d}_V a\|_{2,\mathcal{P}}^2 - C(\kappa+\gamma)\|a\|_2^2, \\ &\|\iota^{\mathsf{T}}\Sigma_1\iota - a_{\mathrm{top}}\|_{2\to 2} \leq C. \end{aligned} \tag{11.i}$$

The constants are independent of the depth, the proper inner geometry and the admitted volume. ∎

Proof. The required flux estimate is

$$\|\mathrm{d}_V Q v\|_{2,\mathcal{P}} \leq C\|\mathrm{d}_U v\|_2 + C\kappa\|v\|_X, \tag{11.ii}$$

where the left side uses the four actual top rows of each retained plaquette. To prove it, first use the flat coefficient formula on the top grid of spacing one and put $N = \eta^{-1}$. Write A_η for the average over the centred N^4 sites and $S_\mu = \eta\sum_{j=0}^{N-1} T_\mu^j$ for the segment mean with ordinary fine shift T_μ. Then $Q_\mu = A_\eta S_\mu$ sampled at the unit-grid origins. The exact identity $T_\mu^N - I = S_\mu(T_\mu - I)/\eta$ gives

$$(\mathrm{d}_c Q v)_{\mu\nu} = A_\eta S_\mu S_\nu (\mathrm{d}v)_{\mu\nu}.$$

The nonnegative flux row has row sum one and counting column sum η^4, also as an upper bound when rows are restricted. In fact the translates of the blocks give that column sum for A_η, and the two segment means preserve it. Thus the flux map is a contraction from the physical fine L^2 norm to the unit-grid L^2 norm.

For a real field choose a common comb frame on the complete fixed-size physical cube about each top plaquette, containing all four fine carriers. All assigned scales in this stock are one. The fine potential is $O_L(\kappa)$, and the coarse bonds of V have angles $O_L(\kappa)$ in the same frame, by the summable raw transport comparison. The actual field row differs there from the flat row by $C_L\kappa\eta^4$ per coefficient. Its physical L^2 functional norm difference is $O_L(\kappa)$. The fine curl differs from the flat curl by $O_L(\kappa)$ as a map on local fine L^2, because its transport difference is $O_L(\eta\kappa)$ before

division by η. The outer curl is a unit-spacing difference and costs a fixed constant on row errors. Insert these estimates in the flat identity. Summing their squares uses the bounded overlap of the complete cubes, and yields eq. 11.ii. Unitary changes of frame preserve these local estimates.

Take $v = H_{1,J}\iota a$. Then $Qv = \iota a$, $\|v\|_X \le C\|a\|_2$, and $K_{1,J}v = Q^{\mathsf{T}}\Sigma_1\iota a$. Subtracting the row-mass form on this field gives exactly

$$\langle a, (\iota^{\mathsf{T}}\Sigma_1\iota - a_{\mathrm{top}})a\rangle = \langle v, (H_W + D_U R D_U^{\mathsf{T}} - B[J] - S^{\mathsf{T}}C_{\omega_0}S)v\rangle_f.$$

The gauge term is nonnegative. The current form, the second-jet contraction and the weighted restoration are bounded by $C\gamma\|v\|_X^2$, using Proposition 7.2, Corollary 7.3 and eq. 6.ii. The Wilson bound eq. 6.iii gives a positive curl square up to $C(\kappa+\gamma)\|v\|_X^2$. By eq. 11.ii, $\|\mathrm{d}_U v\|_2^2 \ge c\|\mathrm{d}_V a\|_{2,\mathcal{P}}^2 - C\kappa^2\|v\|_X^2$. This proves the lower estimate. The normalized row inverse and unit top lengths give the operator upper bound. □

Let q_V be the actual raw one-step plain constraint logarithm on the complete outer stock, and let Φ_V be its pivot graph, with derivative C_V and free restriction the identity. Its second derivative is

$$D^2\Phi_V(0)[u,w] = -h_V D^2 q_V(0)[C_V u, C_V w],$$

with h_V the pivot-supported right inverse. For a subset Y_{out} of the retained free coordinates, e_Y extends by zero on the other free coordinates. We retain all incident pivots when applying $C_V e_Y$, even if they lie outside the selected free subset. A row that reads an internal link of a retained block is incident to that block. A crossing link is read only by its longitudinal face row, since internal contours use no crossing links. Thus zero extension creates no pivot outside the declared incident stock.

Theorem E (The proper outer fluctuation gap)**.** With the preceding proper stock and the hypotheses of Lemma 11.1, define the symmetric outer matrix by

$$\begin{aligned}\langle u, A_Y(\tau)w\rangle = {}& \tau\langle C_V e_Y u, (\iota^{\mathsf{T}}\Sigma_1\iota - a_{\mathrm{top}})C_V e_Y w\rangle \\ & + \tau\langle \iota^{\mathsf{T}}\omega_1, D^2\Phi_V(0)[e_Y u, e_Y w]\rangle \\ & + \langle b_V C_V e_Y u, b_V C_V e_Y w\rangle.\end{aligned} \tag{11.iii}$$

For sufficiently small fixed κ, γ and real τ in a fixed neighbourhood of one, this matrix has bounds $c_L I \le A_Y(\tau) \le C_L I$. On the subspace with zero reconstructed field on each retained internal tree, the same estimates hold without the last term. Its inverse is holomorphic and uniformly bounded on a fixed smaller weighted C^1 complex background and independent-current tube, and for complex τ near one. The bounds are independent of depth, the proper inner region, the free subset, and compatible volume. ■□

Proof. The pivot graph has uniformly bounded first derivative, with $\|C_V e_Y u\|_2 \ge \|u\|_2$ by its free restriction. Its fixed one-step carriers and second-derivative bounds give

$$|\langle \iota^{\mathsf{T}}\omega_1, D^2\Phi_V(0)[e_Y u, e_Y w]\rangle| \le C_L\gamma\|u\|_2\|w\|_2$$

by the second readout bound and bounded overlap. Combine eq. 11.i with the geometric inequality eq. 4.iv on $C_V e_Y u$. Its row term vanishes. The two $O_L(\kappa+\gamma)$ errors are absorbed in the curl-contour gap. For tree-zero fields use the version of that inequality without contours. Tree links are internal and are never pivots, so the tree-zero subspace is fixed in the free coordinates. The upper bound follows from the operator bounds of all factors. An increment of τ costs $C_L|\tau - 1|$ in norm, so the bounds persist.

For the complex domain, the ordered field inverses, normalized coarse Schur matrices, and their current readouts are holomorphic on the common admitted tube. On a fixed smaller tube their changes are $O(\varepsilon)$ in their energy, normalized row and weighted current norms when the joint coefficient increment is ε. This follows by the inverse identity and Cauchy's formula applied to

the bounded normalized operators and readouts, respectively. Restricting to unit top rows turns these into the stated outer norms. The actual hierarchical coarse field varies by $O_L(\varepsilon)$ in raw unit-spacing coefficients, by the summable averaging and node-frame bounds. The outer graph and contour derivative have fixed one-step analytic carriers, so the matrix difference is at most $C_L(\varepsilon + |\tau - 1|)$, uniformly in Y_{out}. A smaller tube gives a norm difference less than half the real gap, proving the claim by a Neumann series. □

11.1. **Identification on the physical chart.** At a given physical critical pair, assume $J = j_U$ is the actual inner Wilson current and $j_U = Q^{\mathsf{T}}\omega$. If the field is critical on the fixed Landau constraint slice, the full stationarity identity follows because $QS = Q$ and the Wilson Ward identity kills the gauge part $I - S$. The row lift proves uniqueness of ω. Both $QH_J = I$ and $QH_{1,J} = I$ then give $\omega_0 = \omega_1 = \omega$. The physical Hessian $H_L = H_W - C_\omega$ uses the same constraint jet. The physical Ward identity for H_L and the second lift's horizontality give

$$H_{1,J}^{\mathsf{T}} H_L H_{1,J} = \Sigma_1 - a_f. \tag{11.iv}$$

Indeed $H_{1,J}^{\mathsf{T}} K_{1,J} H_{1,J} = \Sigma_1$ by its defining equation. The gauge term vanishes, the row penalty contributes a_f, and the remaining form on that slice is H_L.

Let a twice differentiable critical branch for this fixed fine action have coarse parameter B and reduced datum $b(B)$. Use the relative exponential coordinates at the given critical pair U, and freeze its reference slice as $\ker(R_U D_U^{\mathsf{T}})$. The restriction of Q to this slice is onto by the restored row lift. Near zero, its fixed right inverse continues by a finite-dimensional Neumann inverse for $D\mathfrak{q}(a)$ on the same slice. The restricted stationarity multiplier is therefore unique and continuously differentiable along the given branch. Differentiating this stationarity equation yields the constrained Lagrangian saddle with row value $Db(0)$. By uniqueness of that saddle its tangent is

$$H_{1,J} Db(0).$$

The second chain rule then gives the effective action Hessian

$$D^2\mathcal{A}_{\mathrm{eff}}(0) = Db(0)^{\mathsf{T}}(\Sigma_1 - a_f)Db(0) + \langle \omega, D^2 b(0)\rangle_r. \tag{11.v}$$

In fact differentiating the datum constraint twice replaces the fine second-variation term paired with j_U by $\langle \omega, D^2 b\rangle_r$ minus the fine constraint Hessian paired with ω. That subtraction is precisely C_ω.

On a complete interior top stock with all lower-layer data fixed, the specified relative left coarse coordinates give $Db(0) = \iota$ and $D^2 b(0) = 0$ on the variations under consideration. The interior equation uses the same left increment $e^{iB(c)}$ relative to $V(c)$. An outward equation can have an extra endpoint gauge factor from the lower-layer data. It is constant when those data are fixed, whereas an absolute logarithm would still have a second jet at nontrivial $V(c)$. Equation 11.v retains the extra term in such coordinates.

The outer chain rule on the actual pivot graph adds $\langle \iota^{\mathsf{T}}\omega, D^2\Phi_V(0)\rangle$. Together with the relative contour Hessian this identifies $A_Y(1)$ in eq. 11.iii with the physical action-plus-contour Hessian. For $A_Y(\tau)$ the factor τ multiplies the action part, including its multiplier graph term; the contour gauge weight stays fixed. When the reference contours are one, the relative and absolute contour Hessians agree. In the homogeneous geometry this gives the effective curl bound and the next-step Gaussian gap with the same uniform constants. There is also a homogeneous independent-current extension obtained by using ω_0 in the second line of eq. 11.iii. Its gap and continuation have the same proof using the first readout estimate. The two definitions agree on physical pairs by the exact stationarity identity. The linear estimates require only the specified Hessian and chart data; the existence of the critical branch is a separate nonlinear input.

12. Conditional matrix functions and complex Gaussians

Use normalized counting coordinates on the outer free bonds. Let A_Y be the precision on the indicated selected free domain, with zero extension of those coordinates when it is formed from the full outer pivot graph. Thus compression precedes inversion. A subsequent coordinate restriction χ_W, $W \subseteq Y$, takes the covariance marginal after conditioning on Y, at complete parameters. The parameter extensions below have guards that depend on W; away from complete parameters no marginal consistency between different choices of W is asserted. On a tree-zero domain the missing tree coordinates are included in the mask throughout.

For the proper outer form eq. 11.iii, the two premises for the following construction are the real gap of theorem E and a connected family for that same precision. To obtain its family use eq. 9.vi for the second row Schur, and the second-lift rooted readout of theorem C for its pivot curvature. The actual pivot derivative, pivot second jet and relative contour coefficient have fixed one-step carriers. The current root and its fine input cell stay in each coefficient. The compression mask is read on that carrier before the two outer coordinate restrictions are applied. The resulting family has the required row and column sums with local coefficient suprema. The unit outer scale and the bounded number of outer variables per cell allow the coarse short-power construction. In a proper hierarchy the short endpoint balls can be chosen inside the last inner region by increasing the fixed separation after their radii have been chosen. Their background cutoff extension uses the outer unit scale there, and their long words retain all mixed-scale reads. These verifications also apply to a homogeneous inner hierarchy with its specified precision and its own stipulated current lift in the pivot contraction. The coefficient and real-gap premises concern each particular matrix on its own domain.

Theorem F (Filed covariance and logarithm)**.** Let the real symmetric precision A_Y on normalized outer free coordinates have $aI \le A_Y \le b_0 I$, and let its exact connected family, local coefficient tube and selection mask satisfy lemma 9.2. This includes eq. 11.iii on its admitted real and complex domains. For each fixed cube radius $\mathcal{S} > 2$ there are fixed aligned parameter cubes and complete root guards such that the following extensions exist.

Use one common admitted real comparison on Y for the covariance sector. The separate real completions of individual entries establish local supremum bounds; they do not replace this common comparison. The filed inverse and logarithm are normally convergent on $|s_\Delta| < \mathcal{S}$. At all ones they give A_Y^{-1} and the logarithm continued from its positive real value. The logarithm has a uniform diagonal trace per outer cell. For every finite selected coordinate set $W \subseteq Y$ there is a holomorphic symmetric covariance $D_W(s)$ on the same polydisc, with

$$D_W(1) = \chi_W A_Y^{-1} \chi_W^T. \tag{12.i}$$

For a real positive comparison matrix $D_{0,W}$ with $b^{-1} I \le D_{0,W} \le CI$,

$$\|D_{0,W}^{-1/2}(D_W(s) - D_{0,W})D_{0,W}^{-1/2}\| < \tfrac{1}{4} \tag{12.ii}$$

holds uniformly in W, depth and compatible volume on the common smaller coefficient tube. The difference can be made smaller than any prescribed inverse power of the parameter cube width. The logarithm and covariance coefficients retain their declared reads. Zeroing parameters outside a wall gives the corresponding guarded wall dependence and factorization across separated guarded components. ■□

Proof. Use the construction of lemma 9.2 with $H = I - A_Y/b$, $b > 2b_0$, short matrix S and long differences $E_j = H^j - S^j$. In particular S_* at the real comparison data is symmetric and $0 \le S_* \le q_* I$ with $q_* < 1$. A prefix S^j is filed on the full endpoint balls before absolute summation. At a common real datum its row and column sums are at most $C(1 + B_j)^2 q_*^j$. With independent local data before the sums use $C(1+B_j)^4 (q')^j$, $q_* < q' < 1$, with the real-gap collars and the smaller complex tube there. For fixed R_0 these are respectively quadratic and fourth-power polynomial

factors in j, with constants depending on R_0. Both forms are uniform in the inner mesh. The normalized coordinate count here is the count of outer free slots.

Take $R_0 = Dn$ and file the combined nth power $H^n = S^n + E_n$. Mark each of its coefficients by the product of its declared cube parameters, with parameters of the held guard set to one, and denote the resulting matrix by T_s. The new cube width M_c can be taken comparable to a fixed multiple of $M_0 n(R_0+1)$. The short combined coefficients touch a bounded number of these cubes. The old long carrier reserve pays the other marks by eq. 9.iv. Thus on the whole parameter polydisc

$$\|T_s\|_{\mathrm{Sch}} \le C_{\mathcal{S},L}(1+B_n)^4 (q')^n + CnK^n e^{-cDn} =: r_n.$$

Use D fixed sufficiently large and then n large. The bound tends to zero exponentially up to a polynomial in n. The same estimates apply to the parameterized E_j, $j < n$, with bounds $CjK^j e^{-cDn}$. The geometry can be increased after these finite choices, and all the normalized local estimates retain their fixed margins.

For the logarithm use the joined words in

$$\log A_Y = (\log b) I - \sum_{j=1}^{n} H^j \sum_{v\ge 0} \frac{(H^n)^v}{nv+j}. \tag{12.iii}$$

The identity at complete parameters follows by grouping the absolutely convergent scalar series of $I - H$ in the matrix. Away from complete parameters the right side means the chosen filed extension, with the parameters of each joined carrier. Its absolute convergence follows from the grouped contraction and the finite prefix bounds. Each diagonal trace is a sum of closed carriers through that outer coordinate, and the family bound and bounded number of coordinates per outer cell prove the trace assertion. The inverse uses the analogous grouped expansion eq. 9.v. The two algebraic identities at all ones use the full matrices.

For the covariance construction take the complete radius-B_n guards around the selected roots W, including every short-prefix read and its domain and geometry collars. Hold the short prefixes at their unmarked values, and for $j < n$ put $A_{j,s} = S^j + (E_j)_s$, with $E_0 = 0$. The coefficients of $(E_j)_s$ retain their own joined parameter carriers. Use $c_\ell = 4^{-\ell}\binom{2\ell}{\ell}$ and put

$$\begin{aligned} R_W(s) &= b^{-1/2} \chi_W \sum_{j<n} A_{j,s} \sum_{v\ge 0} c_{nv+j} T_s^v, \\ D_W(s) &= R_W(s) R_W(s)^T. \end{aligned} \tag{12.iv}$$

These are ordinary matrix products of the indicated marked blocks, and T is bilinear. The powers in this definition can repeat a cube parameter. This is a matrix-function extension with held prefixes. The distinct-cube marking of an already filed operator family in corollary 9.4 is the other convention. Both retain the same support union for every product. By $0 < c_\ell \le 1$ and $r_n < 1/2$ the series in eq. 12.iv converges normally with the weighted local sums. At all ones $A_{j,1} = H^j$, $T_1 = H^n$. Grouping the binomial series of $(I-H)^{-1/2}$ gives $R_W(1) = \chi_W A_Y^{-1/2}$. The square with T proves eq. 12.i.

For the comparison sector put $R_* = b^{-1/2} \sum_{j<n} c_j S_*^j$. Since $S_* \ge 0$, $R_* \ge b^{-1/2} I$. Its norm is bounded by $b^{-1/2} \sum_{j\ge 0} c_j q_*^j$. Hence $D_{0,W} = \chi_W R_*^2 \chi_W^T$ has both claimed real bounds, by zero extension of a vector in W. The long prefixes and the $v \ge 1$ terms in eq. 12.iv have norm at most a fixed polynomial in n, R_0 times $r_n + nK^n e^{-cDn}$, since the short-prefix sums have polynomial bounds. The finite unmarked prefix differs from the real comparison prefix by at most $C \sum_{j<n} \Delta_j^+$. Decrease the common coefficient tube after fixing n, so this term is bounded by the same tail budget. It follows that $R_W(s) - \chi_W R_*$ is small in operator norm, uniformly in W. Expand $R_W R_W^T - \chi_W R_*^2 \chi_W^T$, use the uniform bound for R_* and the lower bound for $D_{0,W}$, and obtain eq. 12.ii. The polynomial growth of M_c in n gives the claimed stronger budget by taking n larger before decreasing the tube.

On a wall, a marked factor survives only if every one of its parameter cubes is allowed, with the held guards included. Its full filed carrier remains available there. The unmarked prefix starting at W stays in its root guard. Connectedness then forbids a surviving path between distinct components once intersecting guarded read sets have been joined. The covariance on their selected roots is block diagonal and its normalized integrals factor. Agreement of complete wall and guard reads proves coefficient equality by corollary 9.3. For a small period use the common cover and the deck average from the family proof, retaining all lifted root balls and parameters on their periodic images. Descend the square-root matrix before selecting W. Its periodic sector is reducing for it and its bilinear transpose. With isometric pullback I_{per} this gives

$$I_{\mathrm{per}}^{T} R_{\mathrm{cov}} R_{\mathrm{cov}}^{T} I_{\mathrm{per}} = R_{\mathrm{base}} R_{\mathrm{base}}^{T}.$$

Thus the subsequent selection gives the covariance of the actual base precision at complete parameters. The absolute row and column sums descend with no sheet-count factor. □

Corollary 12.1 (The finite-range first-step precision)**.** In the actual one-step chart of corollary 4.3, let K_A be the Wilson and constraint-curvature part of eq. 4.v and let $K_G = (b_V C_V)^T (b_V C_V)$. For τ near one the precision $K_c = \tau K_A + K_G$, compressed to any admitted selection of its free coordinates, has the local inverse root, covariance and logarithm of theorem F. Here one can use localization cubes of side $O_L(n)$, and the local background and independent current variables are admitted on fixed absolute small tubes in the one-step units. ■□

Proof. The centered independent-current action in the exact pivot chart Φ_V is

$$A_V(\Phi_V(u)) - A_V(0) + \langle J - j_V, \Phi_V(u) \rangle - \langle J, C_V u \rangle, \qquad j_V = DA_V(0).$$

Its first derivative is zero, and its second derivative is $C_V^T H_W(V) C_V + \langle J, \Phi_V''(0) \rangle$. Since $\Phi_V''(0) = -h_V D^2 q_V(0)[C_V \cdot, C_V \cdot]$, this is K_A with its actual constraint curvature. In the contour term use the product of the contour mean at $\Phi_V(u)$ and the inverse of the reference contour mean. Its value is the identity at zero, so the Hessian of the trace gauge term is the Gram form K_G. At the real reference with $\tau = 1$, the gap and upper bound are those of corollary 4.3.

The real gap for a field supported on a free coordinate subset uses only the fixed complete block neighborhoods of its support. Its pivot reconstruction uses one-step carriers, and the curl-contour inequality is the sum of estimates on these blocks and their neighbors. Its real-frame comparison and small curvature errors therefore require a fixed additional collar $R_{\mathrm{gap}} = C_L$. Compression is evaluation on the zero extension within the original pivot chart, including the full crossing rows. This gives the same real bounds on every such local coefficient matrix. Real contour gauges act unitarily on the physical free coordinates, so the same real bounds hold in those coordinates.

The pivot derivatives, Wilson stencil and relative contour map in eq. 4.v have bounded finite range $R_K = C_L$ in one-step units, including all data reads. Each entry is holomorphic in a fixed finite stock of the actual local connection and independent current. If $\mathbf{V} = e^{iE} V$ and $\mathbf{J} = J + Z$, bounded derivatives of these chart coefficients on a smaller local tube give the positive entry envelope

$$\|K_c - K_r\|_{\mathrm{Sch,sup}} \le C_L(\|E\|_\infty + \|Z\|_\infty + |\tau - 1|) =: \delta_K^{+}.$$

Here the supremum is taken before summing the endpoints; bounded incidence gives the same C_L in both sums. This bound uses the relative contour completion and the constraint-curvature term in K_A. The increment of τ multiplies just K_A.

Choose $b > 2\|K_r\|$ uniformly, set $H_r = I - K_r/b$, $H_c = I - K_c/b$, and $r = 1 - \gamma/b < 1$ from the real lower gap γ. Take the full-carrier cutoff above the range of the primitive entries. Then $E_j = 0$, $S = H_c$, and the endpoint balls for H_c^j have radius at most $(j+1)R_K + R_{\mathrm{gap}}$. Combining the powers before taking absolute values gives $C_L(1+j)^2 r^j$ at one common real pair, and $p_j^{+} = C_L(1+j)^4 r^j$ with separate local suprema. The latter bound follows entry by entry by taking the indicated

local real compression with its complete gap collar. If C_H bounds the positive envelopes of H_r, H_c, telescoping their powers gives the coefficientwise difference bound

$$\Delta_j^+ \le b^{-1}\delta_K^+ jC_H^{j-1}.$$

After fixing the finite n, choose the absolute field, current and τ radii so $\Delta_n^+ \le r^n$ and $\sum_{j<n}\Delta_j^+ \le r^n$.

Take the cube side to be a power of L above a fixed multiple of $(n+1)R_K + R_{\mathrm{gap}}$, and keep the union of the corresponding padded cubes at each root as its held guard. The side is $O_L(n)$; the number of cubes met by one combined nth-power entry is bounded independently of n. Its parameterized norm is at most $2\mathcal{S}^{C_L}p_n^+$. The sum of the prefix norms is bounded by $\sum_{j\ge 0}p_j^+ + 1$. These are precisely the grouped-step and held-prefix bounds used in eq. 12.iv and eq. 12.iii. Their exponentially small error beats every fixed inverse power of the new cube width, on the stated common smaller absolute tube.

These bounds also hold when the real and complex pair is given only on an output domain and its declared reads. For a combined entry rooted at y, the compression to the radius-nR_K endpoint ball contains every length-n coefficient path from y. Hence its nth power has the same y-row as the original polynomial on that row's support. The ball's real-gap collar is already in the entry carrier. For the held reference prefix on W, take one common compression to the radius-nR_K neighborhood of W inside its guard. The same path observation identifies every prefix row and gives the real comparison bounds there. Thus deleting entries whose complete carriers lie outside the wall reduces the positive majorants, and the fixed-root comparison uses only the supplied real data in the guard. □

Corollary 12.2 (Normalized local Gaussian)**.** In the sector eq. 12.ii, on $d = \dim W$ real coordinates the holomorphic Gaussian density is

$$d\mu_{D_W}(u) = (2\pi)^{-d/2}\det(D_W)^{-1/2}\exp(-u^T D_W^{-1}u/2)\,du.$$

Its determinant branch is the one continued from $D_{0,W}$. Its integral is one and its total variation is at most C^d, uniformly on the stated complex tube and parameter polydisc. Integration against any fixed bounded Borel test on these real coordinates gives a holomorphic function. For $W = \varnothing$ the integral is one. ■□

Proof. Write $D_W = D_{0,W}^{1/2}(I+E)D_{0,W}^{1/2}$ with $\|E\| < 1/4$. Use the convergent matrix logarithm of $I+E$ to define the determinant factor. The singular value lower bound gives $|\det(I+E)|^{-1/2} \le (4/3)^{d/2}$, and $\|(I+E)^{-1} - I\| \le 1/3$. After the real change $u = D_{0,W}^{1/2}v$, the absolute value of the exponential is bounded by $\exp(-|v|^2/3)$. Thus its integral is absolutely bounded by C^d and is holomorphic in the complex symmetric entries on every smaller sector. It equals one on the open real positive slice, so it equals one throughout the sector by holomorphic uniqueness in the complexification of that real space. The same dominating Gaussian gives holomorphic polynomial moments and parameter derivatives on smaller tubes. The trace bound $|\operatorname{Tr}\log(I+E)| \le 2d\|E\|$ follows from the logarithm series. Since a wall cube contains at most $C_L M_c^4$ outer coordinates, a sector budget $o(M_c^{-4})$ yields the corresponding small log-determinant error per wall cube. The count is on selected outer free coordinates, independent of the inner fine mesh. □

13. Additional rows at a retained inward seam

The proper determining set has no inward rows. An optional convention retains some inward bonds. At each additional seam node choose one as a tying bond, oriented toward that node. Its equation determines the node factor. The independent remaining equations are the quotients of the other inward bonds by the tying bond. We call them edge rows. A node with only one retained inward bond contributes no edge row.

In this section all constraint coordinates are raw logarithms, and both row spaces use counting measure. Let K_p and K_e be the differentials of the proper and edge rows. Their transposes map counting row covectors to the physical fine-field dual. If an edge e is the quotient of b_i by its tying bond b_1, the quotient rule gives

$$K_e A(e) = \widehat{\dot{a}_q[A]}(b_i) - \mathrm{Ad}(V_0(e))\widehat{\dot{a}_q[A]}(b_1). \tag{13.i}$$

Here a_q is the actual reduced logarithm at level q, $V_0(e) = W_q(b_i)W_q(b_1)^{-1}$, and $W_q = \mathcal{M}^q U_0$. A reversed stored bond uses $\widehat{a}_q(\bar{b}) = -\mathrm{Ad}(W_q(b)^{-1})a_q(b)$.

Fix an odd $L \geq 13$. Leave $\Omega'_0 = \Omega_0$. For $q \geq 1$, retain precisely the native aligned level-q blocks $B \subset \Omega_q$ with $\mathrm{dist}_\infty(B, \Omega_q^c) \geq 3\xi_q$. Their union is Ω'_q. Distances are periodic; a full torus has distance infinity from its empty complement and is unchanged. No additional localization grid is needed. Choose the original separation floor larger than five, and larger still as required by the fixed proper-inverse collars. Then

$$\Omega_{q+1} \subset \Omega'_q \subset \Omega_q, \qquad \xi'(x) \leq \xi(x) \leq L\xi'(x). \tag{13.ii}$$

Indeed an original level-q block meeting Ω_{q+1} loses at most ξ_q from that region's separation from Ω_q^c, so it is retained. Every discarded point is within $4\xi_q$ of Ω_q^c. For a point in Ω'_{q+1}, a shortest path to Ω_q^c first travels at least $3\xi_{q+1}$ before exiting Ω_{q+1} and then crosses the original separating layer. Subtracting the discarded strip gives a remaining separation of at least $(\rho + 3L - 4)\xi_q$. Thus the peeled sequence retains the required separation. A site loses at most one level, and complete peeled pillars subdivide the old ones.

Proposition 13.1 (Uniform edge Schur correction)**.** Suppose U_0 is real and the original pair (U_0, j_{U_0}) and peeled pair $(U_0, 0)$ obey the respective regular-pair coefficient hypotheses of theorems A and D, with a common small radius ε_s. The first current is the actual inner Wilson current; the second is an independent zero current. Require these admissions on every complete local box used by either proper inverse. Set

$$\begin{aligned} H_{\mathrm{seam}} &= H_W + DRD^{\mathsf{T}}, & T &= H_{\mathrm{seam}} + K_p^{\mathsf{T}} K_p, & G_p &= T^{-1}, \\ H_p &= G_p K_p^{\mathsf{T}} (K_p G_p K_p^{\mathsf{T}})^{-1}, & C_p &= G_p - H_p K_p G_p, & M_e &= K_e C_p K_e^{\mathsf{T}}. \end{aligned}$$

Theorem A gives $T > 0$ with $a_f = \Xi_f^{-2}$: because $K_p = \Xi_f Q$ uses counting row coordinates, $K_p^{\mathsf{T}} K_p = Q^{\mathsf{T}} \Xi_f^{-2} Q$. The proper field and lift estimates used below are the zero-current case of theorem D. There are $c, C, s > 0$, independent of mesh, depth, volume and the retained subset, such that

$$cI \leq M_e \leq CI, \qquad |M_e^{-1}(e, e')| \leq Ce^{-sd(y_e, y_{e'})}. \tag{13.iii}$$

Define

$$\begin{aligned} H_a(b_p, b_e) &= H_p b_p + C_p K_e^{\mathsf{T}} M_e^{-1}(b_e - K_e H_p b_p), \\ C_a &= C_p - C_p K_e^{\mathsf{T}} M_e^{-1} K_e C_p. \end{aligned} \tag{13.iv}$$

For $K = (K_p, K_e)$ these maps satisfy

$$KH_a = I, \quad KC_a = 0, \quad C_a H_{\mathrm{seam}} = I - H_a K, \quad \ker C_a = \mathrm{ran}\, K^{\mathsf{T}}. \tag{13.v}$$

They have the proper raw-datum-to-field and force-to-field pointwise, first-difference and local gradient Hölder bounds, with a smaller positive metric rate. If the edge space is zero, the correction is zero. On a common smaller complex tube the matrices G_p and M_e^{-1} remain bounded and holomorphic. The maps H_a, C_a continue holomorphically with the stated norm bounds and algebraic identities. ∎

Proof. Let H'_p be the proper raw lift on the peeled sequence. For a field set

$$\|A\|_{\mathcal{E}}^2 = \sum_y \max\{\xi_y \|A\|_{\infty,B_y}, \xi_y^2 \|\nabla_{U_0} A\|_{\infty,B_y^+}\}^2.$$

Use the fixed complete collars from the proper estimates. Their overlap is bounded. The exponential lift bounds and eq. 2.xi give $\|H_p b\|_{\mathcal{E}} + \|H'_p b'\|_{\mathcal{E}'} \le C(\|b\|_2 + \|b'\|_2)$. By eq. 13.ii, every peeled cell meeting an old enlarged cell has a comparable scale, with constants depending only on L and the fixed enlargement. Bound the old maximum by the sum of squares over all those peeled cells. Their bounded overlap gives $\|A\|_{\mathcal{E}} \le C_L \|A\|_{\mathcal{E}'}$.

We construct trial data for an edge row. Write its nontying inward bond as (y_i, x), where $x = y_i + \sigma \xi_q e_\mu$. In the removed strip the three consecutive bonds

$$(y_i, x), \quad (x, x + \sigma\xi_q e_\mu), \quad (x + \sigma\xi_q e_\mu, x + 2\sigma\xi_q e_\mu)$$

are proper interior rows of the peeled geometry. Their three successive inner endpoints lie in the first native level-$(q+1)$ block entered by the inward bond, since $L \ge 13$. That boundary block is removed from Ω'_{q+1}. The original separation places this block and y_i in Ω'_q. Give these three rows the data $(1, -2, 1)v$ in the frames transported along that line. Use the covariant minus sign if a stored orientation is reversed. Call this assignment $J_e v$. Distinct directions at a seam node use distinct bonds. Parallel inward bonds on the same block line are at least $L - 1$ bonds apart. Thus, for $L \ge 13$, the three-bond supports are disjoint and avoid tying bonds. Hence $\|Jb\|_2^2 = 6\|b\|_2^2$ for $Jb = \sum_e J_e b(e)$.

At a flat reference the straight-path average of this pattern vanishes at the next level. The paths through the first endpoint block meet its initial segments of lengths one, two or three. Their sums are $1, -1, 0$; the two nonzero sums occur once each. For the next coarse bond the two nonzero sums are $-1, 1$. The averaging coefficients are equal, so each pair cancels. The transverse averaging factor multiplies zero. This cancellation uses the boundary placement of the pattern, not an arbitrary three-bond translate.

At a curved reference, compare the paired transports in their complete double block. Their difference is a loop with scaled curvature $O(\varepsilon_s)$, so the transport comparison costs $C_L \varepsilon_s$. The derivative of the actual straightened one-step mean differs from that straight-path map by the same amount in its finite row and column sums. To see this, work in the admitted local comb frame and differentiate the mean equation and both endpoint factors. At zero curvature the endpoint derivatives cancel the internal contour derivatives. The remaining derivatives are analytic on a fixed smaller finite chart, so the curvature error is $O_L(\varepsilon_s)$. The derivative of the implicit mean has a uniformly bounded inverse on that chart. Real gauge covariance returns the estimate to the original frame.

Put $A_b = H'_p Jb$. Complete reads shared by the two geometries give the same reduced logarithm, by the exact averaging recursion. Thus eq. 13.i reads b on each nontying bond and zero on its tying bond. An old proper row shared with the peeled set reads zero. If one of an old row's level-j endpoint blocks survives in Ω'_j, the row remains an interior or outward row. An omitted old row therefore has both endpoint blocks outside Ω'_j. Every original bare row is shared, so an omitted row has $j \ge 1$. Its full predecessor double block lies in Ω'_{j-1} by the original separation, so its predecessor rows are proper interior rows there. It reads the preceding pattern through that complete double block. The cancellation just proved and the uniform actual row-composition bounds therefore give

$$K_e A_b = b, \qquad \|K_p A_b\|_2 \le C_L \varepsilon_s \|b\|_2, \qquad \|A_b\|_{\mathcal{E}} \le C\|b\|_2. \tag{13.vi}$$

Bounded double-block overlap supplies both Schur sums. No factor depending on the number of levels is introduced.

Correct into the proper kernel by $A = A_b - H_p K_p A_b$. The local edge-row bound and eq. 13.vi give $K_p A = 0$, $\|A\|_{\mathcal{E}} \le C_A \|b\|_2$ and $\|K_e A - b\|_2 \le C_e \varepsilon_s \|b\|_2$. Choose $C_e \varepsilon_s \le 1/2$. On this kernel the row mass vanishes, and the Wilson curl comparison and orthogonality of R give $\langle A, TA\rangle_f \le C_T \|A\|_{\mathcal{E}}^2$.

The physical cell volume contributes ξ_y^4, cancelling the four inverse length powers from the squared first differences. Adjacent scales have bounded ratio.

The constrained inverse of T has the variational formula

$$\langle f, C_p f\rangle_f = \sup_{K_p u=0} \{2\,\mathrm{Re}\langle f, u\rangle_f - \langle u, Tu\rangle_f\}.$$

Take $f = K_e^{\mathsf{T}} b$ and $u = tA$. The real pairing $\langle b, K_e A\rangle$ is at least $\|b\|_2^2/2$. Choosing $t = (2C_T C_A^2)^{-1}$ proves $M_e \geq (4C_T C_A^2)^{-1} I$. Its upper bound and exponential kernel follow by composing the proper constrained inverse with the local edge maps. The counting-to-physical transposes include the fine density.

For the inverse decay, conjugate M_e by $e^{s\min\{d(y_e, y_{e_0}), N\}}$. The exponential kernel and its first distance moment bound the conjugation error by Cs, uniformly in N. Choose $s > 0$ so that $Cs \leq c/2$ and so that the retained kernel rate has the reserve in eq. 2.xi. The conjugated inverse has norm at most $2/c$. Apply it to an edge coordinate at e_0 and let N increase. This proves eq. 13.iii. If necessary increase the fixed separation so that the final smaller rate is also summable. There are only finitely many edge species at a top.

Direct multiplication in eq. 13.iv proves the first three identities in eq. 13.v. The form T is positive on $\ker K$, and C_a is its constrained inverse there. Its kernel is therefore the physical annihilator of $\ker K$, namely $\operatorname{ran} K^{\mathsf{T}}$. Composing the proper exponential kernels proves the asserted field bounds. For a local gradient Hölder estimate, regard each correction as $C_p f$ and apply the corresponding proper force estimate. The row datum is not differentiated.

Finally the proper inverse and local rows have small uniform perturbations on a common smaller complex tube. The same is true of M_e in the normalized weighted row norm. A Neumann series preserves invertibility and the displayed algebraic formulas. Their transposes are bilinear, so the continuation is holomorphic. □

Corollary 13.2 (The corrected seam saddle)**.** Suppose the real reference is constrained critical for the combined raw constraint $\mathfrak{q}$, with $\mathfrak{q}'(0) = K$ and actual multiplier $\omega_0 = H_a^{\mathsf{T}} j_{U_0}$. Let B be defined by

$$\langle v, Bu\rangle_f = \langle \omega_0, \mathfrak{q}''(0)[u, v]\rangle.$$

Fix $0 < \alpha < 1$ and use the strong field norm $\|u\|_{\mathcal{A}_\alpha} = \sup_y \|u\|_{\mathcal{A}_{\alpha,y}}$ on the complete boxes above. Suppose the combined raw constraint uses the actual admitted mean and quotient charts with their two-leg moment bounds. If the scaled actual current on the touching stock is sufficiently small, the saddle $(H_{\mathrm{seam}} - B)u - K^{\mathsf{T}}\nu = f$, $Ku = b$ has the unique solution

$$u = (I - C_a B)^{-1}(C_a f + H_a b), \qquad \nu = H_a^{\mathsf{T}}((H_{\mathrm{seam}} - B)u - f). \tag{13.vii}$$

It obeys the proper force, raw-datum and multiplier estimates, also in a fixed smaller exponential weight. ■□

Proof. Here is the bound for the added quotient rows. At a row of length ξ_q, write the two oriented reduced raw logarithms as a_i, a_1. Its relative edge logarithm is

$$i^{-1}\log(e^{ia_i}e^{-i\,\mathrm{Ad}(V_e)a_1}).$$

The second differential consists of the difference of the two second jets and the symmetric cross commutator of their first jets, with coefficient $-i/2$. Each raw first jet has absolute sum at most $C\xi_q$ and each one of its coefficients has norm at most $C\xi_q(\eta/\xi_q)^4$. Each raw second jet has fixed-leg sum at most $C\xi_q^2(\eta/\xi_q)^4$. The cross commutator has the same fixed-leg bound by multiplying the first two estimates. These are the actual row moments used in eq. 7.iv, converted to raw rows. The physical fine transpose divides by η^4. Inserting $|u| \leq C\xi_q^{-1}\|u\|_{\mathcal{A}_\alpha}$ therefore gives a force bounded by $C\xi_q^{-3}|\omega_0(e)|\|u\|_{\mathcal{A}_\alpha}$. At one level only a bounded number of full seam carriers read a given fine bond. Different levels have separated seam boundaries, so this incidence bound is uniform in depth. Thus the combined Hessian has the same force bound as the proper constraint Hessian.

For a nonbare or edge unit raw datum, $KH_a = I$ makes its lift vanish on every nontouching bare bond. Its current readout therefore uses only the touching current. The column bound for H_a and the physical scale cancellation $\xi^{-1}\xi^{-3}\xi^4 = 1$ give $|\omega_0(e)| \le C\varepsilon_s$ on every row whose second jet is nonzero. Bare proper rows have zero second jet. The force-to-field estimate for C_a now yields $\|C_aB\|_{\mathcal{A}_\alpha\to\mathcal{A}_\alpha} \le C_\alpha\varepsilon_s$. A fixed reduction of the current ceiling makes this at most $1/2$. The same calculation uses the smaller exponential rate because every quotient reads its complete local carrier.

Apply C_a to the first saddle equation and use eq. 13.v. This gives the first formula in eq. 13.vii. Conversely that formula gives $Ku = b$, and its first-row residual lies in $\ker C_a = \operatorname{ran} K^{\mathsf{T}}$. Since $KH_a = I$, applying H_a^{T} gives the second formula and the unique multiplier.

The Neumann series gives the field bound. To estimate the multiplier without taking a second pointwise derivative of u, use the variational pairing: one curl difference falls on H_ae and one on u. For the gauge term use $\langle D^{\mathsf{T}}H_ae, RD^{\mathsf{T}}u\rangle_f$. The two first-difference amplitudes and physical cell volume again cancel their scale powers. The remaining curvature and constraint terms have their force bounds. The column estimate controls $H_a^{\mathsf{T}}f$. The same proof with one exponential weight on the changed factor proves the weighted estimate. □

14. Restrictions and physical gauge null directions

14.1. **A marginal precision retains the gap.** Work first in real physical orthonormal coordinates on a finite index set. Let $A = A^*$ satisfy $a_-I \le A \le a_+I$, where $a_- > 0$, and assume that its entries have uniformly bounded two-sided exponential sums at rate $\mu > 0$. Split the indices into I and O. Empty blocks have their usual zero-dimensional meaning. The covariance restricted to I has precision

$$K_I = ((A^{-1})_{II})^{-1} = A_{II} - A_{IO}A_{OO}^{-1}A_{OI}. \tag{14.i}$$

This is different from the conditional precision A_{II}.

Proposition 14.1 (Uniform restrictions and weighted perturbations)**.** The bounds $a_-I \le K_I \le a_+I$ hold for every subset I. There are $s_0, C > 0$, independent of that subset and the admitted volume, with the following property. For every bounded real one-Lipschitz function ψ and $|s| \le s_0$,

$$\|e^{s\psi}K_Ie^{-s\psi} - K_I\| \le C|s|. \tag{14.ii}$$

If a possibly complex perturbation B_I satisfies $\|e^{s\psi}B_Ie^{-s\psi}\| \le \varepsilon$, then

$$\operatorname{Re}(v^*e^{s\psi}(K_I - B_I)e^{-s\psi}v) \ge (a_- - C|s| - \varepsilon)\|v\|^2. \tag{14.iii}$$

In particular, $C|s| + \varepsilon \le a_-/2$ gives an inverse bound $2/a_-$ in that weighted space. ■

Proof. The positive block A_{OO} is invertible. Solving the second block equation in $A(u, w) = (f, 0)$ gives $w = -A_{OO}^{-1}A_{OI}u$ and proves eq. 14.i. Equivalently,

$$\langle u, K_Iu\rangle = \inf_w \langle (u, w), A(u, w)\rangle.$$

The lower gap follows by dropping $\|w\|^2$ from the lower bound on A. Taking $w = 0$ proves the upper bound.

Let $A_s = e^{s\psi}Ae^{-s\psi}$. The entrywise inequality $|e^{s(\psi(x)-\psi(y))} - 1| \le |s|d(x, y)e^{|s|d(x,y)}$ and the exponential Schur bound imply $\|A_s - A\| \le C_0|s|$ for $|s| \le \mu/2$. Compression preserves that estimate. Choose s_0 so that $C_0s_0 \le a_-/2$. The resolvent identity then gives

$$\|(A_s)_{OO}^{-1}\| \le 2/a_-, \qquad \|(A_s)_{OO}^{-1} - A_{OO}^{-1}\| \le 2C_0|s|/a_-^2.$$

Conjugation commutes with the block decomposition. Insert these bounds and the bounds for the two off-diagonal blocks in eq. 14.i. Expanding the difference of the three-factor products proves eq. 14.ii. Taking the real part and using the norm bound for B_I proves eq. 14.iii. □

Taking $\psi(x) = \min\{d(x,y), N\}$ in the same argument gives pointwise exponential bounds for the restricted inverse, uniformly in N. Sending N to infinity removes the truncation. When a two-sided cell sum is required, the metric growth must also be paid for as in eq. 2.xi. A small unweighted norm of B_I alone gives no uniform bound for its conjugates.

14.2. **The complete Hessian on an actual fibre.** Let $\mathfrak{q}(a)$ be a finite analytic raw constraint, with $\mathfrak{q}(0) = 0$, surjective derivative P, and pivot right inverse h, so $Ph = I$. Let C identify the free coordinates with $\ker P$. The actual pivot chart has the form $\Phi(u) = Cu + hz(u)$, where $z(0) = Dz(0) = 0$ and $\mathfrak{q}(\Phi(u)) = 0$. Let $\mathcal{A}$ be an invariant action that is constrained critical at the reference. Its gradient is $P^{\mathsf{T}}\omega$. The reduced action $\mathcal{S}(u) = \mathcal{A}(\Phi(u))$ therefore has Hessian

$$K_A[v,w] = D^2\mathcal{A}(0)[Cv, Cw] - \langle \omega, D^2\mathfrak{q}(0)[Cv, Cw]\rangle. \tag{14.iv}$$

Indeed twice differentiating the constraint gives $D^2z(0)[v,w] = -D^2\mathfrak{q}(0)[Cv, Cw]$. Pairing the second chart derivative with the ambient action gradient proves eq. 14.iv.

For the gauge splitting use the complete one-step outer fibre chart. Its residual site gauges fix exactly the block centres, and its contour coordinates are indexed by the remaining sites. This residual action preserves the exact constraint fibre. Let $\mathfrak{h}$ be its site gauge space, $T : \mathfrak{h} \to E$ its tangent in the free chart, and $\Gamma : E \to \mathfrak{h}$ the derivative of its contour coordinate at a block-axial real reference. The contour's centre endpoint is fixed and its other endpoint contributes the inverse gauge factor. Differentiating this endpoint law gives

$$\Gamma T = -I. \tag{14.v}$$

This identity concerns the contour mean on its actual branch. It does not require an inverse Hessian.

Lemma 14.2 (Gauge null splitting)**.** For the complete outer fibre and its action-preserving residual gauge action just specified, the complete fibre Hessian satisfies $K_A T = 0$. Moreover,

$$E = \ker\Gamma \oplus \operatorname{ran} T. \tag{14.vi}$$

For a positive gauge weight a, the form $K_G = \Gamma^{\mathsf{T}} a \Gamma$ vanishes on $\ker\Gamma$. The two summands in eq. 14.vi have zero mixed terms for both K_A and K_G. ■

Proof. The residual gauge acts on the exact constraint fibre. In its free chart, differentiate the action-invariance identity once in the gauge parameter and once in a free direction at zero. The derivative of the gauge tangent is multiplied by $D\mathcal{S}(0) = 0$. The remaining term is $K_A[v, T\lambda]$, which is therefore zero. The chart-curvature term in eq. 14.iv is included in this differentiation.

Every u decomposes as $(u + T\Gamma u) - T\Gamma u$. By eq. 14.v, the first term lies in $\ker\Gamma$. If $T\lambda \in \ker\Gamma$, then $-\lambda = \Gamma T\lambda = 0$, proving that the sum is direct. The remaining assertions follow from the Gram form of K_G. □

The splitting is a decomposition for the quadratic forms; it need not be orthogonal in the original free-coordinate Euclidean metric. It proves no physical gap on $\ker\Gamma$ by itself. That lower bound is supplied by the contour and outer-fibre coercivity estimates. At the block-axial reference the contour gauge coordinate vanishes, so the gauge Hessian is exactly $\Gamma^{\mathsf{T}} a \Gamma$ without a missing second-chart term.

14.3. **The two Gaussian powers of the action weight.**

Corollary 14.3 (Physical gauge splitting and the determinant)**.** Use the actual fibre and gauge decomposition of lemma 14.2. Let $d = \dim E$, $r = \dim \operatorname{ran} T$, and assume the complete physical precision $K = K_A + K_G$ is positive. Retain the reference gauge weight and vary only the action weight, putting $Q_\tau = \tau K_A + K_G$ for $\operatorname{Re}\tau > 0$. With branches continued from $\tau = 1$,

$$\det Q_\tau = \tau^{d-r} \det K, \tag{14.vii}$$

$$\int_E e^{-u^T Q_\tau u/2}\, du = \tau^{-(d-r)/2} \int_E e^{-u^T K u/2}\, du, \tag{14.viii}$$

$$\int_E e^{(\tau-1)u^T K_G u/2}\, d\mu_{(\tau K)^{-1}}(u) = \tau^{r/2}. \tag{14.ix}$$

Here $\mu_{(\tau K)^{-1}}$ is the normalized Gaussian on the full free space, before insertion of fluctuation tests. The two logarithmic normalizations are $-\frac{d}{2}\log\tau$ and $\frac{r}{2}\log\tau$, with sum $-\frac{d-r}{2}\log\tau$. ■□

Proof. Choose a basis of $\ker\Gamma$ followed by one of $\operatorname{ran} T$, and denote the combined coordinate map by B. In these coordinates $B^T K_A B = \operatorname{diag}(A, 0)$ and $B^T K_G B = \operatorname{diag}(0, G)$. The complete positivity gives $A > 0$ and $G > 0$ on their respective blocks. Consequently $B^T Q_\tau B = \operatorname{diag}(\tau A, G)$. The factor $(\det B)^2$ is the same in both determinants, proving eq. 14.vii even when the chosen splitting is oblique in the original coordinates.

The real part of Q_τ is at least $\min(1, \operatorname{Re}\tau)K$ as a real quadratic form. The Gaussian integrals are therefore holomorphic on the right half-plane by dominated integration on its compact subsets. Real Gaussian integration in the two sectors proves eq. 14.viii for $\tau > 0$, and hence for the whole half-plane. Finally $-\tau u^T K u/2 + (\tau-1)u^T K_G u/2 = -u^T Q_\tau u/2$. The normalization of the precision τK contributes $\tau^{d/2}$ against the reference integral. Inserting eq. 14.viii proves eq. 14.ix. Empty blocks use the zero-dimensional integral and determinant equal to one. □

On a full periodic one-step fibre with n coarse sites, there are $4L^4 n$ fine positive-bond occurrences and $4n$ pivots. Each carries three colour coordinates, so

$$d = 12(L^4 - 1)n, \qquad r = 3(L^4 - 1)n, \qquad d - r = 9(L^4 - 1)n.$$

The value of r counts the noncentral sites, with injectivity of their gauge tangents given by eq. 14.v. Consequently the compensating expectation depends only on these dimensions and is constant across the admitted full physical fibres. On another domain the determinant formula requires its actual splitting in lemma 14.2; the full-period counts do not apply to an arbitrary mask.

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